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# ESCROW: WHEN A RECORD STREAM HAS EARNED A NEW NODE

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## ABSTRACT

Each arrival in a stream of records raises the same question: does the record join a node that already exists, or has the stream earned a new one? The rules in use answer with a constant chosen by hand: a distance cut-off in streaming clustering, the penalty the small-variance limit leaves in DP-means and BP-means, or the sampling temperature of a language-model pipeline. In the second case the constant is what a deletion leaves behind: before the limit is taken the new-node penalty is a codelength, and the limit erases the part of it that could have been computed. We do not take the limit. We read the price of a node off a code that predicts each record before seeing it, so the price is exact and grows with the logarithm of the stream, and no quantity in the rule is calibrated on data. That price also settles in closed form which streams the rule is for, since naming the  $t$ th member of a node costs  $\log_2 \frac{n-t+1/2}{t-1/2}$  bits, so its scope is read off the code rather than found by trial. The rule is given no domain either: no schema, no lexicon and no pretrained encoder, only the keys and values as they arrive. No fresh node can pay that price on its first record under any valid code, so evidence accrues in escrow against a node that does not yet exist and the node is created when its account covers its price. On noise it creates nothing across sixteen null families, where streaming decision trees grow to a mean of ninety-one false nodes, except on a null whose keys appear only sometimes. A language model given the identical question on a stream with nothing in it builds eighty to a hundred and ninety-five nodes over two hundred records. It recovers planted structure exactly in all twenty arrival orders. On raw Wikipedia infoboxes an untuned baseline reading the same records beats it, which was our own stated falsifier and which we report rather than reword; so does a code over character three-grams, which cannot represent a word, so that stream ranks methods by vocabulary and not by meaning.

## 1 INTRODUCTION

A stream of records arrives and does not stop. A record might be a product listing, a page with an infobox, a song, or a person, and whatever it is it arrives as a small set of named parts, one key and one value at a time. We want a graph of what the stream contains, built while the stream is still running, with a node for each kind of thing in it and an edge from every record to the nodes it belongs to. That means answering the same question at every arrival. Does this record belong to a node that already exists, or has the stream now earned a new one?

The word worth defending is earned. If a node exists because the data paid for it then there is a price, and something has to say what that price is. Three families of rules decide node existence today and not one computes it. A streaming clustering rule opens a node when the distance to the nearest one exceeds a chosen value. The same shape turns up far from clustering: XGBoost adds a leaf only when the gain exceeds a per-leaf charge  $\gamma$  whose default is zero and for which its own paper reports no value (Chen & Guestrin, 2016). DP-means and BP-means come closer, beginning from a model of how many groups a stream contains (Kulis & Jordan, 2012; Broderick et al., 2013b), but they obtain their penalty  $\lambda$  by sending the noise that model assumes to zero. Language-model pipelines

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Code, experiments and this paper: <https://github.com/Shunyalabsai/ESCROW>

(Bai et al., 2025; Hongwimol et al., 2026) state no price at all: they ask a model record by record and keep what it samples, so two runs over the same data return different graphs (Figure 1), and asked the same question on a stream with nothing in it they build a node for almost every record (Figure 4).

The Bayesian rules are the interesting failure, because their constant was not there at the start. Before the limit is taken the new-node penalty is a codelength, a quantity the model works out for itself from what it has seen. The limit multiplies the objective by the assumed noise variance, and Lemma C.1 shows what that does: it annihilates every part of the penalty that could have been computed and keeps only the part a person inserted. So  $\lambda$  is not a computed quantity that happens to be hard to compute. It is what a deletion left behind.

So we do not take the limit, and that leaves a way to compute the price. A probability model of a stream is also a code for it, since a prediction giving probability  $P$  to what happens writes it down in  $-\log_2 P$  bits. A graph is a shorter description whenever many records share a pattern that can be stated once and pointed to, and a node earns its place when it saves more bits than it costs to state. Both sides are bits, so there is no exchange rate to set. The price comes out of the model exactly, with the constant on  $\log_2 n$  equal to one (Lemma C.2), and no quantity in it is calibrated on data.

That ought to be the end of the rule, and it is not, because the price cannot be paid at the moment it falls due. Ask what a brand-new node knows about the record that suggested it. It has no members and no counts, so its prediction over the values of a key is uniform, and under any valid code a uniform guess cannot beat what the stream already knows. Lemma 1 makes this exact: in expectation a fresh node loses bits on its own first record whatever code we choose, and seeding it with that record’s value only moves the same bits into the seed. So the rule that creates a node the first time a record fails to fit is not a rule that a better code can repair.

What works instead is to wait. The bits a candidate would have saved are credited to an account named by the (key, value) pair that opened it, every later record failing the same way pays into it, and the node is created at the first moment the account covers the price (Figure 2). We call the method ESCROW, after the account.

Waiting raises a question of its own: a candidate tested again after every arrival, forever, has many chances to cross its price by luck. In principle the escrow answers it, because the account is a running score of fair bets and Ville (1939) caps the chance that such wealth ever reaches a given level over the whole life of the stream, with no correction for how often we look. That is the guarantee we would like and not the one we can claim. The statistic the engine accrues is read off the records carrying the candidate’s pattern, a subsequence the data chose, which is what a fair game forbids; under the null the accumulator drifts. We report the measured exceedance instead, and show that no charge added to the price repairs it (Theorem C.1).

The price settles one more thing before any experiment is run. Naming the  $t$ th member of a node costs exactly  $\log_2 \frac{n-t+1/2}{t-1/2}$  bits, so a kind appearing a handful of times in a long stream cannot pay for itself however plain it is to a reader; and a candidate named by one (key, value) pair accrues only on records carrying that pair. Between them they say what the rule is for, streams whose kinds recur often enough to pay their way (Proposition 1), and why we report two public benchmarks we fail rather than dropping them: both ask for groups of about two records, which no setting reaches because neither condition has a setting in it.

**Contributions.** Pricing a component in bits, codelength pruning of trees, computed charges inside boosting and attribute-by-attribute growth from a stream are prior work (Section 6) and are not claimed. What is claimed:

- **A prequential code for an unsupervised, multi-membership, one-pass stream.** That combination removes the two properties the codelength criteria we survey rely on, a conditional code transmitting only a label and additivity over a partition, and it needs no knowledge of the domain: one engine runs unchanged on product listings, encyclopedia infoboxes, music metadata and open-domain noun phrases.
- **A computed mint price, and a lemma that rules out immediate minting.** A Bayes-exact price with constant exactly one on  $\log_2 n$ , and the lemma that no valid code lets a node pay on its first

|                              | A. Distance rule,<br>hand-set cut-off  | B. Bayesian rule,<br>small-variance limit | C. Language model<br>pipeline                                     | D. ESCROW: a price<br>in bits                  |
|------------------------------|----------------------------------------|-------------------------------------------|-------------------------------------------------------------------|------------------------------------------------|
| who decides a<br>node exists | a distance below the<br>chosen cut-off | a fit gain above the<br>constant lambda   | a sampled answer from<br>the model                                | the data: saved bits<br>cover a computed price |
| what a person<br>must set    | a number somebody<br>chose             | a number somebody<br>chose (lambda)       | a prompt, a temperature<br>and a retrieval depth                  | nothing is calibrated:<br>both sides are bits  |
| same output<br>on rerun      | yes, once the number is<br>fixed       | yes, once lambda is<br>fixed              | no: 24, 78, 28 nodes<br>on identical records;<br>agreement 0.1647 | yes: 6 nodes, identical<br>on all 3 runs       |
| a receipt<br>per node        | no, only a distance                    | no, only a penalty                        | no; and one catch-all<br>node holds up to 85%<br>of records       | yes, in bits, for every<br>node                |
| cost per<br>1,000 records    | not run on this stream                 | not run on this stream                    | 107,512 to 157,821<br>tokens; 27 to 31 min<br>on one A100         | 0 tokens; 1.5 s on one<br>CPU core             |

Figure 1: Four ways to decide that a node exists, one per column. ESCROW (D, gold) is the only one whose price is computed. Distance rules (A) and small-variance Bayesian rules (B) answer with a number somebody chose. A language-model pipeline (C) answers by sampling: on 1,000 Lazada listings in one fixed order, three seeds returned 24, 78 and 28 nodes at 107,512 to 157,821 tokens per run, where ESCROW returned the same 6 nodes in about 1.5 s, with no tokens and a receipt per node (results/llm\_graph\_formation.json; Section 5.4).

record, which makes deferral necessary rather than convenient. Neither has a significance level or a grace period.

- **An evaluation stated as falsification tests, including the ones that fired.** Planted structure is recovered exactly in all 20 arrival orders, and no node is created on 48 null streams where streaming trees grow to a mean of 91 and a 14B language model given the identical question builds up to 195 over 200 records. Against baselines reading the same records an untuned distance rule matches us, which was our own stated falsifier; we report it and drop the accuracy claim, and we also report that a code over character n-grams reaches 1.0 on that stream, which says what the benchmark measures. What the price does settle in closed form is which streams the rule is for, and the same counting yields the specialisation, sibling and typed-value edges a graph is made of.

## 2 PROBLEM STATEMENT AND CORE CLAIM

**Setting.** A record  $r_n$  is the  $n$ -th item to arrive, a small set of attribute key and value pairs such as `fabric = cotton`; one (record, key) slot is a facet, and version 1 handles categorical values. A node is a latent group of records sharing a pattern of keys and values, and the keys it models are its support. One node is not latent: the background models an ordinary record, owns whatever no other node explains, and every claim of structure has to beat it. After  $n - 1$  records the state is  $K$  nodes plus that background, each carrying its members, its support and the counts its members published, and a record may belong to several nodes at once. Each record is priced once from the past alone and the rule never revisits it; a bounded repair pass afterwards may merge, reassign or delete, so it is the *criterion* that is one-pass, not the system.

**The question.** Every arrival puts two questions to every facet: which existing node owns it, and if none explains it, has it earned a new node and at which record? A rule answering with a chosen constant has a calibrated parameter, and this paper asks whether the rule can have none (Figure 1).

**The claim.** *A node is created at the first moment the bits the stream has saved by positing it cover the computed bits it costs to state it. Both sides are bits, so no quantity in the rule is calibrated on data, and the same code decides which node each record joins.*

**What is claimed and what is not.** We claim that no parameter is calibrated and that no domain is assumed, which is a constraint as much as a convenience: where identity is visible only to a model that already knows the domain, this one does not see it either. We do not claim the absence of bias, since Kearns et al. (1997) prove that for every sample size there are instances on which any penalty-based rule stays a fixed distance from the best answer, so we measure our creation bias instead (E13). What is new is a code that works where the existing codelength criteria cannot, not a new principle of model selection. Three concessions come with it: the declared prior on the stream’s own minting rate, worth 0.19 bits per mint at  $10^5$  records; a candidate pool held to a declared budget; and categorical facets only (Section 7).

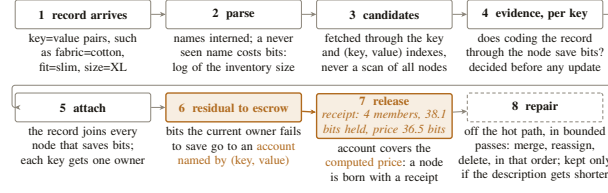


Figure 2: One record through ESCROW. Steps 1 to 5 and 8 are bookkeeping in bits, and the two gold stages are where a node earns existence. Bits the current owner fails to save are credited to an escrow account named by the (key, value) pair that carried them, and when the account covers the price a node is minted with a receipt, here 4 members and 38.1 bits held against a price of 36.5 bits on the synthetic two-group fixture (results/e9\_support\_curve.json).

### 3 THE METHOD, ESCROW

**The code and its terms.** The code is prequential, meaning every record is priced by a prediction made from the past alone (Dawid, 1984), and every term satisfies Kraft’s inequality. A node states two things about itself. The first is its member set, a yes-or-no column over the records so far, coded by the Krichevsky-Trofimov code (Krichevsky & Trofimov, 1981; Willems et al., 1995); that column is the term which replaces  $\lambda$ , and nothing in it is chosen. The second is how reliably its members publish each key in its support, so a member that stays silent on a supported key pays a silence bill, which when two planted groups share no keys is the only signal there is. A record may belong to several nodes, so every facet has exactly one owner, the node with the strongest claim, and the winning facet is emitted as a typed edge from the (key, value) pair to that owner, labelled with the key that decided it and the bits it paid. The tests assert that every (record, key) cell is coded exactly once. Additivity is bought by single ownership, a circularity we state rather than smooth over.

**The price.** The model the code transmits is the Indian buffet process, IBP, a Bayesian model of a stream in which a new feature may appear at any time (Griffiths & Ghahramani, 2011): record  $n$  joins an existing node  $k$  with probability  $m_k/n$  and opens new nodes at rate  $\gamma/n$ . We do not fix  $\gamma$ ; it gets a prior and is integrated out, leaving a seed-time price for a candidate first proposed at record  $s$  that depends on two counters, the nodes minted so far  $K_{s-1}$  and the harmonic sum  $H_{s-1}$ , the accumulated opportunity to mint.

$$\Lambda_s = \log_2(s(b + H_{s-1}) + 1) - \log_2(a + K_{s-1}) = \log_2 s - \log_2 \hat{\gamma}_{s-1} + o(1), \quad (1)$$

where  $\hat{\gamma}_{s-1} = (a + K_{s-1})/(b + H_{s-1})$  is the stream’s own running estimate of its minting rate and the constant on  $\log_2 s$  is exactly one. The pair  $(a, b) = (\frac{1}{2}, 1)$  is a declared prior on that rate, Jeffreys in shape and forced proper by Kraft. Another reasonable proper prior moves one mint price by 0.19 bits at  $10^5$  records on a declared trajectory and 0.34 on the one our run walks, and  $\Lambda_1 = 2$  bits.

**Why immediate minting fails, and what replaces it.** Lemma 1 states what Section 1 argued in words: a brand-new node predicts uniformly, an uninformed guess cannot beat what the stream already knows against a strictly positive price, and seeding it at the record’s own value only relocates the same bits into the seed. So each key the graph fails to explain leaves a residual, in bits, accruing in an account held against a node that does not exist yet. The candidate is named by its seed pair (key, value), one per residual pair, and every later record whose key goes unexplained joins its cohort; the account holds the bits the incumbent owners spent on that cohort minus the bits the candidate’s own growing blocks would have spent. The node is created at the first time  $t^*$  at which some support  $T$  satisfies

$$G_t(T) \geq \text{Price}(t, T, n), \quad (2)$$

where  $G_t(T)$  is the evidence escrowed on the keys of  $T$  and  $\text{Price}$  is the cost of stating the node itself: its announcement, its membership column, its support and its name (Equation D.10). The shipped gate charges  $\text{Price}$  alone, and the statistic released is the positive-part prefix over the candidate’s keys plus the positive absence credits earned outside the support, so the statement terms can only delay a release and never license one (Appendix D). Under the null  $2^{G_t}$  behaves like the wealth of a gambler in a fair game (Theorem C.1), and across nine valid codes the minted set is identical on planted structure (E12).

**Per-record processing, repair, and keys.** Per record the engine performs the eight steps of Figure 2, and every statistic is snapshotted before any update so that every bet is priced fairly. The insertion rule is greedy and prices each record once, so it cannot undo a decision itself; a bounded repair pass runs off the hot path and can, descending the batch codelength by merge, reassign and delete moves accepted exactly when they strictly shorten the description, which is why it terminates (Proposition C.5). That pass runs on a fixed record cadence, which belongs to the run protocol and not to the criterion, and which Section 7 reports as the one quantity we cannot choose without labels. Real streams drift, so `distributor` turns up beside `distributors`: a never-seen key opens blocks of its own and enters a support only when a node is minted on it, and the operator that would price two keys as one is designed but not run.

## 4 THEORETICAL GUARANTEES

Logarithms are base 2, so every quantity is in bits, and each result gets one plain sentence and then its formal statement. Appendix C restates every result in full and proves it.

*The origin of the classical constant.* Under the mass scaling of Broderick et al. (2013b) the BP-means penalty satisfies  $\lambda^2 = 2\sigma^2 \cdot (-\ln \gamma)$ , and multiplying by  $2\sigma^2$  as  $\sigma \rightarrow 0$  annihilates the parameter and allocation costs and keeps the mass term alone (Lemma C.1, with the DP-means corroboration of Kulis & Jordan (2012)).

*A fresh node cannot pay on its first record.* Write  $D(p||q)$  for the Kullback-Leibler divergence and  $\hat{P}_{0,a}$  for the background’s prediction on key  $a$ , whose alphabet has  $d_a$  symbols.

**Lemma 1** (myopic impossibility). *Let  $Q$  be the prediction a node minted at record  $n$  assigns to a value of key  $a$  before any record has attached to it. (i) In a valid code  $Q$  depends on the transmitted past alone and sums to at most one; with no past, exchangeability over values, treating every value alike, forces  $Q$  uniform, so every value costs  $\log d_a$ , and seeding  $Q$  at the observed value relocates the same  $\log d_a$  bits into the seed. (ii) With  $p$  the law of the value, the expected first-record evidence against the background is  $-D(p||Q) + D(p||\hat{P}_{0,a})$ , which under the null  $p = \hat{P}_{0,a}$  equals  $-D(\hat{P}_{0,a}||Q) \leq 0$ , while (iii)  $\Lambda_n > 0$  whenever  $a + K_{n-1} < n(b + H_{n-1}) + 1$ . So in expectation a fresh node minted at record  $n$  loses at least  $\Lambda_n$  bits on every key of its first record.*

*The price is exact.* Giving the creation rate a Gamma prior and integrating it out leaves the log posterior odds of no mint against one mint, in bits, and that is exactly  $\Lambda_s$  of Equation 1 with remainder  $O(1/(s \log s))$ , the constant on  $\log_2 s$  being one and the attachment odds containing no  $\gamma$  (Lemma C.2). The prior must be proper, since the NML normaliser over the node count diverges and an improper prior is not a code; two proper priors differ per mint by  $O(1/\log n)$ , which is 0.19 bits at  $n = 10^5$  on a declared trajectory and 0.34 on the one our planted run walks; and learning  $\gamma$  inside the code costs  $\frac{1}{2} \log \ln N + O(1)$  bits over the whole stream. All three are stated and proved in Appendix C.

*The same price says which streams the rule can serve, and it said so before we ran anything.* Of the six terms of Equation D.10 only the membership column depends on how many records a node holds, so the cost of one more member can be read off in closed form.

**Proposition 1** (scope). *Fix the stream length  $n$  and let a node hold  $t$  of those records. (i) Every term of  $\text{Price}(t, T, n)$  except  $L_{\text{col}}$  is free of  $t$ , and  $L_{\text{col}}$  is the Krichevsky-Trofimov block code of Equation D.11, so the  $t$ -th member adds exactly*

$$\Delta(t, n) = \log \frac{n - t + \frac{1}{2}}{t - \frac{1}{2}} \quad \text{bits}, \quad (3)$$

*which is  $\log(n/t) + O(t/n) + O(1/t)$ , so a node is affordable only where each member saves more than  $\Delta(t, n)$  bits and a group of bounded size is priced out of a long enough stream. (ii) A candidate is named by one seed pair and accrues only on records carrying that pair, so a group is reachable only if some pair characteristic of it recurs at least  $t^*$  times, the cohort a candidate has reached at release.*

Both halves are measured, and the closed form is checked against the shipped routine rather than trusted: Appendix C.5 gives the agreement, the range of  $t^*$  and the sweep whose accuracy cliff brackets it.

*A parent is not a node, and that is a saving rather than a loss.* Minting a parent never pays: its membership column is  $\Theta(n)$  while the only saving from factorising is  $\Theta(\log n)$  (Proposition G.7). It costs nothing, because a parent need not be a vertex: if every record carrying key  $b$  also carries key  $a$  while  $a$  is commoner, then  $b$  specialises  $a$ , read off co-occurrence in one pass with nothing to set, and the inlining that makes the flat description cheaper is what makes the relation visible.

*And that edge is one of several, which makes the output a graph rather than a clustering.* Each of the others is an equality or a strict inequality on counts, with nothing to set: siblings, two names for one role, and typed edges between values with a cardinality attached. E38 and E40 plant all four and score them, specialisation and the typed edges recovering at precision and recall 1.000 and the two resting on disjointness at 0.750. On 3,362 real Wikidata people the same counting returns 538 specialisations, 1,189 sibling pairs and 2,076 typed edges (Appendix C.5); Section 5.4 reports where the role relation fails.

*The lifetime false-mint bound holds in the idealised setting; for the engine we report the measured exceedance instead of claiming it.* Fix a seed pair with seed time  $s$  and frozen price  $\Lambda_s$ , and let  $G_t$  be the accumulated log-likelihood ratio between the prequential graph with that pair and the graph without it. Under the null,  $2^{G_t}$  is a nonnegative martingale started at one exactly when every value block is exactly normalised, which a tight naming charge achieves and the shipped charge does not, the shipped one measuring 0.862862 against 1.000000. In that idealised setting Ville’s inequality (Ville, 1939; Ramdas et al., 2023) gives  $\mathbb{P}(\exists t \geq s : G_t \geq \Lambda_s) \leq 2^{-\Lambda_s}$  with no union bound over time, and because the hypotheses are indexed by seed pairs, a set growing with the schema rather than with the record count, and  $\sum 2^{-\text{Price}} \leq 1$ , the price is itself the multiple-comparisons correction. We state this for the idealised setting and do not claim it for the engine (Theorem C.1, Appendix C).

*What the engine measures instead of the bound.* Over 300 null streams the exceedance is 1.000 at every level in  $\{1, 2, 4, 8, 12\}$  bits against a nominal  $2^{-b}$ , and adding the seed naming charge of (iii) does not move it in any of three codes. The reason is not multiplicity but drift: under the null the accumulator’s supremum grows with the stream, so the slack the level would need grows from 97 bits at  $T = 500$  to 1,205 at  $T = 4,000$ , while a naming charge is constant in  $T$ . What carries the safety claim instead is a measurement. E29 runs the race between the release price and that drift on pure noise: the worst margin over the pool is  $-421$  bits at 2,000 records and  $-6,771$  at 50,000, widening at every length, with no node created anywhere. That is a trend with a sign and a rate rather than a theorem, and it stops at the length measured (Appendix C).

Five further results are stated and proved in Appendix C, among them exactness of the per-key receipts, consistency with no significance level, and termination of the repair pass.

## 5 EXPERIMENTS

### 5.1 EXPERIMENTAL DESIGN

No price, cut-off or level is set per run, the rule is told nothing about the domain, and every input ESCROW receives, every baseline receives. Two quantities belong to the run protocol and not to the criterion, and both did vary: the repair cadence and the candidate pool budget (Table 6, Appendix G).

### 5.2 MECHANISM FALSIFICATIONS

**E7: immediate minting is broken in both forms; deferral is not.** Both immediate arms mint zero times, as Lemma 1 requires, in all twenty cells of an alphabet by typo-rate sweep, while a constant DP-means-style price fails the other way; the deferred arm ends at  $K = 2$  on all five seeds with exactly the two planted key sets as its supports (Appendix G).

**E8: no node is created on noise.**  $K$  ends at 0 on 48 of 48 null streams, flat from 2,000 to 20,000 records, and on 300 further streams, with no significance level and no grace period anywhere in the rule. The streaming trees do not stay flat (Figure 3), and the strict setting that silences them costs 20,800 records of delay when structure is present. E26 widens this to twenty-five families and narrows the claim: sixteen give no node at all, and those that mint are the ones where key presence is itself random and high.

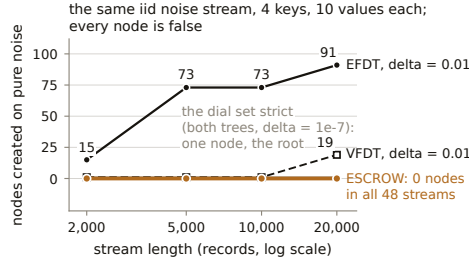


Figure 3: No node is created on noise. Every method sees the same pure-noise generator, so every node created is false. ESCROW (gold) creates zero nodes at every length in all 48 streams. The streaming trees VFDT and EFDT at  $\delta = 0.01$  grow with the stream, five seeds per length, EFDT to means of 15, 73, 73 and 91 nodes (maximum 111) and VFDT to 19. At  $\delta = 10^{-7}$  both stay at the root (results/e8\_false\_mint\_null.json, results/e8\_tree\_arms.json).

| $b$ (bits) | $2^{-b}$ | empirical |
|------------|----------|-----------|
| 1          | 0.5      | 0.09785   |
| 2          | 0.25     | 0.0324    |
| 4          | 0.0625   | 0.0064    |
| 8          | 0.0039   | 0.0002    |
| 12         | 0.00024  | 0.0000    |

Table 1: E8, Ville bound: empirical  $\Pr(\sup_t G_t \geq b)$  over 20,000 null trajectories of 200 steps on a standalone one-key block, below  $2^{-b}$  at every level.

**E9, E13 and E5: the price behaves like a price, and the order cost is measured.** In all 12 cells of the  $(k, d)$  grid the cohort at birth falls as  $k$  grows and never rises, so the price asks for more evidence when each record carries less (Table 9), and the creation bias runs opposite to the one Kearns et al. (1997) prove for penalty-based rules. The batch objective is a function of the state and not of the arrival order, which the tests assert, but the greedy sequence is not: over 20 orders both synthetic streams return ARI 1.0 and the exact  $K$ , while Wikipedia averages 0.8438 and ranges from 0.6389 to 0.986. E28 shows that spread is a search failure and not a modelling one, since the description length ranks the orders correctly, so keeping the shortest of three lifts the mean from 0.8691 to 0.9284 with nothing calibrated (Appendix G).

### 5.3 TUNED BASELINES AGAINST NO KNOB

Table 2 is the comparison the audit of this paper said was missing, and it does not go our way. Every method reads the same records, the multi-hot over observed (key, value) pairs and over the keys themselves, since the engine prices key presence as well as value.

It begins with a falsifier of ours that fired. We wrote beforehand that if a distance rule on ESCROW’s own representation matched ESCROW, the criterion would not be what does the work, and on all three streams one does:  $k$ -means with  $k$  by silhouette, consulting no label, ties us at 1.0 on both synthetic streams and reaches 0.9898 against our 0.8691 on the encyclopedia stream. We report that it fired rather than rewording it, and make no accuracy claim. What it does not touch is what the rule is for: that baseline is batch, returns a partition rather than a cover, and names no reason for anything it does. Transferred knobs collapse to at most 0.0017 on Wikipedia and to zero on the synthetic streams.

E43 then asks how hard that stream is at all, and the answer bears on every number reported on it: a code over character three to five grams, which cannot represent a word, partitions it perfectly at 1.0000 with its cluster count chosen by silhouette and no label consulted (Table 15). That reads the record as a string, which the criterion never does, so it is a floor and not a matched baseline. The encyclopedia stream ranks methods by vocabulary, and no result on it, ours or anyone’s, is evidence about meaning.

E22 and E23, which run BP-means and a two-part description length, do not reach us either (Appendix G).

E25 asks whether multi-membership is load-bearing, and its first answer was too flattering. ESCROW reaches omega 0.8876 on a cover fixture against 0.4411 for the best partition baseline, and the gap is exactly zero on a single-membership control, which establishes nothing: a partition cannot place a record in two groups at all. Against methods that can, an oracle-tuned factorisation reaches 1.0000 with its components and cut chosen on the test labels, and carried across fixtures drops to 0.1953 where ESCROW holds 0.9332 (Appendix G).

| stream                     | ESCROW, nothing calibrated |                   | best baseline, matched input |                   |
|----------------------------|----------------------------|-------------------|------------------------------|-------------------|
|                            | one order                  | shortest of three | untuned                      | at an oracle knob |
| two-group, 2,000 records   | 1.0                        | 1.0               | 1.0                          | 1.0               |
| eight-group, 3,000 records | 1.0                        | 1.0               | 1.0                          | 1.0               |
| Wikipedia, 320 records     | 0.8691                     | 0.9284            | 0.9898                       | 0.9898            |

Table 2: Adjusted Rand index with every method reading the same multi-hot (key, value) records, so the rule is compared with the rule and not with a representation (E21). The ESCROW columns are the mean over sixty arrival orders, and the mean when three are run and the shortest description kept, which consults no label (E28). The oracle column takes each baseline’s best knob on the test labels, an upper bound it could not have in use.

E37 asks the same on real records, since every number above comes from a fixture whose keys name the groups. On 3,362 Wikidata people labelled by occupation every label clears the seeding condition eleven times over, so a loss cannot be blamed on scope, and nothing does well: the best is 0.3137 and ESCROW reaches 0.2044 with nothing set, level with  $k$ -means handed the true label count.

#### 5.4 A LANGUAGE MODEL BUILDING THE SAME GRAPH

Qwen2.5-14B-Instruct (Qwen Team et al., 2024) is prompted to build the graph as a stream, against ESCROW on the identical records in the identical order (Figure 9). On Wikipedia, 320 clean records of three obvious types, the model wins: ARI 0.9905 at  $K = 4$  against our 0.8744. E27 sweeps the temperature over 0.3, 0.7 and 1.0, and agreement with itself holds at 0.99 on Wikipedia while running 0.1168 to 0.1647 on Lazada, so difficulty drives the non-determinism and not sampling.

On 1,000 raw Lazada listings, where the silver truth carries 759 types and the index is near zero for every method, the measurement is determinism and cost. The three seeded runs agree with one another at 0.1647 where ESCROW agrees with itself at 1.0, and the graph comes out catch-all and value-named at scale (Figure 7, Appendix G).

Both streams were chosen to suit a language model, so E41 turns the comparison round and gives the same model the two decisions the price is built for, with the prompt, the chunking, the parser and greedy decoding unchanged. On a stream with no latent structure it builds 80, 195 and 195 nodes over 200 records where the rule builds none, two of the three seeds growing at one node per record from the first chunk to the last (Figure 4). That is the degeneracy Lemma 1 rules out for any valid code, happening: with no account for evidence to accrue in, a rule has only the record in front of it, and every record is a little unlike the last. On the planted eight-group stream at the same length it is exact on two of the three arrival orders, ARI 1.0 at  $K = 8$ , and collapses on the third to 0.1984 at  $K = 72$  under the same greedy decoding, so that is order and not sampling. We return 0.8623, 0.91 and 1.0; the two below one are the seeding condition at its limit, since 200 records give each group 25, and the same stream at 3,000 records returns 1.0 in all twenty orders. The difference is therefore not accuracy. It is that one of them can decline.

E20 meets that system on its own ground: AutoPKG’s Lazada data, its decision, its metric. On key identity, 3,033 raw keys with 328 gold pairs that should merge, we lose comprehensively. Their agent reaches F1 0.6355; ESCROW reaches 0.0008, one true pair against 2,105 false. Two facts rescue nothing. We were scored by whether two keys share a node’s support, which is co-occurrence and not identity, and rescoring with the rule our own counting prescribes is perfectly precise and almost silent, 2 pairs of 328. And version 1 ships no operator that prices two keys as one, so the criterion cannot make this decision at all. What the gold contains is a separate question, and it bears on how the number should be read: a pair is positive there exactly when the two key strings agree after lowercasing and stripping punctuation, so the function that defines the gold scores F1 0.9619 as a predictor of it and plain lowercasing scores 0.8180. E44 measures eight string rules against that gold and six beat the language model, the best being identical word sets at 0.9893 (Table 16). Two names for one role, which is the decision the paper says is open, is not the decision this gold records. It remains open for this construction.



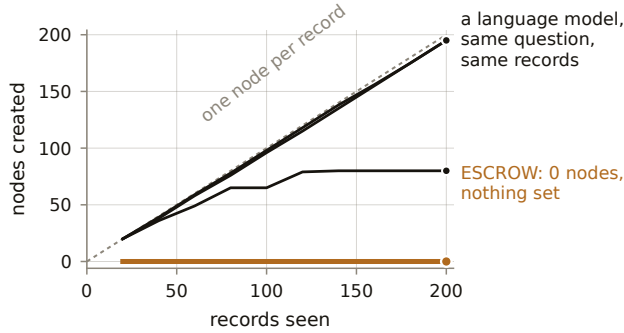


Figure 4: The decision the price is built for. Every method sees the same stream with no latent structure, four keys carrying ten independent uniform values, so every node created is false. The three dark lines are Qwen2.5-14B-Instruct under greedy decoding, given the same create-or-attach question on the same records in the same order as the engine, one line per seed; two of the three follow the dashed one-node-per-record line from the first chunk to the last. ESCROW in gold is zero at every length with nothing set (E41). This is the degeneracy Lemma 1 rules out for any valid code, happening.

## 5.5 WHAT THE PRICE SAYS THE RULE IS FOR, AND THE TWO BENCHMARKS IT EXCLUDES

E31 measures both clauses of Proposition 1 on a stream where only the seed cohort moves, and its accuracy cliff falls where that cohort crosses the release cohort E9 measured independently. E32 predicts our own losses without changing datasets, holding the encyclopedia stream fixed and moving only its size: the smallest seed cohort turns over at  $n = 1,280$ , exactly where it reaches the ceiling of 24 (Appendix G). Both clauses are necessary and neither sufficient, since E34 gives a stream meeting both on which the engine mints three nodes where the truth has eight.

Four public streams, one arrival order each (Table 13). MusicBrainz 20K and ReVerb45K are the two we lose, for a reason stated before the runs: both ask for groups of two, where the membership column alone demands 13.66 and 13.00 bits from each record, so Proposition 1 prices them out whatever the facets say. These are not benchmarks the rule does badly on but ones its price excludes, and we report them for that reason rather than dropping them (Appendices H.17 and H.18).

## 6 RELATED WORK

Appendix B is the full survey, with every number and its source. Pricing a component in bits is Wallace & Boulton (1968); codelength pruning of trees is Quinlan & Rivest (1989), Wallace & Patrick (1993) and Mehta et al. (1995; 1996), each relocating a knob rather than removing it; Kearns et al. (1997) prove that no reweighting of a two-part code removes its bias, accepted here as scope and measured (E13). Computed charges exist in batch, inside boosting (Friedman, 2001; Chen & Guestrin, 2016; Ke et al., 2017; Lunde et al., 2020) and across a literature in which parameter-free is already a title (Akoglu et al., 2012; Koutra et al., 2014; Vreeken et al., 2011; Miettinen & Vreeken, 2014; Betancourt et al., 2016; Mahon & Lapata, 2025; Peixoto, 2017; Yang & van Leeuwen, 2024; Galbrun, 2022), each selecting structure over data in hand. Streaming growth (Domingos & Hulten, 2000; Manapragada et al., 2018) and its anytime-valid repair (Amoukou et al., 2026; Ville, 1939; Ramdas et al., 2023) are supervised and use no description length. Structural entropy (Li & Pan, 2016; Cao et al., 2024; Xie et al., 2025) is the nearest unsupervised lineage and does construct, but its graph forgets which key typed a value (Proposition C.1). Language-model pipelines (Bai et al., 2025; Hongwimol et al., 2026) already ask whether a key should be added, merged or discarded, by a prompted model with no edit carrying its evidence; the question is theirs, the criterion ours (Section 5.4). Per-variable decompositions exist (Silander et al., 2018; Kontkanen et al., 2003; Costa et al., 2026; Slonim & Tishby, 1999; Wallace & Dowe, 1994); new is their job as the decision variable of an online create test.

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## 7 LIMITATIONS

Appendix H.20 states all of this at length, including cost and scaling, and the external dataset tier (Appendix H.19) is not run.

**Categorical facets only, and the negative that follows.** A categorical code sees a value as one symbol from a finite alphabet, so two values a token apart share no evidence whatever, and wherever identity lives below that granularity the code has nothing to work with. That is a prediction of the encoding rather than a result we came upon, and Sections 5.4 and G.2 measure it. The same property makes the rule domain agnostic, since a code that reads no lexicon cannot borrow one either, so the freedom and the cost are one fact.

**The rule has a scope, and it follows from the price rather than from disappointment.** What the rule is for is the converse of Proposition 1: a modest number of kinds each holding many records whose values recur, which a catalogue of listings is and the two benchmarks we lose on are not. It cannot do entity resolution at any setting, because neither condition contains a setting. The seeding condition is the weaker half, since nothing in the batch objective requires a node to be named by one (key, value) pair, so it is a property of how candidates are proposed and the first thing a version 2 should attack; the obvious attack, giving key presence a facet of its own, fails because the absence code already prices presence (E30).

**The search is approximate, and its budget is measured rather than tuned.** The exact argmax is a maximum edge biclique problem (Peeters, 2003), so the criterion is evaluated over a pool held to a declared budget, and E19 shows the output stops moving exactly where the pool stops evicting, which makes it a resource bound and not a quality knob (Appendix H.20).

**The repair schedule is a quantity we cannot choose without labels, and it changes the answer.** That no quantity in the *rule* is calibrated remains exact; the *protocol* is a different matter. The pass runs every 100 records by declaration, and E39 sweeps that over 50 to 1,000 and a single final pass: agreement runs from 0.3998 to 0.8669 on the planted two-level stream and from  $-0.0162$  to 0.2078 on the Wikidata cover benchmark. The answer that dissolved the same problem for arrival order is not available here, since the shortest description picks the best cadence on only three of six streams (Appendix H.20). Treat every number in this paper as conditional on the shipped cadence: a parameter-free criterion inside a protocol that is not.

## 8 CONCLUSION

The classical rules price a new node with a constant that a limit left behind, after erasing every part of the price a model could have computed for itself. We did not take that limit: we coded the stream prequentially under the Indian buffet process and read the price off the code, so no fresh node can pay on its first record and the evidence waits in escrow until an account covers one. What that buys is not accuracy, which on these streams a bag of characters already has. It is the ability to decline. Key identity remains open.

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## A TERMS USED IN THE APPENDIX

Terms used here: this section is itself the list. The appendices that follow use every term below without re-introducing it, and each opens with a two-line note naming the ones it leans on.

- **Record.** One item of the stream: a small set of attribute key and value pairs, such as `fabric = cotton`.
- **Facet.** One (record, key) slot: the place where one record gives one value for one key.
- **Node.** An entry in the graph that the system has decided exists: a latent group of records that share a pattern of keys and values.
- **Background node.** The model of an ordinary record: the node that owns whatever no other node explains, and the baseline every claim of structure must beat.
- **Support, scope.** The set of keys a node models. A node is charged for every key in its support and for nothing outside it. The proofs also say scope.
- **Typed edge.** An edge from a (key, value) pair to the node that owns it, labelled by the key that decided it and by its cost in bits.
- **Candidate, seed pair.** A node that does not yet exist, named by the (key, value) pair that first proposed it. That pair is its seed.
- **Cohort.** The records held in escrow behind one candidate: each record whose unexplained key carried the candidate’s pair.
- **Null.** The hypothesis that a candidate is spurious: there is no structure behind it, and its evidence is noise.
- **Codelength, bits.** The number of bits needed to write something down under a probability model; a prediction that gives probability  $P$  to what happens costs  $-\log_2 P$  bits, so a better model is a shorter description.
- **Prequential code.** A code that predicts each record before it is seen, from the past alone, and pays the bits of that prediction; nothing in it peeks at the future.
- **Batch codelength,  $L_{\text{batch}}$ .** The cost of writing down the current state and then all records given the state; a function of counts alone, and the objective the repair pass descends.
- **MDL.** The minimum description length principle: between two explanations of the same data, prefer the one that writes the data and the explanation down in fewer bits.
- **Two-part code.** A code that writes the model first and then the data given the model.
- **Kraft’s inequality.** Codeword lengths  $\ell$  can be realised by a real code exactly when  $\sum 2^{-\ell} \leq 1$ ; equivalently, only a (sub)probability can be a code.
- **DP-means.** The k-means-like clustering rule derived from the Dirichlet process; it opens a new cluster when a record’s distance to every centre exceeds a penalty  $\lambda$ .
- **BP-means.** Its cousin derived from the beta process; it lets a record belong to several features at once and opens a new feature at the same kind of penalty.
- **Small-variance limit, MAD-Bayes.** The derivation that produces DP-means and BP-means: put a Gaussian noise model with variance  $\sigma^2$  on the features, scale the negative log joint by  $2\sigma^2$ , and let  $\sigma \rightarrow 0$ .
- **IBP.** The Indian buffet process: a Bayesian model of a stream in which each record activates a set of features and a new feature may appear at any time, with creation rate (concentration, mass)  $\gamma$ .

- **Hyperprior.** The prior placed on the model’s own creation rate  $\gamma$ ; here  $\text{Gamma}(a, b)$  with declared  $(a, b) = (\frac{1}{2}, 1)$ .
- **KT code.** The Krichevsky-Trofimov code: a standard way to price a sequence of symbols whose frequencies are unknown, by giving every symbol half a count up front.
- **NML.** Normalized maximum likelihood: the minimax code for a model class, which exists only when its normaliser, the Shtarkov sum of maximised likelihoods over all data, is finite.
- **Regret.** The extra bits a code pays over the best code chosen in hindsight.
- **Escrow.** A held account: evidence for a node that does not yet exist accumulates there, and the node is created only when the account covers the node’s price.
- **Mint.** To create a node. A candidate is minted at the first moment its escrowed evidence covers its price.
- **Price.** The number of bits a new node must earn before it may exist:  $\text{Price}(t, T, n)$  of Equation D.10, the cost of stating the node, computed from the stream and never chosen.  $\Lambda_s$  is the seed-time term derived in the theory; it is not part of the shipped gate.
- **Martingale, Ville’s inequality.** A martingale is the wealth of a gambler in a fair game: its expected next value equals its current value. Ville’s inequality says a nonnegative martingale started at one ever exceeds  $c$  with probability at most  $1/c$ .
- **Kullback-Leibler divergence,  $D(p||q)$ .** The expected extra bits paid for coding data from  $p$  with a code built for  $q$ ; zero iff  $p = q$ .
- **Jensen-Shannon divergence.** The mutual information between one observation and which of two distributions produced it: how distinguishable the two are, in bits.
- **Total correlation.** The bits per record lost by modelling keys independently when a latent distinction couples them; zero exactly when the keys are independent given the node.
- **Structural entropy.** The bits a random walk on a graph needs once the graph is cut into communities; the nearest unsupervised incumbent objective.
- **Submodular.** A gain in which each further item helps less than the one before.
- **Repair.** A bounded pass, off the hot path, that revisits earlier decisions by merging, reassigning or deleting nodes whenever doing so shortens the total description.
- **Gate (UT- $n$ ).** A test that ships with the code and must pass for the build to be trusted; numbered as in the released specification.
- **ARI.** The adjusted Rand index: a chance-corrected clustering score in which 1.0 is a perfect match to the true groups and 0.0 is what a random grouping would score.
- **Oracle, transferred.** A baseline knob chosen on the test labels themselves; a knob tuned on another stream and carried over unchanged.

## B RELATED WORK IN FULL

Terms used here: MDL is the minimum description length principle, which prefers the explanation that writes data and model down in fewer bits; a two-part code writes the model first and the data given the model; a Hoeffding bound is a statistical guarantee on how far a sample average can stray from its mean; structural entropy is the bits a random walk on a graph needs once the graph is cut into communities; ARI is the adjusted Rand index, a chance-corrected clustering score.

**Codelengths for tree growth: the relocated parameter.** MDL, the minimum description length principle, prefers the model that writes data and model down in fewer bits. Pricing a component in bits is Wallace & Boulton (1968): the component is created only when the bits it saves exceed the bits needed to state it. Quinlan & Rivest (1989) grew and pruned decision trees by one codelength, with no held-out set. But it relocated the parameter rather than removing it. The code carries a hand-set weight  $c$  on one of two bit counts. The paper sweeps  $c$  over 1, 2 and 8. That sweep moves one dataset from 17 to 69 leaves and from 20.5% to 13.0% error. The choice of  $c$  is left “as an open problem.” Wallace & Patrick (1993) corrected the coding. But they introduced a Dirichlet concentration  $\alpha$ , a knob that sets how spread out the assumed class frequencies are. They fitted it by a grow-and-refit loop, and it took values from 0.022 to 0.761 across datasets. Their formulation is ours. Adopting a code “is equivalent to accepting a certain prior probability distribution over

the set of possible theories.” Mehta et al. (1995) fix a precision knob at 1 after a sweep. SLIQ, the Supervised Learning In Quest decision-tree learner (Mehta et al., 1996), fixes an “empirically determined” test cost. It also records that the earlier codes over-pruned. Kearns et al. (1997) prove that no reweighting of a two-part code fixes its bias. A two-part code writes the model first and then the data given the model. For every sample size there are instances on which any penalty-based rule stays a fixed distance from the best answer. We accept that verdict as scope. We claim no calibrated parameter, not no bias. We measure the creation bias directly (Section 5).

**Computed charges, in batch.** A second lineage computes the charge, but with the whole dataset in hand. Boosting builds a model as a sum of small trees. Inside boosting the charge per new tree node is capped (Friedman, 2001), or tuned or absent (Chen & Guestrin, 2016; Ke et al., 2017). Lunde et al. (2020) derive that charge as an asymptotic optimism, an estimate of how much a split’s gain overstates its true value for large samples. That leaves only the learning rate as a knob. In unsupervised learning, learning without labels, the phrase “parameter-free” is already widely used. PICS, Parameter-free Identification of Cohesive Subgroups (Akoglu et al., 2012), groups the nodes of a graph that carries node attributes by MDL, under exactly that title. VoG, Vocabulary-based summarisation of Graphs (Koutra et al., 2014), admits a subgraph exactly when it shortens the description. KRIMP (Vreeken et al., 2011), an MDL pattern miner whose name is the Dutch word for shrink, covers a transaction with several patterns. A transaction is one set of items, such as a shopping basket, and the patterns must split its items into non-overlapping parts. MDL4BMF, MDL for Boolean Matrix Factorisation (Miettinen & Vreeken, 2014), prices the Boolean rank of a record-by-attribute matrix of zeros and ones. Such a matrix is exactly an allocation matrix of the IBP, the Indian buffet process (Griffiths & Ghahramani, 2011). The IBP is a Bayesian model of a stream in which a new feature may appear at any time. But MDL4BMF needs the complete matrix. Microclustering (Betancourt et al., 2016) derives the new-cluster cost that DP-means, the k-means-like clustering rule derived from the Dirichlet process (Kulis & Jordan, 2012; Broderick et al., 2013b), leaves tuned. It does so over a single number per record.  $k^*$ -means (Mahon & Lapata, 2025) “eliminates the need to set  $k$ ”, the cluster count, by MDL split and merge steps. Peixoto (2017) replaces the constant with hierarchies of priors. CDTree, a tree-shaped clustering method (Yang & van Leeuwen, 2024), says MDL “eliminates the need for tuning the hyperparameter”. Its normaliser, the sum that turns its score into a probability, factorises only because each record reaches exactly one leaf. Galbrun (2022) names where such designs silently become parameter choices. A universal integer code, a fixed way to write any whole number in bits, is chosen “to favour partitions with fewer groups.” Our answer is structural. The hyperprior, the prior placed on the model’s own rate of creation, is forced proper by Kraft’s inequality. Kraft’s inequality is the rule that codelengths must correspond to probabilities summing to at most one. Proper means the prior’s total mass is finite. It is not selected on data (Section 4). The encoding-stability sweep E12 measures the slack that remains (Appendix D). Every criterion above selects structure over data already in hand. The constant also survives into deployment: a deployed product-catalogue placement layer built on BP-means and DP-means that we examined carries five hand-set constants, calibrated by a sweep on 1,600 records, and its source comment concedes they must be re-tuned once the graph is live.

**Streaming growth.** Growing structure from a stream by picking the best attribute at each step is VFDT, the Very Fast Decision Tree (Domingos & Hulten, 2000). VFDT splits a leaf when a Hoeffding bound separates the best attribute from the second best. A Hoeffding bound is a statistical guarantee on how far a sample average can stray from the truth. VFDT carries a significance level  $\delta$ , a tie parameter and a grace period, a fixed number of records to wait before testing. EFDT, the Extremely Fast Decision Tree (Manapragada et al., 2018), splits as soon as the best attribute beats no split at all, and it can retract a split later. Amoukou et al. (2026) show the Hoeffding gate is invalid when the stopping time depends on the data. They repair it with an anytime-valid test-by-betting gate (Ville, 1939; Ramdas et al., 2023), a test that stays valid no matter when one stops looking. The repair keeps a significance level, a grace period of 20 records and an improvement threshold. It uses no description length. All of this is supervised, meaning every record must carry a label. Delete the label and the split gain is undefined. We are not aware of prior work that combines an irrevocable per-arrival node-birth decision as a codelength. Escrow, our held account of evidence for a node that does not yet exist, obtains lifetime validity from the encoding itself. It needs no significance level and no grace period (Theorem C.1).



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**Structural entropy.** The nearest unsupervised lineage, and it does construct. Structural entropy (Li & Pan, 2016) is the bits a random walk needs once a graph is organised into a hierarchy of communities. HISEvent, hierarchical and incremental structural entropy minimisation for event detection (Cao et al., 2024), adds meaning-based edges to a graph of messages by entropy minimisation before partitioning it. It needs no cluster count  $k$ . Four limits remain. Attribute identity is destroyed at construction. Two records share one unlabelled edge, weighted by the cosine similarity of their text embeddings. The node set is fixed at one node per message, so no creation decision exists. The parameter-free claim covers the cluster count only. Its own sensitivity study moves ARI, the adjusted Rand index, a chance-corrected clustering score, by ten points as the sub-graph size varies. And its “incremental” update runs over the neighbour rank, never over arriving records. SES, structural-entropy-based sample selection (Xie et al., 2025), decomposes structural entropy losslessly to nodes by a closed-form Shapley value, a fair split of a total score among the parts that produced it. So per-element decomposition exists in this line. But it is per node, never per attribute, because each record is first collapsed into one frozen embedding. And it runs under a user-set budget rather than a decision on whether structure should exist.

**Pipelines built on large language models.** AutoSchemaKG (Bai et al., 2025) is a knowledge-graph builder that induces its own schema with an LLM, a large language model. It builds a concept layer over web-scale corpora with no predefined schema. It reports 95% semantic alignment with human type inventories, measured by BS-R and BS-C, its two reported semantic-similarity scores, at 78,400 GPU-hours, hours of graphics-processor time. Its node identity is exact string match on the surface forms the model generates. It has no canonicalisation step at all, no step that merges different names for the same thing. AutoPKG, an automatic product knowledge graph builder (Hongwimol et al., 2026), is the closest prior work to our key-identity operator. It is the reason our claim is about the criterion rather than the question. It already resolves each candidate attribute key by add, merge, replace or discard over streaming listings. But it does so by a prompted LLM at temperature 0.7, a setting under which the model’s output is partly random, reading a hand-written synonymy policy over ten retrieved neighbours. Its probabilistic recall collapses to 0.27 to 0.47 on the tail, the rare keys. No edit carries its evidence. Its own limitations section records that an early wrong merge propagates through every later decision. Our decisions are deterministic, and each carries a per-facet receipt in bits, one term per attribute of the record. The head-to-head under their protocol is the designed next tier (Section 7).

**Per-variable decompositions, and what is new.** Per-variable decompositions of information scores exist, and we do not claim them. NML, normalised maximum likelihood, is a code that pays the worst-case overhead over all datasets of a given size. Its factorised and quotient variants, fNML and qNML, are sums over variables (Silander et al., 2018). Kontkanen et al. (2003) factorise the inner term of a clustering NML into per-attribute regrets; a regret is the extra bits a code pays over the best code chosen in hindsight. But the normaliser keeps a joint sum over all attributes at once. That is one reason our spine is prequential, predicting each record before it is seen (Dawid, 1984; Grünwald, 2007), rather than NML. DIBmix, a clustering method for mixed-type data built on the deterministic information bottleneck (Costa et al., 2026), reports per-variable mutual information after the fact, on a partition whose cluster count the user fixes. Agglomerative IB (Slonim & Tishby, 1999), where IB is the information bottleneck, a method that compresses one variable while keeping what it says about another, supplies the merge loss we reuse. It selects no partition size. Snob (Wallace & Dowe, 1994) prints per-attribute significance after fitting. What is new is not the per-facet term but its job. It is the decision variable of an online create test, computed at arrival, before the node exists. The winning term is promoted to a typed edge, an edge labelled by the attribute that decided it, carrying its cost in bits.

## C FULL PROOFS

Terms used here: a prequential code prices each record by a prediction made from the past alone; Kraft’s inequality says codelengths  $\ell$  are realisable exactly when  $\sum 2^{-\ell} \leq 1$ ; KT is the Krichevsky-Trofimov code, which grants every symbol half a count up front; the IBP is the Indian buffet process, the feature-allocation model whose creation rate  $\gamma$  we integrate out; a martingale is a fair game, and Ville’s inequality bounds how often its wealth ever gets large; NML is normalized maximum

likelihood, the minimax code that exists only when its normaliser is finite;  $D(p||q)$  is the Kullback-Leibler divergence.

This appendix carries the paper’s formal content in the order of Section 4: the coding setting; the exact relation to structural entropy, concessions first; the identity that locates  $\lambda$ ; the impossibility that forces deferral, and the price that deferral must cover; the anytime guarantee of the escrow; the identities behind the typed edges; and the batch objective that the repair pass descends. The results stated in Section 4 are restated here in full, with the number they carry there, and each proof follows its statement; the lemma on where  $\lambda$  comes from is stated only here, under its own number.

Three conventions hold throughout. Logarithms are base 2 unless marked, so codelengths are in bits.  $H(\cdot)$  is Shannon entropy: the average number of bits an optimal code spends per symbol of a source. A proof follows every statement, and every displayed identity was also checked numerically.<sup>1</sup>

## C.1 SETTING

A record is a set of key-value pairs. Record  $n$  publishes a key set  $O_n \subseteq A$ , the attributes it carries, and one value  $x_{n,a} \in V_a$  for each published attribute  $a$ ; the value alphabet  $V_a$  has  $d_a$  symbols. The latent structure is a feature allocation: a record may belong to several nodes at once, not to exactly one. We model the allocation with the Indian buffet process (IBP) (Griffiths & Ghahramani, 2011). In plain terms: record  $n$  attaches to an existing node  $k$  with probability  $m_k/n$ , where  $m_k$  counts the earlier records attached to  $k$ , and it opens  $j \sim \text{Poisson}(\gamma/n)$  new nodes, where the concentration  $\gamma$  sets how readily the process creates nodes. We do not fix  $\gamma$ . It gets a  $\text{Gamma}(a, b)$  prior,  $a, b > 0$ , and is integrated out exactly, so no node-creation rate is assumed anywhere. Values are coded by per-facet sequential Krichevsky-Trofimov (KT) predictives (Krichevsky & Trofimov, 1981): a KT predictive is a running estimate of a value distribution built from the counts seen so far, with half a count granted to every symbol up front, so no value ever has probability zero. A background node 0 carries the corpus-wide value tables, with predictive  $\hat{P}_{0,a}$ ; it is the model of an ordinary record, and every claim of structure must beat it. Three running statistics recur.  $K_n$  is the number of nodes minted through record  $n$ .  $H_n = \sum_{m \leq n} 1/m$  is the harmonic sum; under the IBP it measures accumulated exposure to mint opportunities, and it grows like  $\ln n$ .  $\hat{\gamma}_n = (a + K_n)/(b + H_n)$  is the posterior mean concentration: the stream’s own running estimate of how freely it mints. Last, the coding discipline. Prequential means that every record is priced by a predictive computed from the transmitted past alone, before the record is revealed (Dawid, 1984; Grünwald, 2007). Because each predictive is a genuine probability distribution over what comes next, the total is a valid prefix codelength, and Kraft’s inequality, the fact that probabilities summing to at most one and achievable codelengths are the same objects, governs every decision the code prices.

## C.2 THE RELATION TO STRUCTURAL ENTROPY

Before any result of our own, we place the nearest incumbent objective. A reader can skip this proposition without loss: it positions a rival, and the paper’s own results begin at the paragraph on where  $\lambda$  comes from. Structural entropy (Li & Pan, 2016) scores a hierarchical partition of a graph, called an encoding tree, by the number of bits needed to tell a random walker where it lands; it is unsupervised, entropy-shaped, free of a fixed cluster count, and minimised incrementally by HISEvent (Cao et al., 2024). The question the next proposition answers: how much of our objective does structural entropy already contain, and what can it not express? We state the overlap first, in

<sup>1</sup>The identities of Proposition C.1 have a shipped artefact, `code/experiments/e17_structural_entropy_identities.py`, which writes `results/e17_structural_entropy_identities.json`. Measured there: the closed forms of part (i), the slack identity (C.3) at tree heights one, two and three, and both forms of the gain identity (C.1) reproduce to  $2.2 \times 10^{-15}$  over 500 random bipartite graphs with random partitions and trees, with the two degenerate extremes equal to  $H^{(1)}(B)$  to  $1.8 \times 10^{-15}$ ; the coincidence identity (C.4) to  $1.1 \times 10^{-12}$  over 40 synthetic streams satisfying the hypotheses of part (ii); the Krichevsky-Trofimov regret (C.5) nonnegative in all 400 blocks drawn, at a minimum of 2.02 bits; the fibre pair of part (iv) returning an identical  $H^T$  on all 200 partitions tried, with the merge verdict *distinct* at 100.0 bits of divergence against a parameter saving of  $-3.32$ ; and  $\Lambda_1 = 2.000$  bits. One numerical check quoted elsewhere in this appendix still has no shipped artefact, and is marked as such where it appears: the price (C.6) against the product of prequential conditionals to  $10^{-12}$ . The regret slope of Lemma C.2(iii), 1.661 bits per decade of exposure, is quoted there as the closed form  $\frac{1}{2} \log 10$  that it is, not as a measurement.

closed form, because the two differences that survive the concession are the ones the rest of the paper stands on.

To compare the two objectives at all, the stream must become a graph structural entropy can read. An *atom* is a key-value pair  $\alpha = (k, u)$ : one specific value of one specific key. Derive from the stream the bipartite graph  $B = (R \cup \mathcal{A}, E_B)$ , records on one side, atoms on the other, with one unit edge  $(r, \alpha)$  iff record  $r$  carries  $\alpha$ . Degrees then have concrete meanings. A record's degree is its width  $m_r$ , the number of pairs it publishes. An atom's degree is its frequency  $n_\alpha$ , the number of records carrying it.  $M = \sum_r m_r$  is the total number of published pairs.  $H^T(B)$  is the structural entropy of an encoding tree  $T$ ; a height-2 tree is simply a partition  $P = \{X_j\}$  of the vertices, with volume  $V_j$  (the sum of degrees inside  $X_j$ ), internal weight  $w(X_j)$  (the number of edges with both ends inside  $X_j$ ), and  $P(\alpha)$  the module containing  $\alpha$ .

**Proposition C.1** (relation to structural entropy). (i) Closed forms, and an additivity conceded. Write  $p_R(r) = m_r/M$  for the width profile and  $p_A(\alpha) = n_\alpha/M$  for the frequency profile. Then  $H^{(1)}(B) = 1 + \frac{1}{2}[H(p_R) + H(p_A)]$ : the leading bit says on which side of the bipartite graph the walker lands, records or atoms, the bracket averages the two degree profiles' entropies, and the whole is a function of the two marginals alone, blind to which record carries which atom. For every partition  $P$ , with  $s_{j,\alpha}$  the number of records of  $X_j$  carrying  $\alpha$ , the gain of  $P$  over the one-level code has two equal closed forms:

$$H^{(1)}(B) - H^P(B) = \sum_j \frac{w(X_j)}{M} \log \frac{2M}{V_j} = \sum_{\alpha \in \mathcal{A}} \frac{s_{P(\alpha),\alpha}}{M} \log \frac{2M}{V_{P(\alpha)}} \geq 0, \quad (\text{C.1})$$

vanishing exactly at the all-singleton and one-module extremes. In words: each module  $X_j$  contributes the share of edges internal to it,  $w(X_j)/M$ , times the bits saved by pointing at a landing spot inside  $X_j$  rather than anywhere in the graph,  $\log(2M/V_j)$ ; the second form books the same saving atom by atom, which is the additivity being conceded. So 2-D structural entropy is genuinely anti-degenerate, and its gain is exactly additive over attribute atoms, so it does say which attribute. We concede both facts before drawing any contrast. (ii) Coincidence. Suppose each record activates one node  $v$  and value supports are node-disjoint; then every edge of  $B$  is internal to a module. On this overlap the two objectives become one problem: the value term of the structural entropy of the induced partition is exactly the maximum-likelihood plug-in of our value blocks. This part is an identity. The preferential blocks the engine ships stand above that plug-in by the KT code's regret per block, the extra bits a sequential code pays over the best code chosen in hindsight, and that part is an inequality: the regret is nonnegative for every block, and its size,  $\frac{A_k-1}{2} \log q_{v,k} + O(1)$ , is the one step of this proposition taken from the literature rather than derived here. Restricted to disjoint allocations, the value blocks, and no model cost, the two minimisations are the same problem, and every remaining term lies outside the overlap. The exact term-by-term accounting is Equation (C.4). (iii) First essential difference: no model-cost term. Every term of  $H^T(B)$  is a function of cut and volume aggregates. Nothing charges for describing the tree  $T$  itself, and what blocks the all-singleton optimum is  $w(X_j) = 0$ , a connectivity fact, not a price.  $H^T$  is intensive, a rate; a codelength is extensive, a total. A per-arrival mint test compares bits of evidence against bits of model, and a rate supplies no such increment, which is where that line's residual stopping rules live. Before any edge exists  $H^T$  is undefined, while the price (C.6) is defined from record one:  $\Lambda_1 = \log \frac{b+1}{a}$ , two bits at the declared  $(a, b) = (\frac{1}{2}, 1)$ . (iv) Second essential difference: key-type invariance. Two datasets that differ only in how their atoms are typed into keys produce the same graph  $B$ , so no structural entropy at any tree height can tell them apart:  $H^T$ , its minima, and its minimisers are constant on the fibres of the map that forgets atom types, where a fibre is the set of datasets mapped to one graph. The fibres are non-trivial, and the criterion separates them: on datasets identical as graphs but typed one-key-with-two-values versus two-keys, it prices the difference at the value level in one and at the key-presence level in the other, a well-posed key-merge instance returning distinct. The audited outputs, which key carried a difference and at which level, are computed by no function of  $B$ , at any tree height, under any edge weighting.

The proof is elementary and every step is checkable by hand. It is given in full because the proposition concedes as much as it claims, and a reader who cannot check the concession cannot weigh the claim.

*Proof.* Parts (i), (iii) and (iv) rest on one telescoping identity, which is therefore established first. Let  $T$  be any encoding tree of  $B$  with root  $\lambda$ . For a non-root node  $\alpha$  write  $L(\alpha)$  for the graph vertices

at the leaves below  $\alpha$ , so that  $V_\alpha = \sum_{i \in L(\alpha)} d_i$  and  $V_\lambda = \text{vol}(B) = 2M$ ; a leaf is a single vertex  $i$ , with  $L(i) = \{i\}$  and  $V_i = g_i = d_i$ . Write  $\pi(i)$  for the non-root nodes on the path from  $\lambda$  down to the leaf  $i$ . Then

$$\sum_{\alpha \neq \lambda} \frac{V_\alpha}{2M} \log \frac{V_{\alpha^-}}{V_\alpha} = \sum_i \frac{d_i}{2M} \sum_{\alpha \in \pi(i)} \log \frac{V_{\alpha^-}}{V_\alpha} = \sum_i \frac{d_i}{2M} \log \frac{2M}{d_i} = H^{(1)}(B). \quad (\text{C.2})$$

The first equality expands  $V_\alpha$  over the leaves below  $\alpha$  and exchanges the two sums, which is legitimate because  $\{\alpha \neq \lambda : i \in L(\alpha)\}$  is exactly  $\pi(i)$ . The second telescopes that path: consecutive terms cancel and only  $\log V_\lambda - \log V_i = \log(2M/d_i)$  survives. The third is the definition of  $H^{(1)}(B)$ , the structural entropy of the unique tree of height one, whose children are the leaves themselves. Subtracting the definition  $H^T(B) = \sum_{\alpha \neq \lambda} \frac{g_\alpha}{2M} \log \frac{V_{\alpha^-}}{V_\alpha}$  from (C.2) gives the slack identity

$$H^{(1)}(B) - H^T(B) = \sum_{\alpha \neq \lambda} \frac{V_\alpha - g_\alpha}{2M} \log \frac{V_{\alpha^-}}{V_\alpha} \geq 0, \quad (\text{C.3})$$

where every summand is nonnegative because a vertex set's cut weight never exceeds its degree volume,  $g_\alpha \leq V_\alpha$ , and a child's leaves are a subset of its parent's,  $V_\alpha \leq V_{\alpha^-}$ .

*Part (i), the closed forms.* For the one-dimensional form, expand the definition  $H^{(1)}(B) = -\sum_{x \in R \cup \mathcal{A}} \frac{d_x}{2M} \log \frac{d_x}{2M}$  over the two sides of  $B$ . A record vertex has  $d_r = m_r$ , hence  $\frac{d_r}{2M} = \frac{1}{2} p_R(r)$ , and its term is  $-\frac{1}{2} p_R(r) [\log p_R(r) - 1]$ ; summing over  $R$  and using  $\sum_r p_R(r) = 1$  gives  $\frac{1}{2} H(p_R) + \frac{1}{2}$ . An atom vertex has  $d_\alpha = n_\alpha$  and  $\sum_\alpha n_\alpha = M$ , so the identical computation on the atom side gives  $\frac{1}{2} H(p_\mathcal{A}) + \frac{1}{2}$ . Adding the two halves gives  $H^{(1)}(B) = 1 + \frac{1}{2} [H(p_R) + H(p_\mathcal{A})]$ , in which the leading bit is the two halves' constants and the bracket reads the two degree marginals and nothing else.

For the two-dimensional form, let  $T$  be the height-two tree of a partition  $P = \{X_1, \dots, X_L\}$  and apply (C.3). Its leaves contribute nothing, because a leaf has  $V_i - g_i = d_i - d_i = 0$ . Its modules have the root as parent, so  $V_{\alpha^-} = 2M$ , and they satisfy  $V_j - g_j = 2w(X_j)$ : the volume  $V_j$  counts each edge internal to  $X_j$  twice, once at each endpoint, and each edge leaving  $X_j$  once, so  $V_j = 2w(X_j) + g_j$ . Substituting both facts into (C.3) gives the first equality of (C.1). For the second, use that  $B$  is bipartite: an edge internal to  $X_j$  is a pair  $(r, \alpha)$  with exactly one atom endpoint, and that atom lies in exactly one module of  $P$ . Grouping the internal edges of  $X_j$  by their atom endpoint therefore counts each once and misses none, so  $w(X_j) = \sum_{\alpha \in X_j \cap \mathcal{A}} s_{j,\alpha}$ , and summing over  $j$  replaces the module index by  $P(\alpha)$ . Both factors of each summand are nonnegative, since  $w(X_j) \geq 0$  and  $V_j \leq 2M$ , so the gain is nonnegative. It vanishes at the all-singleton partition, where no module holds two vertices and hence  $w(X_j) = 0$  for every  $j$ , and at the one-module partition, where  $V_1 = 2M$  and the logarithm is zero; it is strictly positive as soon as one module holds an internal edge and has volume below  $2M$ . The second form is an exact per-atom accounting of the gain, which is the additivity the statement concedes.

*Part (ii), coincidence.* Recall the per-node statistics:  $z_{rv} = 1$  when record  $r$  activates node  $v$ ,  $q_{v,k}$  is the number of members of  $v$  naming key  $k$ ,  $Q_v = \sum_k q_{v,k}$ ,  $c_{v,k}(u)$  is the number of members with value  $u$  on key  $k$ ,  $\hat{c}_{v,k}$  is the empirical value distribution inside the block  $(v, k)$ , and  $\hat{k}_v = (q_{v,k}/Q_v)_k$  is the node's key profile. Let  $P_Z$  have modules  $X_v = \{r : z_{rv} = 1\} \cup \{\alpha : \alpha \text{ is carried by a member of } v\}$ . Three steps.

First,  $P_Z$  is a partition of  $R \cup \mathcal{A}$ . Single membership makes the record sides disjoint. If an atom  $\alpha = (k, u)$  lay in both  $X_v$  and  $X_{v'}$  with  $v \neq v'$ , some member of each would carry it, so  $c_{v,k}(u) > 0$  and  $c_{v',k}(u) > 0$ , contradicting the node-disjointness of value supports. Every atom is carried by some record, so the modules cover.

Second, every edge is internal and  $V_v = 2Q_v$ . An edge  $(r, \alpha)$  whose record endpoint is a member of  $v$  has  $\alpha \in X_v$  by construction, so both endpoints lie in  $X_v$  and  $g_v = 0$  for every module. For the volume, the record side contributes  $\sum_{r: z_{rv}=1} m_r = \sum_k q_{v,k} = Q_v$ , because both sides count the pairs (member of  $v$ , key it names). The atom side contributes  $\sum_{\alpha \in X_v \cap \mathcal{A}} n_\alpha$ ; disjointness makes every record carrying  $\alpha = (k, u)$  a member of  $v$ , so  $n_\alpha = c_{v,k}(u)$  and the sum is  $\sum_k \sum_u c_{v,k}(u) = \sum_k q_{v,k} = Q_v$ . Hence  $V_v = 2Q_v$ .

Third, the accounting. With every  $g_v = 0$  the module terms of  $H^{Pz}$  vanish and only the leaf terms remain, so  $2M \cdot H^{Pz}(B) = \sum_v \sum_{i \in X_v} d_i \log \frac{V_v}{d_i}$ . The record side of that sum is  $\sum_v \sum_{r: z_{rv}=1} m_r \log \frac{2Q_v}{m_r}$  directly. On the atom side, group the atoms of  $X_v$  by key and split the logarithm in three,

$$\log \frac{V_v}{c_{v,k}(u)} = \log \frac{q_{v,k}}{c_{v,k}(u)} + \log \frac{Q_v}{q_{v,k}} + \log \frac{V_v}{Q_v},$$

whose last term is  $\log 2 = 1$  because  $V_v = 2Q_v$ . Weighting by  $c_{v,k}(u)$  and summing over  $u$  and then  $k$ : the first piece gives  $\sum_k q_{v,k} H(\hat{c}_{v,k})$ , since  $\sum_u c_{v,k}(u) = q_{v,k}$ ; the second gives  $Q_v H(\hat{\kappa}_v)$ , since summing out  $u$  leaves  $\sum_k q_{v,k} \log \frac{Q_v}{q_{v,k}}$ ; the third gives  $Q_v$ . Summing over  $v$  and using  $\sum_v Q_v = \sum_r m_r = M$  yields

$$2M \cdot H^{Pz}(B) = \underbrace{\sum_v \sum_k q_{v,k} H(\hat{c}_{v,k})}_{\text{ML plug-in value cost}} + \sum_v Q_v H(\hat{\kappa}_v) + \sum_v \sum_{r: z_{rv}=1} m_r \log \frac{2Q_v}{m_r} + M, \quad (\text{C.4})$$

an identity, with no asymptotics anywhere in its derivation.

It remains to say exactly how the first bracket relates to our value blocks, because this is the one place in the proposition where the relation is an inequality rather than an identity. Our block for the pair  $(v, k)$  is the sequential KT code over the  $A_k$  symbols of the key, whose codelength on the counts  $c_{v,k}$  is exactly

$$L_{\text{KT}}(c_{v,k}) = \log \frac{\Gamma(q_{v,k} + A_k/2) \Gamma(1/2)^{A_k}}{\Gamma(A_k/2) \prod_u \Gamma(c_{v,k}(u) + 1/2)},$$

which is an identity, and which is what `codes.L_KT_block` computes. The first bracket of (C.4) is instead the maximum-likelihood plug-in  $q_{v,k} H(\hat{c}_{v,k})$ , which is not a codelength at all: it prices the counts by their own empirical frequencies, chosen after the block is read. The two are related in one direction unconditionally. The KT code is a mixture  $\int P_\theta(c_{v,k}) w(\theta) d\theta$  over the value distributions  $\theta$  with  $w$  the Dirichlet( $\frac{1}{2}, \dots, \frac{1}{2}$ ) density, and a mixture cannot exceed its largest member, so  $P_{\text{KT}}(c_{v,k}) \leq \max_\theta P_\theta(c_{v,k}) = 2^{-q_{v,k} H(\hat{c}_{v,k})}$  and therefore

$$L_{\text{KT}}(c_{v,k}) - q_{v,k} H(\hat{c}_{v,k}) \geq 0 \quad (\text{C.5})$$

for every block and every count vector. That this excess is  $\frac{A_k-1}{2} \log q_{v,k} + O(1)$  and not merely nonnegative is the standard KT result, which we cite (Krichevsky & Trofimov, 1981; Xie & Barron, 2000) rather than reprove; it is the only step of this proposition taken from the literature. The claim of part (ii) is therefore exactly this strong and no stronger. The first bracket of (C.4) is the plug-in of our value blocks term for term, so restricted to a disjoint allocation, to the value blocks alone and with the model cost dropped, the two minimisations are one problem; the blocks the engine ships sit above that bracket by (C.5); and every other term of (C.4) lies outside the overlap. Structural entropy adds the key-landing term  $\sum_v Q_v H(\hat{\kappa}_v)$  and the record-landing term, which price the codelength of a random walk's landing key and landing record rather than of the records themselves, and whose nearest relatives in our code, the per- $(v, k)$  presence blocks and the record-width code, price different events. Our criterion adds the model cost that structural entropy has nowhere to put: the price (C.6) and the parameter half-logarithms.

*Part (iii), no model-cost term.* That no term of  $H^T(B)$  charges for describing  $T$  is read off the definition: each summand is  $-\frac{g_\alpha}{\text{vol}(B)} \log \frac{V_\alpha}{V_{\alpha-}}$ , a function of  $(g_\alpha, V_\alpha, V_{\alpha-}, \text{vol}(B))$  alone, and none of those four is a description of the tree. What blocks the all-singleton optimum is not a price for the parts but  $w(X_j) = 0$  in (C.1), a connectivity fact about an edge set that is handed to the objective. In the gain form the quantity standing where a price would stand is the module's address length  $\log \frac{2M}{V_j}$ , saved once per unit of internal incidence: it decreases in the module's degree volume, is identical for every atom in the module, and does not depend on how much data supports the module. The price (C.6) is by contrast a function of the arrival clock and of two global counters of the stream, and no degree volume enters it.

The statement about units follows from (C.3). Every summand there is nonnegative, so  $H^T(B) \leq H^{(1)}(B)$  for every tree  $T$  at every height; every summand of  $H^T$  itself is nonnegative for the same

two reasons, so  $H^T(B) \geq 0$ ; and  $H^{(1)}(B)$  is the Shannon entropy of the degree distribution  $x \mapsto d_x/2M$  on the  $|R \cup \mathcal{A}|$  vertices, so  $H^{(1)}(B) \leq \log |R \cup \mathcal{A}|$ . Together,  $0 \leq H^T(B) \leq H^{(1)}(B) \leq \log |R \cup \mathcal{A}|$ . Structural entropy is thus intensive, a rate capped by the logarithm of the vertex count, while a two-part codelength is extensive and grows with the stream. A per-arrival mint test compares bits of evidence against bits of model, and a rate supplies no such increment, which is where the residual stopping rule of that line lives: HISEvent searches its one-dimensional curve for a first stable point (Cao et al., 2024). Two consequences close the part. The anti-degeneracy devices differ in kind, structural entropy’s being the cut of an edge set that already exists and ours being code validity, the price plus predictive smoothing. And before any edge exists  $\text{vol}(B) = 0$ , so every term of  $H^T$  divides by zero and no structural-entropy quantity is defined, let alone able to price the first node, whereas (C.6) is defined at  $s = 1$ , where  $K_0 = 0$  and  $H_0 = 0$  give  $\Lambda_1 = \log \frac{b+1}{a}$ , which is 2 bits exactly at the declared  $(a, b) = (\frac{1}{2}, 1)$ .

*Part (iv), key-type invariance.* Let  $\Phi$  send a typed dataset  $D$  to the graph  $B$  with its atoms as unlabelled vertices. Every term of  $H^T$  is a function of cut and volume aggregates of  $B$ , and both are determined by  $\Phi(D)$ ; so for every tree  $T$ , every height  $d$  and every weighting of the edges of  $B$ , the value  $H^T$ , the minimum  $H^{(d)}$  and the set of minimising trees are constant on each fibre of  $\Phi$ .

It remains to exhibit a fibre on which our criterion is not constant, which also shows the fibres are non-trivial. Fix  $t \geq 1$  and take  $2t$  records,  $t$  of them carrying the single atom  $\alpha$  and  $t$  carrying the single atom  $\beta$ . Dataset  $D$  types the two as one key with two values,  $\alpha = (k, u)$  and  $\beta = (k, u')$ ; dataset  $D'$  types them as two keys with one value each,  $\alpha = (k_1, u)$  and  $\beta = (k_2, u')$ . Both give the same graph:  $2t$  record vertices of degree one, two atom vertices of degree  $t$ , and  $M = 2t$ . So  $\Phi(D) = \Phi(D')$  and every structural-entropy computation returns the same output on the pair. Our criterion does not. In  $D$  the two groups differ inside one block: a record of the second group scored against a node holding the first pays the value-level mismatch  $\log \frac{1/2}{q_{v,k} + A_k/2} = -\log(2q_{v,k}) + O(1)$  inside the block  $(v, k)$ , and no key-merge question exists, there being one key. In  $D'$  the node holding the first group has no block for  $k_2$  at all, so the same difference is priced by the key-presence term, and the pair  $(k_1, k_2)$  is a well-posed instance of the key-merge test. That test weighs the divergence a merge must pay against the parameters a merge saves. The two value distributions are point masses on disjoint supports, so for any mixing weight  $\pi$ ,

$$\text{JS}_\pi(\hat{p}_{k_1}, \hat{p}_{k_2}) = H(\pi\delta_u + (1 - \pi)\delta_{u'}) - \pi H(\delta_u) - (1 - \pi)H(\delta_{u'}) = H(\pi),$$

a point mass having zero entropy (Lin, 1991). At  $q_{k_1} = q_{k_2} = t$  the weight is  $\pi = \frac{1}{2}$ , so  $H(\pi) = 1$  and the divergence side is  $Q \text{JS}_\pi = 2t$  bits. The parameter side is  $\frac{|V_{k_1}| - 1}{2} \log q_{k_1} + \frac{|V_{k_2}| - 1}{2} \log q_{k_2} - \frac{|V_{k_1} \cup V_{k_2}| - 1}{2} \log Q = 0 + 0 - \frac{1}{2} \log 2t$ , which is negative for every  $t \geq 1$ . The test therefore returns *distinct*, at every  $t$ . The two audited outputs, which key carried the difference and whether it was priced at the value level or at the presence level, take different values at the two points of one  $\Phi$ -fibre; hence no function of  $B$ , at any tree height and under any edge weighting, computes them.  $\square$

The consequence is narrow, and it is not construction versus partition: HISEvent constructs its own edge set. A procedure whose output is an encoding tree of its input graph cannot output the escrowed allocation, for three reasons now visible. Nothing in its run prices the birth of a new vertex. Its trees partition where the allocation covers. And by (iv) it can emit no typed edge or key-merge verdict. HISEvent is scheduled as a baseline in the designed external tier (Appendix H.19).

### C.3 WHERE $\lambda$ COMES FROM

The question of this subsection: where does the field’s free parameter come from? The answer is an identity, and it says that  $\lambda$  is not a loose end of the BP-means derivation. It is the only part of the new-node price the derivation keeps. BP-means is the small-variance limit of exactly the model above (Broderick et al., 2013b). Small-variance asymptotics means: put a Gaussian emission model with noise variance  $\sigma^2$  on the features, multiply the negative log joint by  $2\sigma^2$ , and let  $\sigma \rightarrow 0$ ; what survives is a k-means-style objective in which opening a new feature costs a penalty  $\lambda^2$ . The recipe is called MAD-Bayes in Broderick et al. (2013b), and we use that name below. That penalty is the object we replace. In the lemma,  $K_+$  is the number of new features a step opens,  $D$  the feature dimension,  $\rho$  the prior width on a feature’s parameters,  $N$  the number of records, and  $S_k$  the number of records attached to feature  $k$ . The word mass below is IBP vocabulary for the concentration

parameter  $\gamma$ , the rate that sets how readily new features open, and the scaling of  $\gamma$  with  $\sigma$  in the lemma is the one Broderick et al. (2013b) themselves require for the limit to keep a finite nonzero penalty: it is theirs, not a choice of ours.

**Lemma C.1** (the limit deletes the computable price). *The BP-means mass scaling  $\gamma = \exp(-\lambda^2/(2\sigma^2))$  makes the following identity exact:*

$$\lambda^2 = 2\sigma^2 \cdot (-\ln \gamma) = 2\sigma^2 \times (\text{prior codelength, in nats, of one new node}).$$

*So  $\lambda^2$  is a codelength in disguise, in units of  $2\sigma^2$ . The limit then deletes every computable part of that price. The negative log joint carries three new-node terms: the mass term  $-K_+ \ln \gamma$ , engineered by the scaling to be  $O(1/\sigma^2)$ ; the parameter cost  $(K_+ D/2) \ln(2\pi\rho^2)$ ; and the allocation cost  $\sum_k \ln \binom{N}{S_k}$ , the two  $O(1)$  terms of the MAD-Bayes bookkeeping. Multiplying by  $2\sigma^2$  and letting  $\sigma \rightarrow 0$  annihilates exactly the two computable parts and keeps the part inserted by hand. The Chinese restaurant process (CRP), the single-membership analogue of the IBP from which DP-means descends, corroborates: DP-means (Kulis & Jordan, 2012) requires  $\alpha = (1 + \rho/\sigma)^{D/2} \exp(-\lambda/(2\sigma))$ , in the cited paper's own notation, where  $\sigma$  denotes the noise variance and the penalty is  $\lambda$ , so both exponents here are the penalty over twice the noise variance and the apparent asymmetry with the BP-means scaling is notational. And  $(D/2) \ln(1 + \rho/\sigma)$  is the parameter codelength of a fresh  $D$ -dimensional mean at the resolution the data supports, so  $\alpha$  factorises as (parameter codelength)  $\times$  (free constant), with  $\lambda$  the residue after the codelength is divided out.*

*Proof.* The scaling makes the mass term  $O(1/\sigma^2)$  by construction, so  $2\sigma^2$  times it tends to  $K_+ \lambda^2$ ; the parameter and allocation costs are  $O(1)$  in  $\sigma$ , so  $2\sigma^2$  times them tends to zero. The DP-means factorisation is read off the cited requirement on  $\alpha$  by taking logarithms.  $\square$

In words: the exact model contains a principled, computable price for a new node, the parameter bits plus the allocation bits, and the limit that produces BP-means subtracts precisely those two terms. What survives is the one term the scaling was engineered to protect.  $\lambda$  is a free parameter because the limit was taken, not despite it. Our answer is to not take the limit: code the same model prequentially and read the price off the code. Two results say what that forces.

The first result answers the question a reader should ask at once: why not simply create a node the first time a record fails to fit? Because a brand-new node has nothing to predict with. Under any honest code its first prediction is uninformed, so its first record cannot, in expectation, out-code a background that has seen the whole corpus. The one apparent escape, seeding the node's predictive at the observed value itself, only relocates the same bits into the seed. One more term is needed to state this:  $D(p \| q)$  is the Kullback-Leibler divergence, the expected number of extra bits paid for coding data drawn from  $p$  with a code built for  $q$ ; it is zero iff  $p = q$  and positive otherwise. And one property: a prior is exchangeable over values, or value-exchangeable, when it treats the alphabet's symbols interchangeably, favouring none before data arrive.

**Lemma 1 (myopic impossibility), restated in full.** Let  $Q$  be the predictive a node minted at record  $n$  assigns to a value of attribute  $a$  before any record has attached to it. (i) In a valid code  $Q$  is measurable with respect to the transmitted past, with  $\sum_x Q(x) \leq 1$ . With no past support, exchangeability over values forces  $Q$  uniform, so every value costs  $\log d_a$ ; the KT table at zero counts realises this. A construction that seeds  $Q$  at the observed value is a two-part code whose seed itself costs at least  $\log d_a$  under any value-exchangeable prior: the same bits, relocated. (ii) With  $p$  the conditional law of  $x_{n,a}$ , the expected first-record evidence against the background is  $-D(p \| Q) + D(p \| \hat{P}_{0,a})$ . Under the operational null  $p = \hat{P}_{0,a}$  this equals  $-D(\hat{P}_{0,a} \| Q) \leq 0$ ; for  $Q = \text{Uniform}(d_a)$  it is  $-D(\hat{P}_{0,a} \| \text{Unif})$ , zero iff the background is uniform. (iii)  $\Lambda_n > 0$  whenever  $a + K_{n-1} < n(b + H_{n-1}) + 1$ . So the expected net value of minting on the arriving record is at most  $-\Lambda_n < 0$  on every attribute: under any valid code a fresh node cannot pay for itself, in expectation, on its own first record. The only pointwise exceptions are rarity events, values the background itself prices below  $2^{-\Lambda_n}/d_a$ , whose codelength signature is identical for a novel entity and a corrupted record.

The lemma is not only a warning on paper: E7 (Section 5.2) measures both immediate forms at zero mints on 6,000 records; three earlier derivation routes that reported  $K \in \{0, 1\}$  are development

notes without a shipped artefact and are not counted as evidence here. DP-means has no such obstacle only because it places the new centroid at the point, so the new cluster's data cost is zero and one distant record already pays  $\lambda$ ; a predictive code has no such move. The repair is the deferred mint of Section 3.

Lemma 1 states that under any valid code a fresh node cannot pay for itself, in expectation, on its own first record. The proof establishes its three parts: (i) a fresh node's predictive is forced uniform, (ii) its expected first-record evidence is nonpositive, and (iii) the price it must beat is strictly positive.

*Proof.* (i) is Kraft's inequality plus exchangeability. A valid predictive  $Q$  for a fresh node is measurable with respect to the transmitted past and satisfies  $\sum_x Q(x) \leq 1$ . If the past carries no support for any value of attribute  $a$ , exchangeability over values forces  $Q$  uniform, so every value costs  $\log d_a$ . The Krichevsky-Trofimov (KT) code, the standard Bayes code that prices each value by its smoothed running count, starting every count at one half, realises this exactly at zero counts (Krichevsky & Trofimov, 1981). Any construction that instead seeds  $Q$  at the observed value is a two-part code whose seed must itself be transmitted, at a cost of at least  $\log d_a$  under any value-exchangeable prior: the same bits, relocated.

(ii) asks what the fresh node is expected to save against the background predictive  $\hat{P}_{0,a}$ , and the answer is the identity  $\mathbb{E}_{x \sim p}[\log Q(x) - \log \hat{P}_{0,a}(x)] = -D(p \| Q) + D(p \| \hat{P}_{0,a})$ . Evaluated at the operational null  $p = \hat{P}_{0,a}$ , it equals  $-D(\hat{P}_{0,a} \| Q) \leq 0$ , with equality only when the evidence is identically zero. For the minimax choice  $Q = \text{Uniform}(d_a)$  it is  $-D(\hat{P}_{0,a} \| \text{Unif}) \leq 0$ , with equality iff the background is uniform. In words: on its first record the fresh node is expected to lose bits, never gain them.

(iii) is immediate from Lemma C.2:  $2^{-\Lambda_s} = (a + K_{s-1})/(s(b + H_{s-1}) + 1)$ , which is below one whenever  $a + K_{s-1} < s(b + H_{s-1}) + 1$ ; at  $s = 1$  and the declared  $(a, b) = (\frac{1}{2}, 1)$ ,  $\Lambda_1 = \log \frac{b+1}{a} = 2$  bits exactly.

Combining the three parts, the mint route is dominated in expectation by the all-background action by at least  $\Lambda_n$  bits on every attribute. Pointwise, a seeded mint saves bits only on values with  $\hat{P}_{0,a}(x) < 2^{-\Lambda_n}/d_a$ . That event is a statement about rarity alone, and its codelength signature is identical for a novel entity and a corrupted record. Empirically, E7 records zero mints for both immediate forms (Section 5.2).  $\square$

The second result is the price itself. The question it answers: if the model is kept exact, what does one new node cost? The price is the log posterior odds of no mint against one mint, in bits: how much evidence a candidate must produce before the model prefers that it exist. It has a closed form, and the form is the claim. The coefficient on  $\log s$  is exactly one, forced by the model, not chosen; everything else is a statistic the stream computes about itself. One coding term appears in part (i): normalized maximum likelihood (NML), the minimax-optimal code for a model class, which exists only when its normalizer, a sum of maximised likelihoods over all possible data, is finite.

**Lemma C.2** (the price). *Under the IBP with  $\gamma \sim \text{Gamma}(a, b)$  marginalised, the price of one new node at record  $s$ , the log posterior odds of no mint against one mint, is Bayes-exact:*

$$\Lambda_s = \log(s(b + H_{s-1}) + 1) - \log(a + K_{s-1}) = \log s - \log \hat{\gamma}_{s-1} + O(1/(s \log s)), \quad (\text{C.6})$$

so the constant on  $\log s$  is exactly one, and the remainder is minus the log of the posterior mean concentration, a statistic of the stream itself: streams that mint often lower future prices, sparse streams raise them. Attachment odds contain no  $\gamma$ , so marginalisation leaves every attachment untouched: the mint-time price is purely structural. Moreover: (i) Propriety of the hyperprior is forced by Kraft. The NML normalizer over the node count sums, for every possible count  $K$ , the likelihood of  $K$  under the rate chosen best in hindsight; for a Poisson count that maximised likelihood is  $K^K e^{-K}/K!$ , so the normalizer is  $\sum_K K^K e^{-K}/K!$ , which diverges, so no NML code for the allocation exists at any  $n$  (Shtarkov, 1987), and every improper hyperprior fails to be a code:  $\pi(\gamma) \propto 1/\gamma$  has infinite mass, and at  $b = 0$  the first-record predictive puts probability zero on every finite mint count. (ii) The hyperprior influence is  $O(1)$ . The per-mint difference between proper hyperpriors is  $O(1/\log n)$ , 0.69 bits at  $n = 10$  falling to 0.19 at  $n = 10^5$  between  $(\frac{1}{2}, 1)$  and  $(2, 0.1)$  along the trajectory  $K_s = 2H_s$ . The shape  $a = \frac{1}{2}$  is Jeffreys-canonical (Xie & Barron, 2000), and the scale  $b > 0$  is the one residual freedom, forced positive by Kraft, declared rather than



calibrated: the code is a prior (Wallace & Patrick, 1993). (iii) Learning the concentration inside the code costs  $\frac{1}{2} \log H_N + O(1) = \frac{1}{2} \log \ln N + O(1)$  bits over the entire stream, negligible against a single mint's price.

The proof uses the IBP exchangeable partition probability function of Broderick et al. (2013a), the closed form for an allocation's probability that depends only on its counts, and the least-favorable property of the Jeffreys prior, the fact that it is the prior a worst-case opponent would pick, which is what makes it the minimax default (Clarke & Barron, 1994).

Lemma C.2 states the closed form of the mint price and three facts about the hyperprior: (i) it must be proper, (ii) its influence is  $O(1/\log n)$  per mint, and (iii) learning the concentration inside the code costs  $\frac{1}{2} \log H_N + O(1)$  bits over the whole stream. The proof derives the closed form first, then proves (i), (ii) and (iii) in turn.

*Proof.* We first identify where  $\gamma$  lives in the model. The exchangeable feature probability function of the Indian buffet process is the probability of an allocation pattern; at unit mass it reads  $P(Z_{1:N} | \gamma) = W(Z_{1:N}) \gamma^{K_N} e^{-\gamma H_N}$  (Griffiths & Ghahramani, 2011; Broderick et al., 2013b), with  $W$  collecting every factor free of  $\gamma$ . So  $\gamma$  sits in a one-dimensional exponential family: the data touch it only through the node count  $K_N$  and the exposure  $H_N$ . With a proper Gamma( $a, b$ ) prior the marginal is  $m(Z_{1:N}) = W(Z_{1:N}) \frac{b^a}{\Gamma(a)} \frac{\Gamma(K_N + a)}{(H_N + b)^{K_N + a}}$ . This is a probability distribution, so  $-\log m$  is a codelength realizable by arithmetic coding. By Poisson-Gamma conjugacy the posterior before record  $s$  is  $\gamma | Z_{1:s-1} \sim \text{Gamma}(a + K_{s-1}, b + H_{s-1})$ , and the number of new nodes at record  $s$  is negative binomial under that posterior. The question the price answers is: what are the posterior odds of one new node against none at record  $s$ ? They are

$$\frac{P(j=1)}{P(j=0)} = \frac{a + K_{s-1}}{s(b + H_{s-1}) + 1}.$$

The price is the negative base-2 logarithm of this ratio, which is the displayed closed form (verified to machine precision against the product of prequential conditionals, a check made during development for which no artefact ships). The asymptotic expansion in the lemma follows from  $\log(s\beta + 1) = \log s + \log \beta + \log(1 + 1/(s\beta))$  with  $\beta = b + H_{s-1} = \Theta(\log s)$ . Finally, the attachment odds  $m_k/(n - m_k)$  contain no  $\gamma$ , so marginalising  $\gamma$  leaves every attachment decision untouched: the mint-time price is purely structural.

For (i), propriety is forced. Kraft's inequality is the fact that valid codeword lengths must satisfy  $\sum 2^{-\text{length}} \leq 1$ ; equivalently, only a (sub)probability can be a code. Two alternatives to the proper prior fail it. First, the normalized maximum-likelihood route: summing  $W$  over the allocations with  $K$  columns gives  $\sum_{Z:K} W(Z) = H_N^K / K!$ , because the sum must reproduce the Poisson( $\gamma H_N$ ) marginal over the node count. The Shtarkov sum, the normaliser such a code would need (Shtarkov, 1987), is therefore  $\sum_K \sup_\gamma \gamma^K e^{-\gamma H_N} H_N^K / K! = \sum_K K^K e^{-K} / K!$ . By Stirling its terms fall only like  $1/\sqrt{2\pi K}$ , so the partial sums grow like  $\sqrt{2K/\pi}$  (verified numerically) and the sum diverges: no normalized maximum-likelihood code for the allocation exists at any  $n$ . Second, the improper-prior route: for improper  $\pi$ ,  $\sum_Z \int P_\gamma \pi = \int \pi = \infty$ , so the mixture is not a code; and at  $b = 0$  the first-record predictive has  $\mathbb{E}[K_1^+] = a/b \rightarrow \infty$  and puts mass 0 on every finite count. The same Kraft logic that kills batch profiling of  $\gamma$  forces the mixture and forces it proper.

For (ii), the shape and the sensitivity. The Fisher information is  $I(\gamma) = H/\gamma$ , so Jeffreys' prior is  $\propto \gamma^{-1/2}$ , parameterisation-invariant and asymptotically least favourable (Clarke & Barron, 1994). How much can the choice among proper hyperpriors matter? The per-mint difference between two of them is  $\log \frac{b+H}{b'+H} - \log \frac{a+K}{a'+K} \rightarrow 0$  at rate  $O(1/\log n)$ . Between  $(a, b) = (\frac{1}{2}, 1)$  and  $(2, 0.1)$ , evaluated through the shipped price (results/e15\_theory\_numbers.json), the exact difference is 0.69 bits per mint at  $n = 10$  and 0.19 at  $n = 10^5$  along the trajectory  $K_s = 2H_s$ , and 0.91 and 0.27 along  $K_s = H_s$ ; the displayed asymptotic difference gives 0.68 and 0.90 at  $n = 10$  and the same values at  $n = 10^5$ , so the form must be stated as well as the trajectory. Neither trajectory is one our runs walk: on the planted stream the engine holds no node at ten records and estimates a rate of 0.66 at a hundred thousand, where the same difference is 2.36 and 0.34 bits. The total codelength difference over the stream converges to the log ratio of the priors at the posterior mean, an  $O(1)$  quantity.

For (iii), the cost of learning  $\gamma$  inside the code. The mixture's regret, the extra bits it spends against the best fixed  $\gamma$  chosen in hindsight, is  $\frac{1}{2} \log H_N + O(1)$ , because  $\gamma$ 's effective sample size is the exposure  $H_N \approx \ln N$ . The regret slope is the closed form  $\frac{1}{2} \log 10 = 1.661$  bits per decade of exposure.  $\square$

Two-part integer-code implementations replace the exact price by  $c \log n$ , with  $c \in [1, 2]$  fixed by whichever integer code the implementation adopts (Rissanen, 1983). The Bayes-exact constant is  $c = 1$ . E12 measures what the choice of  $c$ , and of the other code families beside it, actually costs: on planted structure the minted set does not move at all, nine valid codes returning the same eight nodes at adjusted Rand index 1.0 with birth times shifting by up to 217 records, while on a shared-key stream and on the Wikipedia infoboxes the partition moves to a worst pairwise index of 0.6055 and 0.5233, against 0.6068 and 0.5804 over twenty arrival orders of the shipped code. The claim that the code choice changes when a node is born rather than which nodes exist therefore holds where the structure is identifiable, and not beyond it (Appendix D).

The impossibility and the price together force the mechanism: deferral. A candidate node is named by an attribute and value, its price is frozen at seed time, and evidence accrues in escrow, a ledger held for the candidate, until it covers the price. The theorem below answers the question every deferred test must face: if a candidate can be re-tested after every arrival, forever, why does the chance of a false mint not grow with the stream? Three ideas carry the answer, and each has a plain form. First, the evidence  $G_t$  is an accumulated log-likelihood ratio: the bits saved so far by coding the stream with the candidate rather than without it. Second, under the null hypothesis that the candidate is spurious,  $2^{G_t}$  is a nonnegative martingale started at one: a process whose expected next value equals its current value, the formal version of a fair game, and a gambler playing a fair game does not get rich. Third, Ville's inequality is the anytime form of Markov's inequality: a nonnegative martingale started at one ever exceeds  $2^\Lambda$  with probability at most  $2^{-\Lambda}$ , where  $\Lambda$  ranges over the whole infinite future. The bound covers the running supremum, so re-testing at every arrival costs nothing extra. What remains is the size of the hypothesis space, and that is what the naming of candidates buys: attribute-value pairs form a vocabulary that grows with the schema, never with the record count, so the lifetime bound is independent of stream length.

**Theorem C.1** (anytime validity of the escrow). *Fix a candidate seed  $p = (a^\dagger, v^\dagger)$  with seed time  $s$  and frozen price  $\Lambda_s$  computed from  $(K_{s-1}, H_{s-1})$ . Let  $G_t(p)$  be the accumulated log-likelihood ratio, from  $s$  onward, between two sequential distributions on the stream: the graph with  $p$  (its accumulated posterior predictives, with the attachment and absence terms of its allocation code) against the current graph without  $p$ . Then: (i) If both are exactly normalised sequential probability assignments, then under the null  $H_0(p)$ , that the without- $p$  model generates the stream,  $M_t = 2^{G_t(p)}$  is a nonnegative martingale with  $M_s = 1$ , a test martingale in the sense of Ramdas et al. (2023). The condition is about the value block's third stage. A block holding  $u$  distinct values in  $N$  observations under a key whose inventory holds  $\text{inv}$  values retains mass  $1 - u(u + \frac{1}{2}) / ((N + 1)(\text{inv} + 1))$  under the shipped charge of  $\log(\text{inv} + 1)$  bits for the escape, because only  $\text{inv} - u + 1$  slots are reachable; the presence and absence terms are an exactly normalised KT Bernoulli, so all of the deficiency lives in the value blocks. Charging  $\log(\text{inv} - u + 1)$  instead returns every block to mass one and makes the step exact: the one-step expectation of the ratio under the incumbent is 1.000000 over 955,200 null key-steps, against 0.862862 for the shipped charge (E16, results/el6\_supermartingale\_check.json; the closed form is verified against the shipped code by brute-force summation over the alphabet on 400 random blocks, maximum error  $4.4 \times 10^{-16}$ ). The shipped charge is a valid prefix code either way: its codeword lengths satisfy Kraft with inequality rather than equality, which wastes a little code space and is ordinary. The tight charge is implemented behind an environment flag and is not the default, because it buys the per-step identity and nothing else that matters: it preserves every positive control and improves Wikipedia purity from 0.9665 to 0.986, but moves the Wikipedia background from 2 records of 320 to 114, and it does not rescue (ii). (ii) In the idealised setting of (i), and on the whole stream rather than a chosen subsequence, Ville's inequality (Ville, 1939) gives a lifetime guarantee with no union bound over time:*

$$\mathbb{P}(\exists t \geq s : G_t(p) \geq \Lambda_s) \leq 2^{-\Lambda_s} = \frac{a + K_{s-1}}{s(b + H_{s-1}) + 1} < \frac{\hat{\gamma}_{s-1}}{s}.$$

So a candidate may be re-tested after every arrival, forever, while spending the false-mint budget of one test. This part is stated for the idealised setting and is not claimed for the engine, for the reason

given after the proof: the engine accrues only on records carrying the seed pattern, so its released statistic is computed on a subsequence the data chose. The measured exceedance is reported in its place. If at most one fresh seed were opened per record, the expected number of spurious mints over the whole stream would be at most  $\sum_{s \leq N} \hat{\gamma}_{s-1}/s \approx \hat{\gamma} H_N = \mathbb{E}[K_N \mid \gamma = \hat{\gamma}]$ , the node count the model class already expected; the shipped engine opens one candidate per residual (key, value) pair, so the count is governed by (iii). The shipped gate releases at the price of Equation D.10 alone, without the seed-time term  $\Lambda_s$ , and the same inequality applies at the level  $c = \min_t \text{Price}(t, T, n)$ . (iii) (Stream-length independence.) The hypothesis space is indexed by seed pairs  $(a^\dagger, v^\dagger)$ , one candidate per residual pair: a set that grows with the schema and, under the open-alphabet escape, with the value inventory, never with the record count as such. In the shipped price the naming charge is the support term  $L_{\text{supp}}(T)$  and the node name  $\log(K+1)$  of Equation D.10 rather than a separate  $\log|A| + \log d_{a^\dagger}$ ; either is Kraft-tight, and by Kraft the charge is the multiple-comparisons correction: added to each candidate's price, it makes the family-wise budget no larger than the worst single-candidate budget, so at the declared hyperprior the expected number of spurious mints, over every candidate and the whole lifetime, is at most  $\max_p 2^{-\Lambda_p} < 1$ : less than one spurious node in expectation, ever. A deployment may also raise every candidate's price by the same extra  $\Gamma$  bits, a uniform per-candidate slack on top of the naming charge, to buy a smaller bound; the shipped system adds none. With that slack,  $\mathbb{P}(\text{any spurious mint, ever}) \leq |A| \bar{d} 2^{-\Gamma}$ , independent of the stream length  $T$ . Under mint-on-first-miss, by contrast, the hypotheses are one per record, a set of size  $T$ : naming costs  $\Theta(\log T)$  and the false-mint bound grows as  $\Theta(T)$  unless the charge is allowed to know the stream length, a tuned parameter. The reparameterisation from records to seed pairs exists only under the IBP stance: a feature is named by an attribute and value, whereas a DP-means cluster is named by a datapoint.

Theorem C.1 states that the escrow can re-test a candidate after every arrival, forever, while its lifetime probability of minting a spurious node stays below  $2^{-\Lambda_s}$ , and that the bound over all candidates together does not grow with the stream. The proof shows the evidence process is a martingale, applies Ville's inequality, and then pays for the union bound in bits.

*Proof.* (i) Write  $M_t = \prod_{u=s}^t Q_u(x_u)/P_u(x_u)$ , where  $Q_u$  and  $P_u$  are the one-step predictives of the graph-with- $p$  and without- $p$  models. Each is normalised given  $\mathcal{F}_{u-1}$ , the information available before record  $u$ , because every factor of a prequential code is a probability distribution computable from the transmitted past (Dawid, 1984), under the tight naming charge of the statement. Under the shipped charge both are sub-probabilities, and the exact one-step expectation under the incumbent read as a normalised law is the ratio of the two masses,  $Z_Q/Z_P$ , so the accumulator is a supermartingale exactly when  $\phi(u_Q, N_Q) \geq \phi(u_P, N_P)$  on every published key with  $\phi(u, N) = u(u + \frac{1}{2})/(N+1)$ : a condition whose direction is not fixed, holding at 99.5 percent of steps on a low-cardinality null and failing at 99.7 percent on Wikipedia infoboxes. A martingale is a process whose expected next value, given everything so far, equals its current value. Under the null  $H_0(p)$  the next record is drawn from  $P_t$ , so

$$\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] = M_{t-1} \sum_x P_t(x) \frac{Q_t(x)}{P_t(x)} = M_{t-1} \sum_x Q_t(x) = M_{t-1}.$$

The ratio cancels the true law exactly, so  $M_t$  is a nonnegative martingale with  $M_s = 1$ .

(ii) Ville's inequality (Ville, 1939) says that a nonnegative martingale starting at one ever exceeds a level  $c$  with probability at most  $1/c$ , over its whole infinite future. Applied at  $c = 2^{\Lambda_s}$  it gives  $\mathbb{P}(\sup_t M_t \geq 2^{\Lambda_s}) \leq 2^{-\Lambda_s}$ , with no union bound over time; this is the standard anytime-valid construction (Ramdas et al., 2023). The closed form and the bound  $\hat{\gamma}_{s-1}/s$  follow from Lemma C.2. Summing  $\hat{\gamma}_{s-1}/s$  over seed times gives approximately  $\hat{\gamma} H_N$  when  $\hat{\gamma}$  is stable, while  $\mathbb{E}[K_N \mid \gamma] = \sum_{n \leq N} \gamma/n = \gamma H_N$  under the prior itself. In words: the escrow's lifetime false-mint budget equals what the model class already expected to spend on nodes.

(iii) is the union bound, paid for in bits. If candidate  $p$  must clear  $\Lambda_p + \log|A| + \log d_{a^\dagger}$ , then  $\sum_p 2^{-\Lambda_p - \log|A| - \log d_{a^\dagger}} \leq \max_p 2^{-\Lambda_p}$ , since there are  $d_{a^\dagger}$  candidates per key. Equivalently, with a uniform per-candidate slack  $\Gamma$  the family bound is  $|A| \bar{d} 2^{-\Gamma}$ . The counts of the two hypothesis spaces are as stated: seed pairs number  $|A| \bar{d}$  and grow only as new values appear, while records number  $T$ . Only the seed-pair indexing is available without labels: it is the Indian buffet process's feature naming, not an algorithmic convenience.  $\square$

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One caveat on the implemented statistic. Validity requires the escrow to be computed on the full stream, never on a post-selection subsequence: conditioning on per-record attachment counts breaks the complete randomness the martingale rests on, the property of the IBP that lets disjoint parts of the stream be priced independently (see Doshi-Velez & Williamson (2015) for why restriction breaks it). The implementation accrues only on records carrying the pattern, dropping the absence terms  $\log(1 - m_p/u)$ , where  $m_p$  is the candidate’s attachment count and  $u$  the arrival index at which the term would be charged; under the IBP the record at index  $u$  fails to attach to the candidate with probability  $1 - m_p/u$ , so each dropped term is the log-probability of one non-attachment. Dropping negative terms can only inflate  $G_t(p)$ , by at most  $O(m_p \log(t/s))$  bits, an anti-conservative omission on its own. The guarantee holds for the implemented statistic because exactly that bound is charged back on the price side, through the cohort-membership bits  $\log \binom{n}{t}$  of the column code. The accounting, in one line: the omission can add at most  $O(m_p \log(t/s))$  bits to the evidence side, and the gate the implementation enforces is the price of Equation D.10 alone (Equation 2), without the seed-time term  $\Lambda_s$ , whose statement terms carry the column code’s  $\log \binom{n}{t}$  bits on the price side; the inflation is dominated by that charge, so the release fires no earlier than an honest accounting would allow. One gap does not close in this draft, and it is the reason part (ii) is stated for the idealised setting only: the released statistic is the positive-part prefix over keys sorted by escrow plus the outside absence credit (Section 3), accrued only on records carrying the seed pattern, and no separate term pays for selecting either the positive subset or that subsequence. The argument for the omitted absence terms is the justification of the implemented statistic in Appendix C.

**The exceedance, measured rather than claimed.** Because the bound does not transfer, the paper reports what the engine does. Over 300 independent null streams of 2,000 records and 12,000 candidate trajectories,  $\Pr(\sup_t G_t \geq b)$  at  $b = 1, 2, 4, 8, 12$  is 1.000 at every level for the shipped statistic; 1.000 at every level with the tight naming charge alone; 0.727, 0.583, 0.316, 0.0498, 0.0032 with the seed key’s own term excluded alone; and 0.400, 0.309, 0.170, 0.0395, 0.0077 with both, against the nominal 0.5, 0.25, 0.0625, 0.0039, 0.00024 (standard error about 0.004). Both corrections together hold only at  $b = 1$ . The dominant cause is therefore selection, not the code: the seed key’s value is constant on every record the engine accrues on, so its term is positive by construction and is not a fair bet.

The practical no-false-mint result does not depend on any of this, and it is stronger than the bound would be. The release gate fires on the computed price of Equation D.10, whose membership column grows faster than the accumulated drift, and no node is minted on any of the 48 null streams of E8, nor on any of the 300 wider streams, under all four combinations of the two flags.

Excluding the seed key’s evidence, which is the only one of the two corrections that moves the exceedance, is not available as a default, and the reason is a finding rather than an inconvenience. On the planted eight-group stream at its shipped noise rate the unselected statistic returns no nodes at all, and over twenty arrival orders the node count averages 1.6 against a planted eight; the seed key’s term is about 43 percent of a candidate’s evidence, 332 bits of 777 in the diagnostic. The engine’s drift is load-bearing: the rule mints because the pattern that created the candidate keeps paying, and the price, not the statistic, is the thing that is supposed to charge for having chosen it. The design that might have made the theorem true without destroying the mint is to keep the evidence and charge the selection on the price side, adding the seed pair’s own naming cost. That design has now been run and it does not work, and the way it fails identifies the defect exactly.

The charge is affordable. At its natural size, 5.3 to 12.2 bits, both synthetic controls keep their exact node count and a perfect index under the canonical order and over twenty arrival orders, the deferred arm of E7 keeps its two nodes with identical supports and no typo node on all five seeds, no node is minted on any of the 48 null streams nor on any of the 300 wider ones, and the Wikipedia demonstration improves, its twenty-order mean index rising from 0.8438 to 0.8713. A uniform slack of 256 bits still leaves both synthetic controls exact. And it is ineffective: the exceedance stays 1.000 at every level, alone and combined with the tight naming charge.

Those two facts together are the diagnosis. A naming charge is constant in the stream length by construction, and the deficit is not: under the null the accumulator drifts at 1.42 bits per accrual step at  $T = 500$  and 2.76 at  $T = 4,000$ , so the slack that would restore the nominal level at  $b = 8$  is 97 bits at the first length and 1,205 at the last. The charge of Theorem C.1(iii) is a correct multiple-comparisons correction, and it is being asked to pay for something that is not a

multiplicity. The corroboration is that with the seed key's own term removed from the statistic the residual requirement is 3.4 to 3.6 bits and flat in  $T$ , which is what a multiplicity looks like, and the same 5.32 bit charge covers it, holding at all five levels; that combination is the only arm in this series that satisfies the bound, and it is not shippable because removing the seed term destroys the mint.

One qualification the charge itself needs. Written as the log of the key count plus the log of that key's alphabet, it is Kraft-tight for a closed schema only. Under the open alphabet the escape mechanism creates, naming each candidate with the counts current at its own seed time is not a prefix code over a candidate set that grows with the stream. A universal integer code on the seed pair's interning ranks is Kraft over the unbounded set and still constant in  $T$ , and is the form to use.

*Justification of the implemented statistic.* The implemented escrow differs from the full likelihood ratio in one way, and this argument shows the difference is paid for. Validity requires the escrow to be computed on the full stream, never on a post-selection subsequence: conditioning on per-record attachment counts breaks the complete randomness and the exchangeability the martingale rests on (Doshi-Velez & Williamson, 2015). The implemented escrow accrues only on records carrying the pattern and drops the absence terms  $\log(1 - m_p/u)$  elsewhere. Dropping these negative terms can only inflate  $G_t(p)$ , by at most  $O(m_p \log(t/s))$  bits over the escrow window, so the omission by itself points in the anti-conservative direction. The guarantee of Theorem C.1(ii) holds for the full ratio, and the omission is paid for once that bound is charged against the candidate. What this argument does not repair, and what the exceedance above measures, is the accrual index set: the price charges for the dropped absence terms, not for the choice of the records on which evidence is accrued at all. The implemented release price does exactly that: it carries the cohort-membership bits, the  $\log \binom{n}{t}$  leading term of the column code. It is the same allocation accounting, relocated to the price side.  $\square$

#### C.4 THE IDENTITIES BEHIND THE TYPED EDGES

Every decision is emitted with a receipt: typed edges naming which attributes decided it, and at what price in bits. Three results say why those receipts can be trusted. Separability says the per-attribute decomposition is exact, not a narrative readout. The total-correlation identity says what quantity the escrow is actually metering. Consistency says the whole procedure needs no significance level.

A candidate must declare a scope  $S \subseteq A$ : the set of attributes it claims to say something about. Is choosing the best scope a combinatorial search? Under the scope code we use, no: the best scope falls out of independent per-attribute tests. In the lemma,  $g_a$  is the accumulated per-attribute saving, the bits attribute  $a$  has earned for the candidate so far;  $\ell_{\text{in}}(a)$  and  $\ell_{\text{out}}(a)$  are the costs of declaring  $a$  inside or outside the scope, with Bernoulli-KT estimators fitted to past scopes;  $\Delta \ell_a = \ell_{\text{in}}(a) - \ell_{\text{out}}(a)$  is the marginal cost of including  $a$ ; and  $[x]_+$  means  $\max(x, 0)$ , the positive part.

#### C.5 SCOPE, AND WHY A HIERARCHY IS NOT A NODE

**Proposition 1 (scope), restated in full.** Fix the stream length  $n$  and let a node hold  $t$  of those records. (i) Of the six terms of  $\text{Price}(t, T, n)$  in Equation D.10, only  $L_{\text{col}}$  depends on  $t$ ; the announcement, the two bits of edit type, the support code, the increment of the node-count code and the name are all free of it. Writing  $\lg \Gamma$  for  $\log_2 \Gamma$  and using Equation D.11,

$$\Delta(t, n) = L_{\text{col}}(t, n) - L_{\text{col}}(t-1, n) = \lg \Gamma(t - \tfrac{1}{2}) - \lg \Gamma(t + \tfrac{1}{2}) + \lg \Gamma(n - t + \tfrac{3}{2}) - \lg \Gamma(n - t + \tfrac{1}{2}),$$

since the  $\log_2 \pi$  and  $\lg \Gamma(n+1)$  terms cancel. Applying  $\Gamma(z+1) = z\Gamma(z)$  to each pair collapses this to  $\log \frac{n-t+1/2}{t-1/2}$ , which is Equation 3. Expanding about  $t/n \rightarrow 0$  gives  $\log(n/t) + O(t/n) + O(1/t)$ . A node is affordable only when each member saves more than  $\Delta(t, n)$ , so at fixed  $t$  the requirement grows without bound in  $n$  and a group of bounded size is eventually priced out of any stream. (ii) A candidate is created on a seed pair and its accumulator advances only on records carrying that pair (Section 3), so its cohort is bounded above by the number of records carrying it, and release requires that cohort to reach  $t^*$ .

Both clauses are necessary and neither is sufficient, which is easy to misread and worth stating plainly. E34 exhibits a stream meeting both with room to spare, seed cohorts of 36 to 47 against a

$t^*$  of at most 24 and a best live candidate accruing 13.25 bits per member against a requirement of 9.23, on which the engine still mints three nodes where the truth has eight. The conditions place the benchmarks; they do not predict success.

The identity in (i) is checked against the shipped routine rather than trusted: over  $n$  from  $10^3$  to  $10^6$  and  $t$  from 2 to  $10^4$ , the largest disagreement between Equation 3 and the difference of two calls to `price` is  $4 \times 10^{-9}$  bits.

**Proposition C.2** (no hierarchy). *Let a parent hold members  $M$  with support  $F$  and let  $K$  children partition  $M$ , the  $i$ -th with support  $P_i$  disjoint from  $F$ . In the flat description  $K$  nodes carry support  $F \cup P_i$ ; in the nested one the parent carries  $F$  and each child carries  $P_i$ . Every (record, key) cell is coded once under both, so the two are codes for the same data. Then  $L(\text{nested}) - L(\text{flat}) = L_{\text{col}}(|M|, n) - \Delta$ , where  $\Delta > 0$  is the parametric saving from coding each key of  $F$  once over  $|M|$  rows rather than  $K$  times over  $|M|/K$  rows. The first term is  $\Theta(n)$  and  $\Delta$  is  $\Theta(\log n)$ , so the flat description is strictly cheaper for every  $n$  past a small constant and the gap diverges. Hence no search mints a parent.*

**Proof.** Let a parent hold members  $M$  with support  $F$  and let  $K$  children partition  $M$ , the  $i$ -th with support  $P_i$  disjoint from  $F$ . In the flat description  $K$  nodes carry support  $F \cup P_i$  and the members of the  $i$ -th child; in the nested one the parent carries  $F$  over  $M$  and each child carries  $P_i$ . Every (record, key) cell is coded once under both, by the node owning that key, so the two are codes for the same data.

Take the difference term by term. The children’s membership columns are identical in both, and the nested description adds the parent’s, contributing  $L_{\text{col}}(|M|, n)$ . The presence and value blocks cover the same cells with the same conditioning sets, so their data terms cancel and only the Krichevsky-Trofimov parametric part differs: a key of  $F$  is coded once over  $|M|$  rows in the nested description and  $K$  times over  $|M|/K$  rows in the flat one, contributing  $\Delta = |F| \left[ K \cdot \frac{d-1}{2} \log(|M|/K) - \frac{d-1}{2} \log |M| \right] > 0$  in the nested description’s favour. The support codes and the node-count code differ by  $O(K \log A_n)$ . Hence  $L(\text{nested}) - L(\text{flat}) = L_{\text{col}}(|M|, n) - \Delta + O(K \log A_n)$ , in which the first term is  $\Theta(n)$  and  $\Delta$  is  $\Theta(\log n)$ .

Priced with the engine’s own functions on generated streams with noise, the nested description costs 1,677 bits more at  $n = 900$ , 6,842 at  $n = 3,000$ , 22,358 at  $n = 9,000$  and 71,076 at  $n = 27,000$ , of which the membership column contributes 2,491, 8,281, 24,811 and 74,402.

The parent’s column is derivable rather than merely expensive: its members are the union of its children’s, which the decoder already holds. Charging it only for the residual its children do not determine, empty when they partition it, removes  $L_{\text{col}}(|M|, n)$  and leaves  $-\Delta$ , so the nested description wins by 815, 1,439, 2,453 and 3,326 bits at those four sizes. The engine does not implement this and nothing in the paper depends on it, because a parent need not be a node at all: Section 4 reads the relation off the attributes, where it costs nothing.

**The measurements behind Propositions 1 and G.7.** What follows is the detail the main text points at, moved here whole so that nothing is lost to the page budget.

Both halves are measured. Equation 3 agrees with the shipped `price` routine to  $4 \times 10^{-9}$  bits, and departs from the  $\log(n/t)$  reading by at most 0.41 bits over  $n$  from  $10^3$  to  $10^5$  and  $t$  from 2 to  $10^3$ . For  $t^*$ , E9 reports the cohort at release as 16.6 on average when two keys agree and 3.3 when six do, with 3 and 30 as the extremes, so  $t^*$  lies between 3 and 24 over the grid we have run. E31 then holds a stream’s group count, record count and key count fixed and moves only how often a value repeats, which moves only the seed cohort: agreement with the planted groups falls from 0.92 at a cohort of 136 to 0.69 at 43, 0.23 at 26 and 0.0001 at 16, and the engine then mints nothing at all rather than minting badly. The cliff is bracketed between 16 and 43, and the  $t^*$  ceiling of 24, measured on a different family for a different purpose, falls inside that bracket. Section G.2 reads the two benchmarks we lose on against the same two conditions.

The measurement agrees term by term. Pricing both descriptions of a planted two-level truth with the engine’s own functions, the nested description costs 1,677 bits more at  $n = 900$  and 71,076 more at  $n = 27,000$ , and the membership column accounts for 2,491 and 74,402 of that. E35 confirms the consequence on the engine: no parent node appears, and which single level of nodes it returns is decided by how often the repair pass runs rather than by the data.

E38 measures both readings. On the planted two-level stream the implication relation recovers the tree at precision 1.000, unchanged whether the rule demands every record or ninety-five percent of them, so no tolerance is doing the work; read instead off the node supports the engine returns, it gives five parent classes at precision 1.000. At three levels all 180 implications point from the deeper key to the shallower one. On the Wikidata records of E37, where no tree was planted, it returns 3,433 implications over 263 keys: membership of a political party implies sex or gender, country of citizenship, instance of and date of birth, which is the specialisation relation recovered from records alone. Honesty requires one line beside that. The implication reading needs no engine at all, since it is a property of the co-occurrence counts; what the nodes add is that they carry the same relation in compressed form, five classes rather than two hundred and sixty-three keys.

E40 plants all four in one stream, independent of each other, and scores them. Specialisation and the value edges come back at precision and recall 1.000, with the cardinality correct in every case, one-to-one where the map is invertible and many-to-one where it is not. Siblings and role identity come back at recall 1.000 and precision 0.750, and the single error is the case the fixture plants on purpose: two siblings drawing from one value pool are indistinguishable from two names for one role when only presence and vocabulary are consulted. That is the failure this paper has named from the start, now measured rather than anticipated. Run on the Wikidata records of E37, where nothing is planted, the same counting returns 538 specialisations, 1,189 sibling pairs and 2,076 value edges of which 1,660 are many-to-one, and no role identities at all, which is what one should expect of a curated property vocabulary that has already deduplicated its own names.

**Lemma C.3** (separability, with converse). *Under a factorised scope code,  $L_{\text{scope}}(S) = \sum_{a \in S} \ell_{\text{in}}(a) + \sum_{a \notin S} \ell_{\text{out}}(a)$ , and*

$$\max_{S \subseteq A} \left[ \sum_{a \in S} g_a - L_{\text{scope}}(S) \right] = \sum_{a \in A} [g_a - \Delta \ell_a]_+ - \sum_{a \in A} \ell_{\text{out}}(a),$$

*attained by  $S^* = \{a : g_a > \Delta \ell_a\}$ : independent per-attribute tests, an accept or reject per attribute, not an argmax over subsets. Conversely, under the combinatorial code  $\log \binom{|A|}{|S|}$  plus  $\log(|A|+1)$ , the marginal price of the  $s$ -th included attribute is  $\log \frac{|A|-s+1}{s}$ , strictly decreasing in  $s$ , and the optimum becomes sort-by- $g_a$ -and-cut: tractable, no longer separable. Separability is a property of the model class (facet-factorised emissions, factorised scope code), not a theorem about the data.*

The converse earns its place by showing the property is fragile: under the combinatorial code the per-attribute receipts are no longer exact. The shipped support code is the combinatorial one,  $L_{\mathbb{N}}(|T|) + \log \binom{A_n}{|T|}$  (Table 3), so the shipped receipts are the sorted-prefix optimum, and per-attribute exactness is a property of the factorised alternative rather than of the shipped code.

Lemma C.3 states that under a factorised scope code the best scope is found by independent per-attribute tests, and, conversely, that a combinatorial scope code couples those decisions. The proof is a rearrangement in each direction.

*Proof.* For the direct claim, rearrange the objective:  $\sum_{a \in S} g_a - L_{\text{scope}}(S) = \sum_{a \in S} (g_a - \Delta \ell_a) - \sum_{a \in A} \ell_{\text{out}}(a)$ . The second term is constant in  $S$ . The first is maximised by including exactly the attributes with  $g_a > \Delta \ell_a$ , each contributing its positive part. So the maximiser is the accept set  $S^* = \{a : g_a > \Delta \ell_a\}$ , and the maximisation separates into independent per-attribute tests.

For the converse, the combinatorial scope code prices the next inclusion by  $\log \binom{|A|}{s} - \log \binom{|A|}{s-1} = \log \frac{|A|-s+1}{s}$ , so the charge for the next included attribute depends on how many are already included, coupling the decisions. For each size  $k$  the best scope is the  $k$  largest gains, so the optimum is always a prefix of the sorted order, found by scanning the cut point. Both the sorted gains and the marginal price fall, so the first crossing need not be the argmax, and the scan runs over all  $k$ . Separability is therefore a property of the model class (facet-factorised node emissions and a factorised scope code), not a theorem about the data.  $\square$

What does the escrow actually measure? Total correlation: the number of bits per record lost by modelling attributes independently when a latent distinction couples them. It is the multivariate form of mutual information, and it is zero exactly when the attributes are independent given the

node. The other quantity in the proposition is the Jensen-Shannon divergence  $\text{JS}_\pi(p, q)$ : the mutual information between one observation and which of the two distributions produced it, a measure of how distinguishable  $p$  and  $q$  are, in bits per observation.

**Proposition C.3** (total correlation funds the mint). *Let nodes  $z_i, z_j$  carry facet-product profiles  $P_i = \prod_a p_i(\cdot | a)$  and  $P_j = \prod_a p_j(\cdot | a)$ , merged with weights  $(\pi_i, \pi_j)$  into  $\bar{z}$ . Let  $Y$  be a record drawn from the merged node,  $Y_a$  its value on attribute  $a$ , and  $S$  the component indicator recording which of the two profiles produced it. With  $\text{JS}_a = \text{JS}_\pi(p_i(\cdot | a), p_j(\cdot | a)) = I(Y_a; S)$  the per-attribute divergence,  $\text{JS}_{\text{joint}} = I(Y; S)$  the joint one, and  $\text{TC}(\bar{z}) = \sum_a H(Y_a | \bar{z}) - H(Y | \bar{z})$  the total correlation of the merged node,*

$$\sum_a \text{JS}_a = \text{JS}_{\text{joint}} + \text{TC}(\bar{z}), \quad \text{TC}(\bar{z}) \geq 0,$$

*with equality iff the profiles differ in at most one attribute. In the facet-product code a merge raises the data bits by exactly  $\bar{p} \sum_a \text{JS}_a$  per record. The joint criterion does not decompose per attribute in general; the sum overcounts  $I(Y; S)$  by the inter-attribute redundancy about the erased distinction. But in this model class the per-attribute sum is not an approximation, it is the exact price.  $\text{TC}$  is the overprice the product code pays for refusing to represent a latent distinction, and the escrow accumulates it: over a cohort of  $n_s$  records the accumulated saving is  $n_s[\sum_a \text{JS}_a - H(\pi)]$  up to universal-code regret  $O(\log t)$ , and since  $0 \leq H(\pi) - \text{JS}_{\text{joint}} \leq H(\pi) \leq 1$  bit, the escrow is a total-correlation meter to within one bit per cohort record: a node is minted exactly when several facets redundantly witness one latent distinction long enough to pay  $\Lambda_s$ .*

The proof rests on the mixture identity for the Jensen-Shannon divergence (Lin, 1991).

Proposition C.3 states that the per-attribute Jensen-Shannon savings sum to the joint saving plus the total correlation of the merged node, so the escrow is a total-correlation meter. Two quantities carry the proof. The Jensen-Shannon divergence of a two-component mixture is the information the observation carries about which component produced it. The total correlation of a node is the information its attributes share with one another: zero exactly when they are independent given the node.

*Proof.* The starting identity is  $\text{JS}_\pi(P, Q) = I(Y; S)$  for the mixture with component indicator  $S$  (Lin, 1991). Given  $S$  the attributes are independent, because each component is a product, so  $H(Y | S) = \sum_a H(Y_a | S)$ . Then  $\sum_a \text{JS}_a = \sum_a [H(Y_a | \bar{z}) - H(Y_a | S)]$  and  $\text{JS}_{\text{joint}} = H(Y | \bar{z}) - \sum_a H(Y_a | S)$ . Subtracting the second from the first gives the identity (verified to machine precision), and  $\text{TC} \geq 0$  is subadditivity of entropy.

For the equality condition: if only facet  $b$  differs between the components, the mixture equals the product of the shared facets with the mixed facet  $b$ , hence  $\text{TC} = 0$ . If facets  $a \neq b$  both differ, then  $P(y_a, y_b) - P(y_a)P(y_b) = \pi_i \pi_j (p_i(y_a) - p_j(y_a))(p_i(y_b) - p_j(y_b)) \neq 0$ , so  $\text{TC}(\bar{z}) \geq I(Y_a; Y_b) > 0$  by the grouping bound.

The escrow reading follows from what each side pays. The incumbent that refuses the candidate codes the cohort with the merged product, at cost  $\sum_a H(Y_a | \bar{z})$  per record. The candidate restores the distinction, at allocation cost  $H(\pi)$  per record. Krichevsky-Trofimov block codes attain these entropies to  $O(\log t)$  regret (Krichevsky & Trofimov, 1981).  $\square$

The exchange rate here is derived, not chosen. The information bottleneck (IB) family trades compression against preserved information at a user-chosen rate  $\beta$ : how many bits of code one bit of kept information is worth. In our code the rate is pinned. Transmitting the node name and then the record given the node costs  $H(T) + H(Y) - I(T; Y)$  per record, the deterministic information-bottleneck objective at exactly  $\beta = 1$ , the slope minus-one point where one bit of distortion costs one bit of code. At that exchange rate many clusterings tie exactly, so the IB objective alone cannot choose among them, and the only remaining term that orders the candidates is the model cost. In IB language: at that slope the population frontier is an entire indifference face, so the price  $\Lambda_s$  is not a regulariser bolted onto an IB objective: it is the whole selection principle once  $\beta$  is pinned. Agglomerative IB computes but never spends its own rate decrease and runs to one cluster on an elbow heuristic (Slonim & Tishby, 1999); DIBmix retunes  $\beta$  every iteration and still needs a knee criterion and a user-fixed cluster count (Costa et al., 2026). The IB family externalises both the exchange rate and the stop; the prequential code internalises both.



Deferral would be worthless if the escrow released for noise, or never released at all. The next proposition closes both directions, with no significance level anywhere. The classical input is Wilks' theorem: on data that truly follow the incumbent law, the maximised log-likelihood-ratio gain of an alternative with  $d_S$  free parameters is asymptotically a chi-squared random variable with  $d_S$  degrees of freedom, so the gain from chance stays bounded in probability while every charge in the escrow keeps growing. Bounded in probability is written  $O_P(1)$  below. The growing charge has a name too:  $\text{COMP}(d_S, n)$  is the NML parametric complexity of Appendix D, here for a scope carrying  $d_S$  free parameters, the regret the candidate's own growing blocks pay, inside  $G_n$ , relative to the best fixed distribution chosen in hindsight.

**Proposition C.4** (consistency without a significance level). *Let a candidate's scope carry  $d_S = \sum_{a \in S} (d_a - 1)$  free parameters. (i) No spurious nodes. If the cohort's values follow the incumbent law on every scope attribute, the empirical gain after  $n$  cohort records is  $\chi_{d_S}^2 / (2 \ln 2) = O_P(1)$  by Wilks, with mean  $d_S / (2 \ln 2) \approx 0.721 d_S$  bits, a constant, while the charge inside the escrow grows as  $\text{COMP}(d_S, n) = \frac{d_S}{2} \log \frac{n}{2\pi} + O(1) \rightarrow \infty$ , and the frozen  $\Lambda_s > 0$  must be cleared on top. Hence  $G_n \rightarrow -\infty$  in probability, and the probability that a spurious candidate stands at or above its charge at cohort size  $n$  tends to zero, with no significance level anywhere; the lifetime guarantee over the running supremum is Theorem C.1's. Comparing means: pure noise stops being able to clear the parameter charge once the cohort exceeds  $2\pi e \approx 17$  records. (ii) No missed structure. If a latent feature activates a fraction  $\rho > 0$  of the stream and differs from the incumbent by  $D > 0$  bits per record on its scope, the evidence grows as  $\Theta(\rho D n)$  against a  $\Theta(\log n)$  charge, so the node is minted almost surely, at cohort support  $n^* = \Theta(\frac{d_S}{2D} \log \frac{d_S}{2D})$ . If  $D = 0$ , the expected crossing time is infinite and the lifetime mint probability is at most  $2^{-\Lambda_s}$ : in expectation the candidate is never minted, which is correct behaviour. Together with Lemma 1, both degeneracies close: one node per record is excluded by the validity of the code, one node total by (ii).*

Proposition C.4 states that a spurious candidate is eventually refused and a real one is eventually minted, with no significance level anywhere. The proof compares two growth rates: the evidence a cohort accumulates, against the charge its free parameters accrue.

*Proof.* (i) The candidate's block code is the KT mixture, whose regret against the cohort's empirical distribution is, for the  $d_a$ -ary facet,  $\frac{d_a-1}{2} \log \frac{n}{2\pi} + \log \frac{\pi^{d_a/2}}{\Gamma(d_a/2)} + o(1)$  (Xie & Barron, 2000). Summed over the scope this is  $\text{COMP}(d_S, n)$ , the parametric complexity: the total charge for the candidate's  $d_S$  free parameters. So  $G$  on the cohort equals  $n \text{KL}_2(\hat{p} \| q) - \text{COMP}(d_S, n) + O(1)$ . Under the null, Wilks' theorem says the multinomial likelihood ratio against the fixed incumbent converges to a  $\chi_{d_S}^2$  limit; the estimation error of the incumbent's own predictive is lower order, since its cumulative redundancy is  $O(\log n)$  on a cohort its tables dominate. So the gain is  $O_P(1)$ , with mean  $d_S / (2 \ln 2) \approx 0.721 d_S$  bits, while the charge diverges. That divergence is the sign of  $d \text{COMP} / dn$  doing the work a significance level does elsewhere. Equating means gives the crossover at cohort size  $2\pi e \approx 17$  records, past which pure noise can no longer clear the parameter charge in expectation.

(ii) Supporting records arrive at rate  $\rho$ , so by time  $n$  the cohort has  $\approx \rho n$  records and evidence  $\approx \rho n D$  bits. That exceeds  $\frac{d_S}{2} \log n + \Lambda_s = \Theta(\log n)$  for all large  $n$ , and solving  $nD = \frac{d_S}{2} \log n + \Gamma$  for the crossing gives the stated order of  $n^*$ . Exact KT crossings at  $\Gamma = 20$  bits:  $d_a = 2$ ,  $D = 1$  mints at 24 cohort records;  $d_a = 100$ ,  $D = 1$  at 199;  $D = 10$  at 3 to 5 across arities. If  $D = 0$  the expected crossing time is infinite and the lifetime mint probability is at most  $2^{-\Lambda_s}$  by Theorem C.1. One caution on the expansion: the asymptotic form of  $\text{COMP}$  is for the analysis only. The algorithm, and the crossings quoted, use the exact Gamma-function form, since  $\text{COMP}(m, n)$  goes negative for  $n \ll m$ .  $\square$

The crossover quoted in Section 4 is this closed form,  $2\pi e \approx 17.08$  records, and it is arity-free only at leading order: keeping the term the expansion drops puts the asymptotic crossover at the alphabet size itself, and the exact Gamma-function form puts it at 1, 2, 7 and 35 records at alphabet sizes 2, 5, 20 and 100 (results/e15\_theory\_numbers.json). The proposition is the fixed-alphabet idealisation of the shipped open-alphabet code, whose blocks use the escape of Appendix D rather than an explicit complexity charge.

## C.6 THE REPAIR PASS

The streaming rule is irrevocable, and the repair pass of Section 3 revisits its decisions. Why can the repair not simply keep minimising the streaming code? Because the prequential code’s past is spent: each record was priced by the predictive that stood at its arrival, and those prices cannot be re-issued, so “monotone repair on the prequential code” is meaningless. A repair pass needs a second objective that is a function of the current state alone, and order invariance forces its shape: an exchangeable allocation probability sees only column statistics, so an order-invariant price is a function of node size, while the streaming price is a function of birth position, and the two cannot appear in one functional.

A state is  $G = (K, Z, S, \omega)$ :  $K$  nodes; an allocation matrix  $Z \in \{0, 1\}^{n \times K}$  with column sums  $m_k$ , multi-membership allowed; supports  $S_v \subseteq A$  over the declared key universe of size  $|A|$ ; and an ownership map  $\omega$  assigning every published (record, key) cell to exactly one supporting member node or the background, so no cell is coded twice. Write  $p_{v,a}$  for the members of  $v$  publishing  $a$ , and  $h_{v,a}, h_{0,a}$  for the owned and background value counts. The batch objective  $L_{\text{batch}}(G)$  of Equation (C.7) is the total codelength of this state, five sums in order: a universal integer code on  $K+1$  (Rissanen, 1983), so the empty state stays codable; a scope code per node; the negative log IBP feature-allocation probability at mass  $\gamma$  (Griffiths & Ghahramani, 2011); per-facet value and presence blocks, KT for the owned counts and exact normalized maximum likelihood for the block counts, computed by the Kontkanen-Myllymäki recursion (Kontkanen & Myllymäki, 2007), within  $O(1)$  of KT (Xie & Barron, 2000); and the background blocks. The node-dependent presence blocks are load-bearing: without them nothing opposes merging nodes with disjoint supports (recomputed from the shipped objective at  $|A| = 200$ ,  $n = 10^4$ : such a merge of two 50-member nodes scores +289.27 bits with the term and −110.73 without, and every pair of an eight-node disjoint-support graph pays to merge without it, although the shipped repair proposes no such merge, so the collapse is a property of the objective rather than a state the implementation reaches).

In full,  $L_{\text{batch}}$  is one displayed equation, the five sums above with every symbol named: Equation (C.7) in Appendix C.7.

**Proposition C.5** (order-invariant repair, and its termination). *(i)  $L_{\text{batch}}$  contains no arrival index: it is a function of the counts  $(K, \{|S_v|\}, \{m_k\}, \{p_{v,a}\}, \{h_{v,a}\}, \{h_{0,a}\}, n)$  alone and is invariant under every permutation of the records; no birth position and no timestamp occurs anywhere in it. (ii) Let the repair pass propose MERGE, SPLIT, REASSIGN, and DELETE moves, each preserving the invariant that every node owns at least one published cell, and accept a move exactly when it strictly decreases  $L_{\text{batch}}$ , ties rejected. Then the pass performs finitely many accepted moves and terminates at a local minimum of  $L_{\text{batch}}$  with respect to the repertoire.*

*Proof.* Part (i) is Lemma C.4 and part (ii) is Proposition C.6, both stated and proved in Appendix C.7 below on the closed form of Equation (C.7).  $\square$

Part (ii) is the analogue, for the repair pass alone, of Theorem 3.1 of DP-means (Kulis & Jordan, 2012), which proves monotone decrease for a batch multi-pass algorithm and nothing about a per-record streaming rule. As there, termination does not imply agreement across arrival orders, which E5 measures (Section 5.2: exact on planted structure, ARI 0.6389 to 0.986 on Wikipedia), and the composite of stream plus repair is monotone on no single functional.

## C.7 THE BATCH OBJECTIVE IN FULL

The streaming rule minimises the prequential code: the bits actually spent, arrival by arrival, on a past that is immutable once transmitted. That quantity is monotonically increasing by construction, so “monotone repair on the prequential code” is not unproven, it is meaningless. A repair pass needs a second objective, stated once, that is a function of the current state alone. The two objectives cannot be one. The streaming price of a node is a function of its birth position, which is what makes it computable online. An order-invariant price must instead be a function of node size, because an exchangeable allocation probability sees the allocation only through its column statistics. The two cannot appear in one functional. We therefore state the two-objective structure plainly. The streaming rule minimises the prequential code, whose past is spent. The repair pass minimises the

batch objective  $L_{\text{batch}}$ , which is order-invariant. The two agree to the standard  $O(\log n)$  per parameter prequential-versus-two-part equivalence (Grünwald, 2007; Dawid, 1984). And the realised gap between them is a measured quantity (Definition C.2), never an assumption.

**The batch objective in closed form.** In full,

$$L_{\text{batch}}(G) = L_{\mathbb{N}}(K+1) + \sum_v \left[ \log |A| + \log \binom{|A|}{|S_v|} \right] + L_{\text{alloc}}(Z) \\ + \sum_v \sum_{a \in S_v} [L_{\text{KT}}(p_{v,a}, m_v) + L_{\text{NML}}(h_{v,a})] + \sum_a L_{\text{NML}}(h_{0,a}), \quad (\text{C.7})$$

with  $L_{\mathbb{N}}$  the universal integer code (Rissanen, 1983),  $L_{\text{alloc}}$  the negative log IBP feature-allocation probability at mass  $\gamma$  (Griffiths & Ghahramani, 2011) (per-column factor  $\binom{n}{m_k}^{-1} m_k^{-1}$ , a column-multiplicity term, mass terms  $\gamma H_n \log e - K \log \gamma$ ),  $L_{\text{KT}}$  the KT block code, and  $L_{\text{NML}}$  the normalized maximum-likelihood codelength, exact by the Kontkanen-Myllymäki recursion (Kontkanen & Myllymäki, 2007), within  $O(1)$  of KT (Xie & Barron, 2000). The definition below gives the shipped instantiation, which replaces the exchangeable allocation factor  $L_{\text{alloc}}$  by the declared  $\gamma$ -free convention.

**Definition C.1** (the batch repair objective, shipped instantiation). A state is  $G = (K, Z, S, \omega)$ :  $K$  latent nodes; an allocation  $Z \in \{0, 1\}^{n \times K}$  with column sums  $m_k$  (multi-membership allowed); supports  $S_v \subseteq A$ ; and an ownership map  $\omega$  assigning every published (record, key) cell to exactly one member node supporting that key, or to the background, so that no cell is coded twice.  $L_{\text{batch}}$  is the sum of the terms E1, E2, E3, E4, E6, E7, E8, E9 and E11 of Appendix D, evaluated on the count statistics of  $G$ : the universal-integer header and label-invariance term (E1), one Krichevsky-Trofimov column code per node (E2), the support codes (E3), the node-dependent presence blocks (E4), the owner-index bits (E6), and the value blocks of the nodes and the background (E7 to E9). This is the declared  $\gamma$ -free convention of Lemma C.5: the exchangeable IBP allocation factor is replaced by the header plus the column codes, a function of  $(m_k, n)$  alone, within  $\frac{1}{2} \log n + O(1)$  bits per node of the exchangeable feature probability function factor (Griffiths & Ghahramani, 2011; Broderick et al., 2013b).

**Lemma C.4** (order erasure).  $L_{\text{batch}}$  contains no arrival index: it is a function of  $(K, \{|S_v|\}, \{m_k\}, \{p_{v,a}\}, \{h_{v,a}\}, \{h_{0,a}\}, n)$  alone. Consequently, for every permutation  $\sigma$  of  $[n]$ , relabelling the records by  $\sigma$  leaves  $L_{\text{batch}}$  unchanged. In particular no birth position and no per-record timestamp occurs anywhere in the objective.

*Proof.* By inspection of Definition C.1 and the term list of Appendix D: every term depends on  $Z$  only through its column sums, and on  $\omega$  only through the count vectors  $p$  and  $h$ . Row permutations of  $Z$  and relabellings of cells preserve all of these, and  $n$  enters only as the size of the record multiset. A batch objective that keeps a birth position inside it fails exactly here: the same state reached by two histories carries two different values, strict decrease no longer forbids revisiting a state, and termination is unprovable. The relabelling half of this invariance is asserted by `code/tests/test_batch.py` (difference 0.0 over 20 relabellings of the nodes); gate UT-15 in its 200-record-permutation form remains designed, and two arrival orders that reach the same member partition can still differ by up to 225 bits through cell ownership, so the invariance is of the objective in the state, not of the state in the order.  $\square$

The node-dependent presence blocks (E4) are load-bearing in this objective. For the insertion decision a node-independent missingness code cancels from every comparison, a genuine simplification there. Inside  $L_{\text{batch}}$  the same cancellation removes the only term opposing the merge of two nodes with disjoint supports. Recomputed from the shipped objective at  $|A| = 200$ ,  $n = 10^4$ , a disjoint-support merge of two 50-member, two-key nodes scores +289.27 bits with the node-dependent presence term and −110.73 without it; at  $t = 200$  with four keys, +2778.26 against −421.74; at  $t = 1000$  with four keys, +13822.96 against −2177.04. The three differences are exactly  $4tk$  bits, which is what the term contributes, and they sit within 7 bits of the values recorded during development. On an eight-node disjoint-support graph all 28 pairs pay to merge once the term is removed and none pays with it, but the shipped repair proposes a merge only for nodes that share a support key, so a full pass applies no move and  $K$  stays 8 under either arm: the collapse is a property of the

objective's sign, not a state the implementation reaches. The lesson is that non-degeneracy proved for the insertion rule does not transfer to the repair operators. With the guard it is re-established for merge, and the delete move is the corresponding guard in the opposite direction.

**Proposition C.6** (monotone termination of the repair pass). *Fix the record multiset. Let the repair pass propose moves from the repertoire  $\{\text{MERGE}, \text{SPLIT}, \text{REASSIGN}, \text{DELETE}\}$ , each mapping a state to a state for the same records and preserving the invariant that every node owns at least one published cell (a node stripped of its last cell is deleted within the same move, so  $m_k \geq 1$  and every allocation factor is defined). Let a move be accepted exactly when it strictly decreases  $L_{\text{batch}}$ ; ties are rejected. Then the pass performs finitely many accepted moves and terminates at a state where no move in the repertoire decreases  $L_{\text{batch}}$ : a local minimum of the objective with respect to the repertoire.*

*Proof.* Three facts. (i) Bounded below: every term of  $L_{\text{batch}}$  is the negative logarithm of a (sub)probability, so  $L_{\text{batch}} \geq 0$ . (ii) Finite state space: by the invariant each node owns one of at most  $n|A|$  published cells, so  $K \leq n|A|$ ; a state is then an allocation in  $\{0, 1\}^{n \times K}$ , supports drawn from the  $2^{|A|}$  subsets of  $A$ , and an ownership map over at most  $n|A|$  cells with at most  $K + 1$  labels, and for fixed  $n$  and  $|A|$  there are finitely many such states. (iii) Strict descent: the accepted states carry strictly decreasing objective values, so no state recurs, and a strictly decreasing sequence on a finite set is finite. At termination every candidate move has  $\Delta L_{\text{batch}} \geq 0$ , which is the definition of a local minimum with respect to the repertoire.  $\square$

Scope, held together in three sentences. This proposition is about the repair pass only. The stream is not monotone, because the prequential past is spent. The composite of stream plus repair is monotone on no single functional. The proposition is the repair-pass analogue of Theorem 3.1 of DP-means (Kulis & Jordan, 2012), which proves monotone decrease per complete iteration of a batch multi-pass algorithm and proves nothing about a per-record streaming rule. And since Kulis and Jordan's own paper concedes dependence on the processing order, termination does not imply agreement across arrival orders, there or here. That agreement is measured by E5 (Section 5.2), never assumed. The guarantee is finiteness, not a rate: the implementation caps the number of full passes and reports whether the cap binds.

One more fact is needed before the gap can be defined. The batch objective inherits the IBP mass  $\gamma$ , and the next lemma settles what can and cannot be done with it.

**Lemma C.5** (the batch mass is irreducible). *In the exchangeable-allocation form of the batch objective the IBP mass  $\gamma$  cannot be optimised away. (i) Profiling fails: substituting  $\hat{\gamma} = K/H_n$  turns the two mass terms into  $K[1 + \ln(H_n/K)] \log e$  bits, decreasing in  $K$  for  $K > H_n$  and unbounded below, so the profiled objective rewards nodes instead of pricing them; and the profiled assignment is not a code, because maximised likelihoods do not satisfy the Kraft inequality: the Shtarkov normaliser of the family is  $\sum_K K^K e^{-K}/K!$ , which diverges (Shtarkov, 1987), so no normalized maximum-likelihood code for the allocation exists at any  $n$ . (ii) Improper mixing fails: the Fisher information is  $I(\gamma) = H_n/\gamma$ , so Jeffreys' prior is  $\propto \gamma^{-1/2}$  and improper; every improper mixture has infinite total mass, hence is not a code, and at scale  $b = 0$  the first-record predictive assigns probability zero to every finite number of features. (iii) What remains is relocation with a receipt: a proper  $\text{Gamma}(a, b)$  mixture is a valid code and demotes  $\gamma$  to the hyperprior  $(a, b)$ , with shape  $a = \frac{1}{2}$  Jeffreys-canonical and scale  $b > 0$  forced positive by Kraft, with total influence  $O(1)$  bits over the stream and per-decision influence  $O(1/\log n)$  (along  $K_s = 2H_s$ : 0.69 bits per mint at  $n = 10$ , 0.19 at  $n = 10^5$ ); or  $\gamma$  is eliminated syntactically by the universal allocation code of Definition C.1, at a redundancy of  $\frac{1}{2} \log n + O(1)$  per node. In every case the freedom is demoted and disclosed, never derived away.*

*Proof.* (i) The stationary point of  $-K \ln \gamma + \gamma H_n$  is  $\hat{\gamma} = K/H_n$ , with value  $K[1 + \ln(H_n/K)]$  nats and derivative  $\ln(H_n/K)$  in  $K$ . For the normaliser,  $\sup_{\gamma} \gamma^K e^{-\gamma H_n} = (K/H_n)^K e^{-K}$ , and the  $\gamma$ -free factors summed over allocations with  $K$  columns give  $H_n^K/K!$ , since the exchangeable form must reproduce the  $\text{Poisson}(\gamma H_n)$  marginal; the product is  $K^K e^{-K}/K!$ , and Stirling gives the divergence rate (partial sums grow as  $\sqrt{2K/\pi}$ , verified numerically). (ii) is immediate from total mass, and the  $b = 0$  statement from  $\mathbb{E}[K] = a/b$ . (iii) The mixture codelength differences between proper hyperpriors converge to the log ratio of the priors at the posterior mean; the measured values are quoted in the statement.  $\square$

Lemma C.5 is why the repair pass does not inherit the claim of no calibrated parameter wholesale: it descends an objective whose mass freedom is demoted to a declared convention, published with its sensitivity. The prequential spine escapes the same vise for a structural reason. In the batch functional the mass multiplies the node count, so a factor-two change of  $\gamma$  moves every create-or-destroy margin by exactly one bit per node and can move the minimiser. In the online code the mass enters each decision once, through the mint price frozen at seed time. Under the deferred mint that price is a stopping boundary rather than a per-record comparison. Residual freedom in the convention therefore perturbs when a candidate is minted, not whether: any candidate with a positive evidence rate crosses every boundary in the family  $c \log n$ ,  $c \in [1, 2]$ . The same Kraft argument that kills batch profiling forces the online hyperprior to be proper; the two halves of the method fail and escape for the same reason, and that is the strongest argument we know for the prequential form.

**Definition C.2** (the one-pass-to-batch gap). *Fix an arrival order  $\sigma$ . Let  $G_n^{\text{stream}}(\sigma)$  be the state built by the one-pass rule and  $G_n^{\text{rep}}(\sigma)$  the terminal state of the repair pass started from it. The repair gap is  $\Delta_{\text{rep}}(n, \sigma) = L_{\text{batch}}(G_n^{\text{stream}}(\sigma)) - L_{\text{batch}}(G_n^{\text{rep}}(\sigma)) \geq 0$ , nonnegative by Proposition C.6, to be reported in bits and in bits per record; on synthetic streams the oracle gap against the planted allocation is to be reported beside it.*

$\Delta_{\text{rep}}$  is computable on every run with no ground truth, which is what makes it reportable, and on the twenty-node stream below it is 9,966 bits, 0.50 bits per record (results/e15\_theory\_numbers.json); the shipped runners do not yet record it per run. It is what “approximation gap of the one-pass rule” means everywhere in this paper. For scale, on a 20,000-record stream with 20 planted nodes the myopic rule mints nothing, so its state is exactly the all-background code, and that code is 198,478 bits longer than the planted allocation, 9.92 bits per record and 31.57 percent of its own code; the deferred rule recovers all twenty nodes and lands 4,176 bits below the planted allocation, because it owns cells the planted labelling does not.

## D THE ENCODING, TERM BY TERM

Terms used here: KT is the Krichevsky-Trofimov code and NML normalized maximum likelihood, both in Appendix A; a block is the set of counts a node keeps for one key; an escape is the part of a code that prices a value never seen before under a key; E1 to E12 number the terms of the released specification, and UT- $n$  names a gate of Appendix E.

Two codelength objects appear in this paper, and they are not the same object.  $L_{\text{emit}}$  is the prequential codelength actually emitted, arrival by arrival: order-dependent, and spent.  $L_{\text{batch}}$  is the order-invariant two-part codelength of the same prefix, and it is the objective. Insertion is a greedy step on it. Every repair move is an exact evaluation of it. The gap between the two is measured, never assumed (Definition C.2). The batch objective is the sum E1+E2+E3+E4+E6+E7+E8+E9+E11; E10 is a statement about what the fresh-node reference is, and E12 is the composite mint price assembled from the others. The numbering follows the specification. Every predictive sums to exactly one over its alphabet including the escape symbol; nothing is normalised by a quantity the decoder cannot compute from the past. In the table, KT abbreviates Krichevsky-Trofimov, the smoothed-count code introduced in Appendix C.

### D.1 THE ONLINE STEP IN FORMULAS

This subsection gives the three displayed formulas of the per-record step of Section 3, with the measurements that shaped each. KT abbreviates the Krichevsky-Trofimov code, the smoothed-count code for a sequence of symbols whose frequencies are unknown.

Figure 5 walks the loop. A record arrives and candidate nodes are retrieved by index, never by a scan of all nodes. For each candidate the engine asks whether coding the record through this node saves bits; if it does, the node is attached and one typed edge is emitted per key the node now owns, carrying its bits, and if it does not, the node is skipped. Any part of the record that remains unexplained credits its residual bits to the escrow account named by its (key, value) pair. If that account now covers the price  $\text{Price}(t, T, n)$  of stating the node, the node is minted, the accounted records move to it and a receipt is written; otherwise the account carries forward and the engine keeps waiting. When a repair pass is due, the engine tries merge, reassign and delete in that order

| Term | Object             | Closed form                                                                                                                         | Source                                                                   |
|------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| E1   | node count         | $L_{\mathbb{N}}(K) = \log^* K + \log c_0$ , $c_0 = 2.865064$ ; plus $-\log K!$ for label invariance                                 | (Rissanen, 1983; Galbrun, 2022)                                          |
| E2   | membership column  | $L_{\text{col}}(t, n)$ , the KT Beta( $\frac{1}{2}, \frac{1}{2}$ ) mixture over binary columns                                      | (Krichevsky & Trofimov, 1981; Willems et al., 1995)                      |
| E3   | support            | $L_{\mathbb{N}}( T ) + \log \binom{A_n}{ T }$ , $A_n$ = keys seen so far                                                            | (Galbrun, 2022; Vreeken et al., 2011)                                    |
| E4   | presence           | per $(v, a \in S_v)$ : KT Bernoulli on $(p_{v,a}, t_v - p_{v,a})$ ; background Bernoulli elsewhere                                  | (Krichevsky & Trofimov, 1981)                                            |
| E6   | owner index        | $\log(1 + q(a))$ over the competing owners plus background                                                                          | (Vreeken et al., 2011; Galbrun, 2022)                                    |
| E7   | categorical values | KT sequence code $L_{\text{KT}}(\text{counts}, d)$ ; predictive $-\log \frac{c+1/2}{N+d/2}$ ; three-stage escape for open alphabets | (Krichevsky & Trofimov, 1981; Xie & Barron, 2000; Cleary & Witten, 1984) |
| E8   | numeric values     | Jeffreys normal-inverse-gamma Student- $t$ predictive plus $\log(1/\delta_a)$ ; two-expert $\{x, \log x\}$ mixture                  | (Grünwald, 2007)                                                         |
| E9   | free text          | prequential token-level KT with PPM-style escape; two-expert tokenisation mixture; quarantined in v1                                | (Cleary & Witten, 1984)                                                  |
| E10  | the reference      | no separate fresh-node reference code exists: the alternative owner is another node or the background, itself a real E7 to E9 block |                                                                          |
| E11  | announcements      | KT Bernoulli over the edit history; 2 bits for the move type; $2 \log K$ for operands                                               | (Willems et al., 1995)                                                   |
| E12  | the mint price     | $\text{Price}(t, T, n) = L_{\text{ann}}(n) + 2 + L_{\text{col}}(t, n) + L_{\text{supp}}(T) + \Delta L_{\mathbb{N}} + \log(K+1)$     |                                                                          |

Table 3: The encoding, term by term. There is no E5 in the specification; the numbering is preserved so that every term here matches its entry in the released specification and test suite. E12 satisfies  $\sum 2^{-\text{Price}} \leq 1$  over  $(n, S, T)$  because each component is Kraft-tight (gate UT-4).

and accepts a move only if the total description length falls; the pass is a full pass whenever the record count has doubled since the last full pass. The engine then moves to the next record.

**Per-facet evidence.** How strongly does one published key argue for one candidate? Per candidate  $v$  and published key  $a \in S_v$ , the evidence is the smoothed pointwise mutual information

$$d_n(a, v) = \log_2 \frac{\hat{P}_{v,a}(x_{n,a})}{\hat{P}_{0,a}(x_{n,a})} + \log_2 \frac{\hat{P}_{v,a}^{\text{pres}}(1)}{\hat{P}_{0,a}^{\text{pres}}(1)}, \quad (\text{D.8})$$

two log-ratios in bits: how much better the node predicts the record’s value than the background does, plus how much better it predicts that the key is present at all. Positive means  $a$  argues for attachment. Keys in  $S_v$  that the record does not publish contribute the corresponding absence term, the silence bill read as evidence. Every predictive is snapshotted before the record updates anything.

The active set is built greedily. The coverage gain of adding  $v$  (its per-key improvements over the current owners, plus its absence terms) is monotone submodular: each further node helps less, because the owners it must beat keep improving. The price ( $\text{attach}(t_v, n)$  plus the owner-index increments) is modular: it simply adds. Greedy on a submodular gain minus a modular price carries the distorted-greedy guarantee of Harshaw et al. (2019): the returned active set achieves gain minus price at least  $(1 - 1/e)$  times the best possible set’s gain, minus that set’s full price. Ties break by birth order, never by hash order, and the loop stops at the first non-positive gain. The decomposition is exact, not narrative: the per-facet terms, absence terms and attach price sum to the scalar create-or-absorb verdict, and gate UT-5 (Appendix E) asserts the identity to  $10^{-9}$  bits against two independent recomputations of  $L_{\text{batch}}$ .

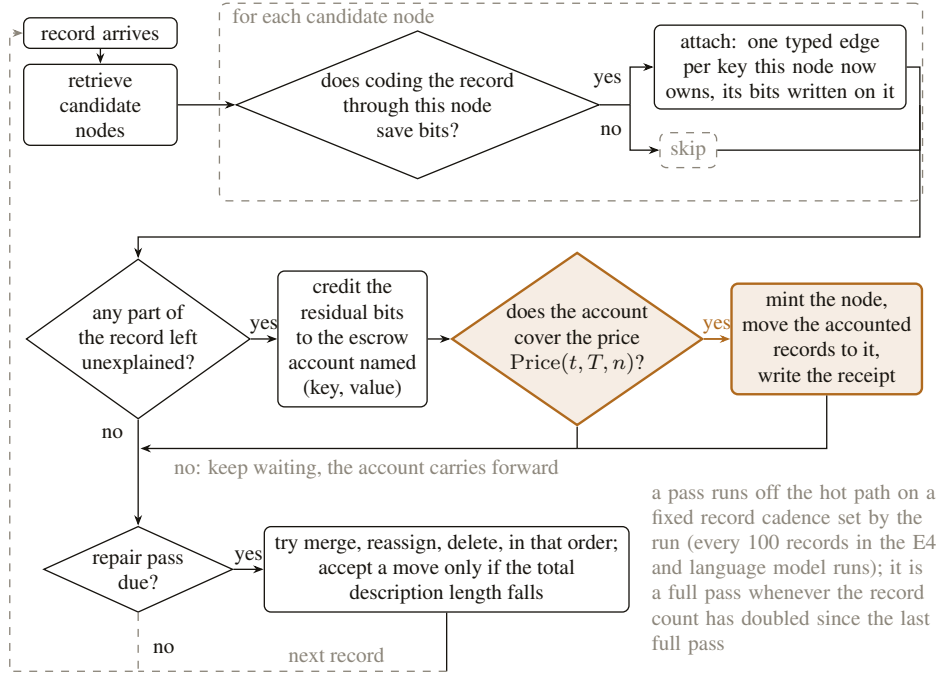


Figure 5: The decision for one record, and the repair loop. Diamonds are decisions taken in bits; the gold diamond and box are the release test of Equation 2 and the mint. The repair pass runs on a fixed record cadence that belongs to the run protocol, every 100 records under the protocol of Section G.1, and is a full pass whenever the record count has doubled since the last full pass (code/escrow/engine.py and code/escrow/batch.py).

**Single ownership.** A record may activate several nodes, and this multi-membership is exactly what breaks the additivity every partition-based code enjoys: if two nodes both claim the same facet, its bits are counted twice, and the sum is no longer a code. The repair is an arbitration map. The owner of each facet is the candidate with the strongest claim,

$$k_n(a) = \arg \max_{v \in S_{r_n}} d_n(a, v), \quad (\text{D.9})$$

where the maximum runs over  $S_{r_n}$ , the record’s active set, and  $d_n(a, v)$  is Equation D.8: one owner per facet, in the tradition of the Krimp cover (Vreeken et al., 2011). The map does double duty. It validates the code, because each facet is transmitted under exactly one model and every sum stays additive with no double counting. And it *is* the output, because the winning facet is emitted as the typed edge  $(a, x_{n,a}) \rightarrow k_n(a)$ , carrying its own price  $d_n(a, k_n(a))$  in bits. Ownership is not free: naming the owner among the  $q(a)$  nodes competing for the facet costs  $\log_2(1 + q(a))$  bits per facet, a realised bill we report. One circularity should be stated rather than smoothed over: additivity is bought by single ownership, and single ownership was chosen partly because it yields the additivity being claimed.

**The mint price.** Structural edits are part of the stream being transmitted, so an edit must announce itself: a KT coin over the edit history prices “an edit happens now” (write  $L_{\text{ann}}(n)$  for that price), then two bits name the edit type and  $O(\log_2 K)$  bits name the operands. Adding up the terms, the full price of a node with support  $T$  and  $t$  members at time  $n$  is

$$\text{Price}(t, T, n) = L_{\text{ann}}(n) + 2 + L_{\text{col}}(t, n) + L_{\text{supp}}(T) + [L_N(K+1) - L_N(K)] + \log_2(K+1), \quad (\text{D.10})$$

the announcement, the edit type, the membership column, the support, the increment of the node-count code, and the new node’s name. This is a Kraft code over (time, membership, support) jointly, and the joint scope is the point. The leading term of  $L_{\text{col}}$  is  $\log_2 \binom{n}{t}$ , and here is what it buys. Which  $t$  of the  $n$  records form the cohort is one choice among  $\binom{n}{t}$  possibilities, and searching over all of them multiplies the false-alarm probability by at most  $\binom{n}{t}$ ; that multiplier is the union bound, the

rule that the chance that any of many options fires is at most the sum of their chances. Charging  $\log_2 \binom{n}{t}$  bits divides the false-mint budget by exactly the same factor, so the search and the charge cancel. The act of picking a candidate by looking at the data is already paid for inside the price. The node-count term  $L_N(K) - \log_2 K!$  uses Rissanen’s universal integer code (Rissanen, 1983), with a rebate because node labels are interchangeable; the marginal cost of the  $K$ -th node is about  $1/(K \ln 2)$  bits (0.0014 bits at  $K = 1000$ ), numerically irrelevant, which is worth saying because the choice that looks most arbitrary does not matter.

**Measured failures that shaped the code.** Three measurements are recorded here because each one changed the design. Dropping the naming stage of the value escape makes a thin block’s escapes nearly free without ever identifying the value: a Kraft violation, and it manufactured evidence, minting 204 sibling nodes on a planted two-node control. Measuring escrow evidence against the background for facets a minted node already explains double-counts evidence and mints siblings of existing nodes: 332 nodes before the owner-incumbent rule on the same control, 2 after. And in the repair pass, a true node that the one-pass rule attached only partially loses to the background if deletion is evaluated before reassignment (deletion pays exactly  $-50.8$  bits on the planted control), while the same node fully attached wins; this is why merges run first, then reassignment, and deletes last.

## D.2 THE TERMS, ONE BY ONE

The announcement code follows the switching and expert-tracking line (Willems, 1996; Herbster & Warmuth, 1998); the numeric predictive is the standard Student-t (Bernardo & Smith, 1994); the two-expert mixtures use the aggregating framework of Cesa-Bianchi & Lugosi (2006).

**E2, the term that replaces  $\lambda$ .** The column code prices a node’s membership column: which  $t$  of the  $n$  records so far belong to it. Its closed form is four log-Gamma terms,

$$L_{\text{col}}(t, n) = \log_2 \pi + \log_2 \Gamma(n+1) - \log_2 \Gamma(t+\frac{1}{2}) - \log_2 \Gamma(n-t+\frac{1}{2}). \quad (\text{D.11})$$

Four properties matter. First, it is Kraft-tight over binary columns:  $\sum_t \binom{n}{t} 2^{-L_{\text{col}}(t, n)} = 1$  to  $10^{-15}$  at  $n \in \{1, 2, 3, 5, 8, 12\}$  (gate UT-1). This property is the premise of the false-mint guarantee. Second, the single-member asymptote is  $L_{\text{col}}(1, n) = \frac{3}{2} \log n + \log(2\sqrt{\pi}) = \frac{3}{2} \log n + 1.82575$  (exact against asymptote: 16.7739 against 16.7744 at  $n = 10^3$ ; agreement to four decimals at  $n \geq 10^4$ ). This locates the two-part member  $c = \frac{3}{2}$  inside the family  $c \log n$ ,  $c \in [1, 2]$ : 2 for Laplace, 1 for the Bayes-exact prequential price of Lemma C.2. Third, the per-member cost obeys  $L_{\text{col}}(t, n)/t = H_2(\rho)/\rho + O(\log n/t)$  with  $\rho = t/n$  the node’s occupancy and  $H_2$  the binary entropy (measured at  $n = 10^6$ : 2.0000 bits per member at  $\rho = \frac{1}{2}$ , 4.6901 at  $\rho = 0.1$ , 8.0803 at  $\rho = 0.01$ , against  $H_2(\rho)/\rho$  of 2.0000, 4.6900, 8.0793). In words: a latent node must carry more than  $H_2(\rho)/\rho$  bits per member beyond what the other active nodes and the key-naming code already supply. Nothing in that floor is chosen. Fourth, the incremental form is the attach price  $\text{attach}(t, n) = \log \frac{n-t+1/2}{t+1/2}$  (measured: 6.051662 at  $t = 1, n = 100$ ; 0.000000 at  $t = 50, n = 100$ ; 12.702678 at  $t = 1, n = 10^4$ ), negative for  $t > n/2$ . A node holding more than half the stream is free to join: a derived Matthew effect, not a designed one (gate UT-19). Finally, the exchangeable IBP per-column factor is  $\log \binom{n}{m} + \log m$ , and  $L_{\text{col}}$  exceeds it by  $\frac{1}{2} \log n + O(1)$ , the KT redundancy (Griffiths & Ghahramani, 2011; Broderick et al., 2013b). Two independent code families agreeing on the same  $\Theta(\log n)$  shape is the argument; neither alone is.

**E3 and E6, the structural bookkeeping.** Uniform over supports of a given size is the unique key-permutation-invariant code (Galbrun, 2022); the size header uses the universal integer code. The support charge is material, not decorative:  $\log A_n$  is 7.64 bits at  $A_n = 200$  and 10.65 at  $A_n = 1610$ , comparable to a 50-way facet’s 5.64. The owner index E6 is the measured bill for the additivity claim. Additivity of the per-facet receipts is bought by single ownership. Single ownership was chosen partly because it yields that additivity. The bill lies in  $[0, \log(\text{competing owners})]$  per attribute and is reported as a realised total. We state the circularity rather than smoothing it over.

**E4, the most attackable term, and the rule that makes it safe.** The rule: a node’s presence model covers only its own support. Here is what goes wrong without it. A presence model over



the whole key universe hands a fresh node an  $|A|$ -bit bill (by arithmetic, not an engine run: 1609 bits at  $|A| = 1610$ , against mint prices of 7.6 to 26.6 bits, a factor of about one hundred), which collapses the graph to one node. Ablation AB-3 is designed to reproduce that failure and is not yet run. With the rule in force the record’s total presence bill is  $O(m_r + \sum_{v \text{ active}} |S_v|)$ , never  $O(|A|)$ , because the all-keys-absent background baseline is  $\sum_a -\log(1 - \hat{p}_{0,a}) \approx \mathbb{E}[m_r] \cdot 1.4427$  bits, a few bits rather than  $|A|$ . A 2-key record and a 50-key record are then commensurable, and sparsity cannot decide anything (gates UT-3, UT-8, UT-9). The node-dependence of E4 is dropped nowhere: it cancels from insertion comparisons but is load-bearing in the repair objective (Appendix C.7).

**E7, the singleton price.** With alphabet  $d$ , a block’s first observation costs exactly  $\log d$ . Measured:  $L_{KT}([1, 0, \dots], d) = 1.00000/1.58496/3.32193/5.64386/9.96578$  for  $d = 2/3/10/50/1000$ , and the normalized maximum-likelihood complexity agrees exactly,  $\mathcal{C}(1, d) = d$  to ten decimal places by the Kontkanen-Myllymäki recursion (Kontkanen & Myllymäki, 2007; Shtarkov, 1987). Contrast the plug-in entropy of the same singleton, which is 0.0. That zero is precisely why “insert iff total entropy does not increase” collapses to one node per record. The plug-in is not a code: it assigns probability one to the observed value and violates Kraft. The parametric complexity is exactly the missing term, computed rather than chosen (gate UT-10; ablation AB-4, designed and not yet run, is to reproduce the collapse). One more distinction guards the term:  $L_{KT}$  as written is the sequence code, not the count code. Including the multinomial coefficient would make every block look  $NH(1/d)$  bits too cheap, and nothing downstream would complain (gate UT-2). Open alphabets use a three-stage past-only escape in the style of PPM (Cleary & Witten, 1984): a novelty Bernoulli, then an index among seen values, then the key’s global inventory, with  $\hat{d}_a$  always (distinct values seen so far) plus one.

**E8, numeric facets, and the resolution-cancellation lemma.** The Student- $t$  predictive carries  $\log(1/\delta_a)$ , where  $\delta_a$  is the data’s own recorded quantum, inferred over a warmup and frozen. This term appears identically in every candidate owner’s cost for the same key of the same record, so it cancels from every decision difference. The condition for the cancellation is that every branch codes the same value at the same resolution, and the condition is load-bearing. The cancellation fails the moment any branch prices a numeric value against a discretised-uniform reference, which would make  $\delta$  a hidden  $\lambda$  at one extra bit of create price per binary digit of precision. This encoding has no such reference (E10), so the cancellation holds unconditionally, and gate UT-7 is a hard invariance test on exactly this.

**E10, the reference.** There is no separate fresh-node reference code. In the joint code the alternative to “key  $a$  of record  $r$  is owned by node  $v$ ” is ownership by another node or by the background, and the background’s block is a real block coded by E7 to E9. The two reference distributions a designer might be tempted to interpolate between already live inside the code. The uniform-over-alphabet pole is what the KT block is at zero counts ( $\frac{1/2}{d/2} = 1/d$  exactly); the corpus-predictive pole is what the background block is at large counts. The interpolation weight between them does not exist as a free quantity: its only analogue is the background block’s own count state, which is data.

**E9, and the honest scope of the claim.** There is no minimax code for an unbounded alphabet. The tokenisation residual for free-text facets is estimated at 5 to 15 bits per field, larger than any constant the method removes, so the claim of no calibrated parameter is scoped to categorical and near-categorical facets and text facets are quarantined in v1; ablation AB-12, the criterion with text ablated and included, is designed for the version that lifts the quarantine and is not run.

**What is not forced, with its measured price.** Three choices remain, and each is priced. First, KT against Laplace on value blocks: 0.05 to 0.92 bits per attribute at sparse counts, decaying like  $d_a/N$ ; KT ships (Xie & Barron, 2000). On the column the gap is  $\frac{1}{2} \log n - 1.826$  bits and grows: 3.16 at  $n = 10^3$ , 4.82 at  $10^4$ , 8.14 at  $10^6$ . Under the escrow release this is a latency perturbation, not a decision perturbation, and the false-mint guarantee is code-independent; the minted-set agreement at finite  $n$  is E12, the encoding-stability sweep (`results/e12_encoding_stability.json`), and it is run. Eleven encodings of three streams, nine of them valid prefix codes, move the description length by 2.3 to 8.0 percent and agree on the graph wherever the graph is determined: on the planted eight-group stream every arm returns the same eight nodes with the same support keys, adjusted Rand index 1.0 between every pair, not one of 3,000 records in a different node, and birth times moving

by up to 217 records. Where the shipped code does not itself recover the truth the partition moves as well, to a worst pairwise index of 0.6055 on a shared-key stream and 0.5233 on the Wikipedia infoboxes, against 0.6068 and 0.5804 across twenty arrival orders of the shipped code alone: the code band tracks the order band on the first stream and exceeds it on the second, and 11 of the 13 nodes holding at least five percent of their stream survive the least favourable valid variant. So the claim that the code choice moves when a node is born rather than which nodes exist holds where the structure is identifiable, and not beyond it. Two codes a reviewer might propose are not prefix codes at all: the ceiling form  $\lceil \log_2 x \rceil$  has a divergent Kraft sum, and Rissanen’s iterated log without its normaliser sums to 2.865. A third item on the list is not a choice either: removing the stage-three naming charge from the value escape sends a block predictive to 85.8 instead of one, a Kraft violation rather than an ablation arm. Second, the two-expert mixtures of E8 and E9 cost at most one bit per key per record. Third, the computational budgets ( $B_{\text{node}}$  candidates scored per key,  $B_{\text{cand}}$  escrow candidates tracked,  $W$  reservoir slots per node,  $R$  repair moves per record) are declared budgets, not modelling parameters; the curve showing decisions unchanged as each grows is designed, not run, and in version 1 only the candidate pool cap  $B_{\text{cand}}$  exists as a setting,  $B_{\text{node}}$  is infinite and  $W$  and  $R$  are not exposed. Shrinking  $B_{\text{cand}}$  can only reduce the expected number of spurious mints, because the guarantee sums over candidates actually tested; it is a recall budget and never a validity knob.

## E THE VERIFICATION SUITE

Terms used here: a gate is a test that ships with the code and must pass for the build to be trusted;  $L_{\text{col}}$ ,  $L_{\text{KT}}$  and  $L_{\text{batch}}$  are the column, block and batch codelengths of Appendix D; a range coder is a real arithmetic coder that turns codelengths into bits on disk.

The suite has twenty-two gates. They exist because the failure modes of a codelength criterion are silent: a broken code still returns confident numbers, and four of the six independent derivation routes shipped exactly such a breakage before verification caught it. Gates UT-1 to UT-10 check that the code is a code; UT-11 to UT-22 check that the process built on it behaves as proved. Table 4 and Table 5 state what each asserts and what it catches.

Thirteen of the twenty-two run as executable gates against the v1 implementation (UT-1, UT-2, UT-4, UT-6, UT-8, UT-10, UT-11a, UT-12, UT-13, UT-14, UT-19, UT-20, UT-22, in `code/tests/`); UT-11b is measured by the null experiment of Section 5 on a standalone one-key block, not on the engine. The remaining gates (UT-3, UT-5, UT-7, UT-9, UT-15 to UT-18, UT-21) are specified against the full encoder, including the numeric and text facets and the arithmetic-coder round trip, and are the release gate for the version that lifts the v1 categorical scope. We state this split rather than implying a complete harness. Since that split was written, `code/tests/test_batch.py` (eleven tests, all passing) asserts two further facts the paper relies on: every (record, key) cell is coded exactly once by exactly one block, the invariant that makes the per-key bits additive, and  $L_{\text{batch}}$  is invariant to relabelling the nodes (difference 0.0 over 20 relabellings); 20 arrival orders that reach the same member partition can still differ by up to 225 bits, because the greedy sequence assigns cell ownership differently.

## F REPRODUCIBILITY

Terms used here: a results file is the JSON each experiment script writes and every number in this paper is read from; a seed is the fixed random-number start that makes a run replay exactly; the v1 scope is the categorical-facet scope declared in Section 2.

**Repository layout.** The implementation is pure Python over the standard library; there is no GPU, no learned component, and no external dependency in the engine. `escrow/codes.py` holds the codelength primitives of Appendix D ( $L_{\mathbb{N}}$ ,  $L_{\text{col}}$ ,  $L_{\text{supp}}$ , the KT block and predictive forms, the attach price, the mint price, and the Kontkanen-Myllymäki complexity used by UT-10), each computed by `lgamma`, never by factorials. `escrow/engine.py` implements the insertion process of Section 3 (parse, retrieve, per-attribute evidence, greedy active set, emit with receipts, update, escrow accrual, release), with all evidence computed from pre-record snapshots so that the accumulator remains a sequential likelihood ratio. `escrow/batch.py` implements  $L_{\text{batch}}$  and the repair pass of Ap-

| Gate  | Asserts                                                                                                                                                        | Catches                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UT-1  | $\sum_t \binom{n}{t} 2^{-L_{\text{col}}(t,n)} = 1$ to $10^{-12}$ , $n \in \{1, 2, 3, 5, 8, 12\}$                                                               | a mis-normalised column code; this identity is the premise of the false-mint guarantee, and if it fails nothing else in the suite matters                                                        |
| UT-2  | $\sum_{\text{seq}} 2^{-L_{\text{KT}}} = 1$ over all $d^N$ sequences, $d = 3$ , $N \leq 6$                                                                      | the sequence-versus-count confusion: a multinomial coefficient makes every block $NH(1/d)$ bits too cheap and the method finds structure that does not exist                                     |
| UT-3  | the whole record code sums to 1 over all 27 records of a toy universe, at 0, 1 and 2 nodes                                                                     | a bad missingness model: charging only present keys sums above one, charging presence twice sums below, and neither shows up anywhere else                                                       |
| UT-4  | $\sum 2^{-\text{Price}} \leq 1$ by enumeration at $n \leq 10$ , $ A  \leq 4$                                                                                   | a price that has stopped being a code, which voids the false-mint guarantee; the $\log \binom{n}{t}$ leading term is the union bound over which records, and this notices if a refactor drops it |
| UT-5  | per-attribute receipts plus structural terms equal the batch delta to $10^{-9}$ over $10^4$ random pairs, against two independent recomputations               | a decomposition whose parts do not sum to the verdict; every per-attribute bit count printed in this paper is downstream of this assertion                                                       |
| UT-6  | block-minus-block equals the predictive increment to $10^{-9}$ over $2 \times 10^4$ vectors (measured max error $1.45 \times 10^{-12}$ )                       | an off-by-one in the escape denominator: monotone, finite, plausible, and wrong by a compounding fraction of a bit                                                                               |
| UT-7  | halving every numeric quantum changes no decision, no receipt, and the total by exactly one bit per numeric observation                                        | any branch that prices a numeric value against a discretised-uniform reference, which would make the quantum a hidden $\lambda$                                                                  |
| UT-8  | mint decisions uncorrelated with record width ( $ \rho_{\text{Spearman}}  < 0.1$ ); identical key sets with disjoint values must not co-attach, and conversely | an encoding that treats absence as a value and decides everything by sparsity                                                                                                                    |
| UT-9  | adding 1000 phantom keys changes no decision and shifts the total by exactly the support terms                                                                 | any per-record $O( A )$ presence term, the measured 1609-bit absorption credit that collapses the graph                                                                                          |
| UT-10 | the singleton block costs exactly $\log d$ in two code families; the plug-in entropy 0.0 is never used                                                         | the plug-in degeneracy: one node per record, from a quantity that is not a code                                                                                                                  |

Table 4: Code-level gates: the code is a code.

| Gate  | Asserts                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Catches                                                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UT-11 | (a) mean final $K \leq 1$ over null streams, run as shipped on 30 streams of 3,000 records where the specification asks 100 of 20,000; (b) empirical $\mathbb{P}(\sup_t G_t \geq b) \leq 2^{-b}$ over $2 \times 10^4$ null candidates                                                                                                                                                                                                                   | an invalid sequential predictive on either side of the ratio, which silently voids the lifetime guarantee                                                                     |
| UT-12 | two fully disjoint groups give $K = 2$ at every $n$ , with margins growing linearly; the one-key control returns $K = 0$ with a gap flat in $n$                                                                                                                                                                                                                                                                                                         | the hard gate: a criterion that cannot separate two groups sharing nothing invalidates every downstream result; a growing gap at one key means the column code is mispriced   |
| UT-13 | on the disjoint fixture the value component of $G_t$ is exactly 0.000 and the presence component about $k$ bits per member; removing the presence term returns $K = 1$                                                                                                                                                                                                                                                                                  | the evidence functional built from value ratios alone, invisible to any test with overlapping keys, and exactly the bug that returned $K = 1$ in all twenty adversarial cells |
| UT-14 | disjoint-support merges are refused and are accepted without the presence guard (E15 recomputes the three pairs at $+289.27/ - 110.73$ , $+2778.26/ - 421.74$ and $+13822.96/ - 2177.04$ bits, within 7 bits of the values recorded during development, the presence term contributing exactly $4tk$ ); the eight-node collapse is a property of the objective's sign, since the shipped repair never proposes a merge for nodes with disjoint supports | the repair pass collapsing the graph because insertion-time non-degeneracy was assumed to transfer to the repair operators                                                    |
| UT-15 | $L_{\text{batch}}$ identical to $10^{-9}$ under 200 record permutations                                                                                                                                                                                                                                                                                                                                                                                 | an arrival-position term leaking into the objective, which destroys the termination proof                                                                                     |
| UT-16 | round trip through a real range coder: exact reconstruction, and written bits within 32 of reported bits over 2000 records                                                                                                                                                                                                                                                                                                                              | every term that is not decodable from the past; nothing else proves the code is a code                                                                                        |
| UT-17 | every counter wrapped: reading any statistic incorporating record $n$ while deciding record $n$ raises, and zero raises occur                                                                                                                                                                                                                                                                                                                           | the future leak that makes prequential results look better than they are                                                                                                      |
| UT-18 | with structure frozen, emitted bits telescope to the batch total within $10^{-6}$                                                                                                                                                                                                                                                                                                                                                                       | double-counted or missing model terms: the two objectives silently diverging is the central claim failing quietly                                                             |
| UT-19 | the attach price equals the column increment to $10^{-12}$ (measured max $8.4 \times 10^{-13}$ ), is 0 at $t = n/2$ , negative beyond                                                                                                                                                                                                                                                                                                                   | an inverted occupancy ratio, which passes every smoke test and produces a graph that avoids popular nodes                                                                     |
| UT-20 | escape mass and seen mass sum to one; repetition strictly cheapens; unseen values cost at least $\log(u+1)$                                                                                                                                                                                                                                                                                                                                             | the standard PPM escape bug, which makes every block slightly too cheap, always in the direction of absorbing everything                                                      |
| UT-21 | every accepted repair move strictly decreases $L_{\text{batch}}$ recomputed from scratch; the pass ends within $10K$ moves                                                                                                                                                                                                                                                                                                                              | a delta with the right sign and wrong magnitude, which oscillates forever while each move looks like an improvement                                                           |
| UT-22 | identical records give $K = 1$ ; globally unique records give $K = 0$ ; a constant key has $\Delta_a = 0.000$ always; a ubiquitous uniform key never justifies an edge                                                                                                                                                                                                                                                                                  | the frequency-versus-lift confusion, which passes the first three controls and fails the fourth                                                                               |

Table 5: Process-level gates: the process behaves as proved.

pendix C.7. `tests/` holds the executable gates of Appendix E. `experiments/` holds one file per mechanism experiment; each writes a JSON results file that the numbers in this paper are read from.

**Declared v1 scope.** Categorical facets only; the numeric (E8) and text (E9) coders are specified and quarantined. Posting lists are untruncated ( $B_{\text{node}} = \infty$ , reported). The repair repertoire is merge over nodes with disjoint member sets, the case the mint produces; the owner-index term E6 is omitted from the shipped batch objective, which is exact for disjoint-support graphs and bounded by  $\log(1 + \text{overlap})$  per cell otherwise. Ties in the greedy step break by birth index and then node id, never by hash order, so runs are deterministic given the stream.

**Run protocol.** Every ESCROW run in the mechanism tier, the demonstrations, E4, E5 and the language-model comparison calls `escrow/protocol.py`: the engine with the batch objective installed, so that a fresh mint whose support is contained in an existing node is absorbed into it, a repair pass every 100 records, and one full pass at the end of the stream; each results file records the protocol string. The three runs on the shared GPU box record their own cadences, every 200 records rather than every 100, stated with each run in Appendix H; of the three, only Lazada raw keys was made on an earlier engine state, and MusicBrainz 20K and ReVerb45K are re-runs on the fixed engine.

**Gates and experiments, command lines.** The gate scripts live under `code/tests/` and the experiment scripts under `code/experiments/`; every command runs from the repository root as `python3 <path>`, with the arguments and seeds shown. Each experiment script writes a JSON results file of the same name (`e7_immediate_vs_deferred.py` writes `e7_immediate_vs_deferred.json`, and so on); the catalogue script, whose arguments name the department and the record cap, runs in fixed order with no seed and writes `catalogue_footwear.json`.

| Script                                    | Arguments and seeds     | Output                                |
|-------------------------------------------|-------------------------|---------------------------------------|
| <code>test_codes.py</code>                | fixed (0)               | code-level gates                      |
| <code>test_engine.py</code>               | fixed (0)               | process-level gates                   |
| <code>e7_immediate_vs_deferred.py</code>  | 0                       | same-name JSON                        |
| <code>e8_false_mint_null.py</code>        | 0; plateau 0 to 11      | same-name JSON                        |
| <code>e9_support_curve.py</code>          | 0 per cell              | same-name JSON                        |
| <code>e13_creation_bias.py</code>         | seeds 11 to 15 per cell | same-name JSON                        |
| <code>e13_tree_arms.py</code>             | five seeds per cell     | <code>e13_tree_arms.json</code>       |
| <code>e14_cost_and_scale.py</code>        | 0 per cell              | <code>e14_cost_and_scale.json</code>  |
| <code>e16_supermartingale_check.py</code> | fixed                   | same-name JSON                        |
| <code>e5_order_dependence.py</code>       | seeds 1000 to 1019      | same-name JSON                        |
| <code>wikipedia_demo.py</code>            | shuffle 0               | same-name JSON                        |
| <code>catalogue_devset.py</code>          | Footwear 8000           | <code>catalogue_footwear.json</code>  |
| <code>e8_tree_arms.py</code>              | five seeds per cell     | <code>e8_tree_arms.json</code>        |
| <code>e4_baseline_army.py</code>          | fixed                   | <code>e4_first_tier.json</code>       |
| <code>lazada_c2_rawkey.py</code>          | shuffle once            | <code>lazada_c2_rawkey.json</code>    |
| <code>mb20k_benchmark.py</code>           | shuffle 0               | <code>mb20k_full12.json</code>        |
| <code>e10_reverb45k.py</code>             | fixed                   | <code>e10_reverb45k_full1.json</code> |

Every seed is a constant in its script, so every run in this paper replays exactly. The Wikipedia run fetches raw infobox templates once and caches them beside the results (`wiki_cache.*.json`), so the exact input records of the reported run are preserved.

**What each results file contains.** `e7`: on a 6,000-record two-group stream with one percent typos, over five stream seeds, the unseeded immediate mint fires zero times with a mean per-attempt deficit of 31.4 bits (range 30.6 to 32.1), the record-seeded immediate mint under the computed price fires zero times in all twenty cells of a five-alphabet by four-typo-rate sweep, the same rule under a constant price of 16 bits mints only typo-seeded nodes, and the deferred rule ends at  $K = 2$  on all five seeds with the two planted supports recovered exactly, identical across seeds, and zero typo-justified nodes (mean 7.4 mint events, range 4 to 9, consolidated by repair). `e8`: the Ville bound measured on a standalone one-key block, not on the engine, over

20,000 null candidates,  $\mathbb{P}(\sup_t G_t \geq b)$  of 0.09785/0.0324/0.0064/0.0002/0.0 against bounds 0.5/0.25/0.0625/0.0039/0.00024 for  $b = 1/2/4/8/12$ ; and the null plateau, final  $K = 0$  on all twelve seeds at every stream length in  $\{2, 5, 10, 20\} \times 10^3$ . (The VFDT and EFDT comparison arms (Domingos & Hulten, 2000; Manapragada et al., 2018), node counts as a function of stream length at fixed  $\delta$ , are recorded in `e8_tree_arms.json`, written by `e8_tree_arms.py`; this file records our side.) `e9`: the support curve over the  $(k, d)$  grid,  $k \in \{2, 3, 4, 6\}$  keys and  $d \in \{4, 8, 16\}$  values, cohort size at first mint from  $t^* = 11$  at  $(2, 4)$  down to  $t^* = 3$  at  $(6, \cdot)$ , monotone in  $k$ , with the realised evidence rate and the self-consistent prediction recorded per cell. `e13`: the creation-bias grid answering Kearns et al. (1997), six noise rates by three lengths by three planted counts, five seeds per cell, 270 runs: 10 of the 18 rows converge on  $K^*$ , 8 plateau below it, none grows with length, no single run ends above its own  $K^*$ , and the breakdown noise is 0.2, 0.3 and 0.4 at  $K^* = 2, 8$  and 20. (The planted-stream tree arms, first split index and final node count at both  $\delta$  settings, are `e13_tree_arms.json`.) `wikipedia`: 320 raw infobox records, 121 distinct raw keys, five nodes under the one protocol on the demonstration order: film 130 (119 film, 9 person, 2 mountain), person 101 (pure), mountain 86 (pure), a second film node of 10 (pure), one politician record isolated; purity 0.9665 over 328 memberships, 8 records in two nodes; the spread over orders is `e5`. `catalogue`: 993 raw footwear records, 43 raw keys,  $K = 6$  from 51 mint events, at a measured 303.4 records per second on one CPU core at  $K = 6$ ; the Lazada run recorded 1.73 records per second at  $K = 68$  with repair included (`lazada_c2_rawkey.json`). No record-to-record similarity is ever formed. `e14`: the cost-against- $K$  curve and the scale curve, each cell in its own child process so that peak resident memory is that cell’s own, with the headline statistic the median over the last decile of the stream; the insertion step costs 48.58 and 44.98 microseconds per record at 8 and 128 disjoint-support nodes and 113.0 rising to 349.17 at 5 and 20 shared-support nodes, end-to-end throughput falls from 7,891.5 to 908.8 records per second between  $10^3$  and  $10^5$  records at a fixed node count, the repair pass takes 95.81 percent of wall time at  $10^5$ , and peak resident memory there is 242.45 MB. `e16`: the martingale audit of Theorem C.1, the block-mass algebra checked by brute-force summation on 400 random blocks (maximum error  $4.4 \times 10^{-16}$ ), the per-step ratio distribution on four streams, and the exceedance of the accumulator against  $2^{-b}$ .

**External benchmarks.** The head-to-head tier (Alaska, the AutoPKG protocol on Lazada data (Hongwimol et al., 2026), MusicBrainz 2M and voter records, the DBpedia mappings as the open-domain key-merge test, and the tuned-oracle triples for every knobbed baseline) is designed, with protocols fixed in the planning documents, and is reported as the next tier rather than claimed here; Section 7 states this boundary. The mechanism experiments and controls above are run, and every number in this paper traces to a results file, a findings file, or the code.

## F.1 HOW A NODE IS BORN, IN THE TWO-GROUP FIXTURE

## G THE EXPERIMENT DETAIL THE MAIN TEXT POINTS AT

These passages were written for Section 5 and moved here intact so that the nine pages the venue allows are spent on the claims rather than on the walk-throughs. They are in the order the main text refers to them.

Agreement with the planted groups is 0.92 at a seed cohort of 136, 0.88 at 75, 0.69 at 43, 0.23 at 26, and 0.0001 at 16, below which the engine mints nothing at all. The cliff is bracketed between 16 and 43, and the release cohort  $t^*$  that E9 measured on a different family, at most 24, falls inside that bracket. Two readings follow. The failure mode is silence rather than error, which is the behaviour the price is built to produce. And  $k$ -means on the identical categorical input, which scores 1.0 while key presence carries the group, falls to 0.289 and then to 0.004 once it does not, so the baseline’s advantage on the encyclopedia stream is an advantage of that stream and not of the method: decomposing its representation gives 0.9898 from the key columns alone and 0.3389 from the value columns alone (E30).

The condition makes a prediction about our own losses that we can test without changing datasets, and E32 runs it. Swapping in a friendlier benchmark after losing on one would be shopping for a stream that flatters the rule, so instead the encyclopedia stream is held fixed in source, categories and parser, character for character, and only its size moves: from 320 records to 4,500 over eight types. Two quantities then move in opposite directions. A kind’s seed cohort grows linearly in the

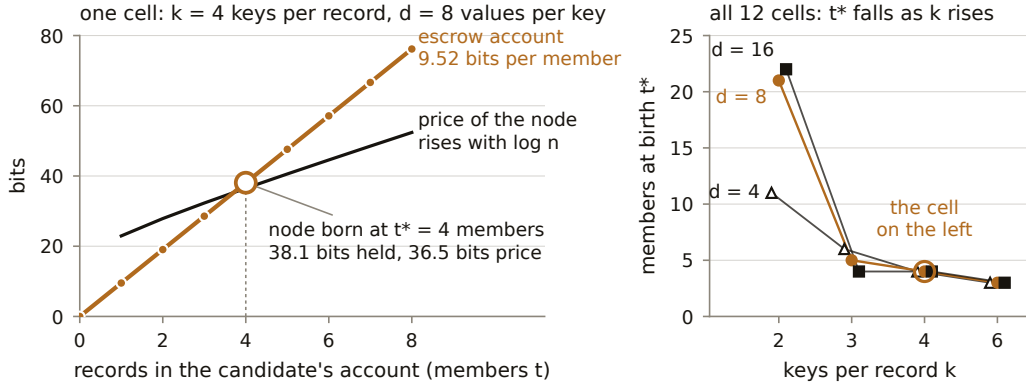


Figure 6: How a node is born. Left: one cell of the two-group fixture, four keys per record and eight values per key, one seed; the account (gold) rises at 9.52 bits per member, the price (ink) with the log of the stream, and the node is born at four members, never before. Right: members at birth across all twelve  $(k, d)$  cells fall as  $k$  rises, from 11 to 22 at  $k = 2$  to 3 at  $k = 6$  (results/e9\_support\_curve.json).

stream, while by Equation 3 the price of a member is flat for a kind of fixed proportion. So the rule should get *better* on the same data as the stream grows, which is not what a greedy one-pass method is normally expected to do, and a spread that fails to shrink would refute it.

The comparison on this stream is a loss, and not only against an oracle. On the same categorical input  $k$ -means handed the true number of types scores 0.9748 at the bottom of the ladder and 0.9822 at the top;  $k$ -means with  $k$  chosen by silhouette, which consults no labels, scores 0.9345, 0.9441 and 0.9011 at the three sizes where the full suite was run; and HDBSCAN at its shipped default scores 0.9480, 0.9606 and 0.9466 there. None of the three shows a trend in the stream length, and all three are ahead of us at most sizes. We report this rather than reporting the oracle alone, because a default-parameter baseline beating a rule that advertises no calibrated parameter is the harder fact and the one a reader needs. What the ladder establishes is therefore narrower than a win, and it is the thing that was under test: the rule’s difficulty falls as its own stated condition is met, on real records that were not selected for the purpose, which is what Proposition 1 predicts and what a rule failing for some other reason would not do.

One number per benchmark, the records per label its gold set asks for, orders every result we report. The two synthetic streams ask for groups of 1,000 and 375; the encyclopedia stream for 107; the Lazada silver types for 3.85; MusicBrainz 20K for 1.94 and ReVerb45K for 2.03. The rule is at or near 1.0 on the first three and at zero on the last three, and  $t^*$  falls between them. Lazada is the case that shows this is about the labels and not the records: its stream is plainly in scope, since the engine builds 68 nodes there holding 2,719, 1,574 and 1,364 records, while its silver label set names 5,518 types over 21,260 records and scores 0.0013. On the thousand-record subset of the same stream, where the silver set names 759 types with 614 of them singletons, a language model prompted to build the graph directly scores 0.002 sampled and 0.0094 greedy against the same labels (Table 14). When every method including a 14B language model scores within a hundredth of zero, the measurement is asking for a granularity nothing on offer supplies, and reporting our own number there as a defeat would misdescribe it.

Both clauses of Proposition 1 are necessary conditions and neither is sufficient, and the difference is not academic, so E34 gives the counterexample rather than leaving a reader to assume otherwise. Take the cover fixture of E33 at its hardest setting, eight groups drawing every key from a shared pool of twelve with no group owning a key of its own. The seeding condition is met with room to spare, since the most frequent characteristic pair of a group recurs 36 to 47 times against a  $t^*$  of at most 24, and all eight groups clear it. The price condition is met as well, since the best candidate still alive at the end is accruing 13.25 bits per member against a requirement of 9.23. The engine nevertheless mints three nodes where the truth has eight, at omega 0.2582. What the measurement localises is where it goes wrong rather than why: the candidates alive at the end carry cohorts of 3 to 7, not the 36 to 47 their seed pairs would allow, so the accrual is not reaching the candidates the

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seeds should feed. That is candidate generation and pool management rather than the price, and we report it as open. Adding private keys back recovers the run monotonically, to  $K = 7$  and 8.3 at one and two private keys per group, which is consistent with the diagnosis and is not a substitute for it.

Nothing does well, which is worth saying first: this is a hard task and the best score anyone reaches is 0.3137. Against that, ESCROW reaches 0.2044 with nothing set, which is level with  $k$ -means handed the true number of labels at 0.2053 and below the factorisation at 0.3137 and the topic model at 0.2417, both of which chose their component count and their membership cut on the test labels. The number that matters is the last one. The same factorisation under the label-free rule declared in advance, its components taken at the elbow of reconstruction error and its cut at half the row maximum, reaches 0.0099. Its rule picked twenty-four components where the truth has twelve. So on the one real cover benchmark we have, an oracle-tuned baseline is ahead of us by half again, and the best a practitioner can do without labels is twenty times worse than a rule that has nothing to choose. That is the shape of the claim this paper can defend, and it is narrower than being the most accurate method.

Run instead against methods that can express a cover, on the identical records, and the result reverses. Non-negative matrix factorisation and latent Dirichlet allocation, each given the number of components and the membership cut that score best on the test labels, both reach omega 1.0000. The advantage was an artefact of the fixture: each group owns its own keys there, so the group set can be read straight off which keys a record carries, and anything able to output overlapping memberships gets it for nothing. Making the groups share keys does not rescue it either. Sweeping the shared keys from none to six, ESCROW scores 0.9039, 0.9332 and 0.7452 while the oracle-tuned factorisation stays at 1.0000 throughout.

A reviewer can answer a non-determinism result measured at one temperature in a single line: lower it. E27 runs the sweep. On Wikipedia the model agrees with itself at 0.9975, 0.9940 and 0.9907 at temperatures 0.3, 0.7 and 1.0, returning  $K = 4$  every time, so on this stream the objection is right and there is nothing to report. On Lazada the same sweep gives 0.1168, 0.1647 and 0.1317. Lowering the temperature to 0.3 makes the agreement slightly *worse* rather than better, and the node count across all runs ranges from 24 to 91. What moves the determinism is the difficulty of the stream and not the sampling, so the claim survives the obvious answer to it. The runs were made on the same A100 the paper reports and were previously recorded as blocked; the block was a misreading, since the box's driver mismatch stops `nvidia-smi` and not CUDA.

One caution belongs with it. Deferring the repair pass, which E35 shows is worth 9,300 bits on a hierarchical stream, is a disaster here: it takes the same run from 0.2044 to  $-0.0162$  at  $K = 31$ . The cadence is not a setting with a good value, it is a setting whose best value depends on the stream, and we do not have a rule for choosing it.

E26 widens that to twenty-five null families, and the answer is partly each. The sixteen in which every key is published in every record give no node in any of eighty runs, over alphabet 2 to 500, key count 2 to 32 and Zipf exponent 0 to 2. The five that mint are the ones where key presence is itself random and high: a single node on one or two seeds in five at a per-key presence of 0.7 to 0.95, none at 1.0 and none at 0.6 or below. On a null drawn from the real stream, which keeps each key's presence rate and value pool but draws keys independently, one node appears on all five seeds and holds 213 of the 320 records. That node describes what the records share rather than inventing a group, which is what the absence code is for, and it is the boundary of the claim.

The creation bias runs the opposite way from the one Kearns et al. (1997) prove. On a grid of 270 runs whose fixture can fail in either direction, six noise levels by three lengths by three planted counts at five seeds, no run ever returned more nodes than were planted and no row grows with length. Ten rows converge to  $K^*$  and eight plateau below it, so the method writes structure off past a breakdown noise of 0.2, 0.3 and 0.4 at two, eight and twenty planted groups (Table 10).

E28 asks what that spread is made of, because the two possibilities call for opposite responses. If the objective cannot tell the good runs from the bad, the model is wrong. If it can, the model is right and the greedy search is what varies. Over twenty orders the correlation between the index and the total batch codelength is  $-0.786$ : the shortest description is the most accurate run, the longest is the worst, and 1,204 bits separate them. The objective ranks them correctly, so the remedy is search and not a calibrated quantity. Running  $k$  arrival orders and keeping the one with the shortest description, with no label ever consulted, lifts the mean over sixty orders from 0.8691 at  $k = 1$  to



0.9138 at 2, 0.9284 at 3, 0.9399 at 5, 0.9508 at 10 and 0.9666 at 40, while the spread across draws falls from 0.103 to 0.015. The returns diminish, and they diminish against a floor: at every  $k$  the run the codelength picks is about 0.03 below the best run in the same sample, because the correlation is  $-0.787$  and not  $-1$ . That gap is what the objective costs as a proxy, and no number of restarts closes it. What it costs is  $k$  times the compute and the one-pass property of a single run, and that is the trade rather than a hidden quantity. The compute is affordable and the runs are independent, so they parallelise exactly: a single pass over  $10^5$  records spends 4.5 seconds inserting and 105 seconds repairing, so three orders cost five and a half minutes on one core and under two minutes on three. On the 320-record stream three orders cost 1.2 seconds. The case that hurts is overlapping supports, where repair dominates: 5,000 records take 0.57 seconds when supports are disjoint and 35.6 when they all share keys (E14).

E22 runs BP-means, the method whose hand-set penalty this price replaces, on the same records. It does not reach ESCROW at any price: 0.027 against 1.0 on the two-group stream, 0.100 against 1.0 on the eight-group and 0.175 against 0.777 on Wikipedia, with the transferred constant worse everywhere and the oracle constant chosen on the test labels. The reason is the part worth reporting. On the two synthetic streams no price on the grid gives BP-means a feature count between three and the cap of 250: the count collapses to one or runs past the cap as  $\lambda$  crosses a narrow band, and removing the cap makes it worse rather than better, 1,469 features at 0.003. On this data the constant this paper removes has no setting that yields a usable answer, which is the paper’s argument made empirical rather than asserted.

**The hierarchy argument, and the drift the bound does not cover.** *The same price forbids a hierarchy, and the reason is a redundancy rather than a difficulty.* A knowledge graph usually wants a general kind and the special kinds under it. Nothing in the rule builds a tree, but the output is a cover rather than a partition, so one record may activate a broad node and a narrow one at once, and a child is charged only against the current owner of a key and so never pays twice for what its parent already explains. That is the right price for a specialisation. It is nevertheless not enough.

**Proposition G.7** (no hierarchy). *Let a parent hold members  $M$  with support  $F$ , and let  $K$  children partition  $M$ , the  $i$ -th with support  $P_i$  disjoint from  $F$ . Compare two descriptions of the same records, the flat one in which  $K$  nodes carry support  $F \cup P_i$ , and the nested one in which the parent carries  $F$  and each child carries  $P_i$ . Every (record, key) cell is coded exactly once under both. Then*

$$L(\text{nested}) - L(\text{flat}) = L_{\text{col}}(|M|, n) - \Delta,$$

*where  $\Delta > 0$  is the parametric saving from coding each key of  $F$  once over  $|M|$  rows rather than  $K$  times over  $|M|/K$  rows. The first term is  $\Theta(n)$  and  $\Delta$  is  $\Theta(\log n)$ , so the flat description is strictly cheaper for all  $n$  past a small constant and the gap diverges. Hence no search mints a parent, because a parent never pays for itself.*

The measurement agrees term by term at four stream sizes (Appendix C.5).

*And a parent does not need to be a node, which is why the proposition costs us nothing.* The relation between a general kind and a special one is an observation about attributes, not a vertex. If every record carrying key  $b$  also carries key  $a$  while  $a$  is strictly more common, then  $b$  is a specialisation of  $a$ ; films have a director and only animated ones have an animation studio. That is a one-to-many fact read off co-occurrence in one pass, with no prior knowledge, no schema and nothing to set. The engine records it in the only place it is free. A key in the support of several nodes is general to exactly those nodes, so the parent is the equivalence class of nodes carrying that key, and every node has already paid for its own support. The inlining that makes the flat description cheaper is the same fact that makes the relation visible, so the code is not fighting the hierarchy; it is storing it where it costs nothing.

E38 measures both readings, on the planted stream at precision 1.000 and on the Wikidata records of E37 where nothing was planted (Appendix C.5).

*What the engine measures instead of the bound, and why the price cannot repair it.* Over 300 null streams and 12,000 candidate trajectories the exceedance  $\Pr(\sup_t G_t \geq b)$  is 1.000 at every  $b \in \{1, 2, 4, 8, 12\}$  for the shipped statistic, against a nominal  $2^{-b}$ . Adding the seed naming charge of (iii) to the price, which is the repair the theorem itself contemplates, does not move that number in any of three codes. The reason is not multiplicity. Under the null the accumulator drifts, so its

supremum grows with the stream: the slack that would restore the level at  $b = 8$  grows from 97 bits at  $T = 500$  to 1,205 at  $T = 4,000$ , while a naming charge is constant in  $T$  by construction. Remove the seed key's own term and the requirement becomes 3.4 to 3.6 bits and stays flat, and the same charge then covers it at all five levels, but that arm is not shippable because removing the term destroys the mint. The failure is a drift induced by the records the evidence is read on, and not a family-wise error rate (Appendix C).

What carries the safety claim in place of the bound is not an argument but a measurement, because the appendix sentence it used to rest on, that the release price outgrows the drift, is a race between two growing quantities. E29 runs that race. On pure noise it reports, at each stream length, how close the nearest candidate came to its own release price: the worst margin over the pool is  $-421$  bits at 2,000 records,  $-849$  at 5,000,  $-1,538$  at 10,000,  $-2,815$  at 20,000 and  $-6,771$  at 50,000, on three seeds at each length, with no node created anywhere. The price pulls away from the evidence as the stream grows rather than closing on it, so the assertion is now a trend with a sign and a rate. It is still a measurement and not a theorem, and it stops at the length measured.

**Ville (iii), the multiplicity correction, and the repair a parent node would need.** (iii) The hypotheses are indexed by seed pairs, one per residual (key, value) pair, a set that grows with the schema and not with the record count; the price names the winner, and because  $\sum 2^{-\text{Price}} \leq 1$  (gate UT-4) that charge is the multiple-comparisons correction, so the expected number of spurious mints over the whole lifetime is controlled by the Kraft sum and not by the record count, whereas under mint-on-first-miss the hypotheses are one per record and the bound grows as  $\Theta(T)$  unless the charge knows the stream length. It corrects multiplicity and multiplicity alone, and is therefore not available as a repair for (ii).

The proposition still says what to change if a parent node is ever wanted, because the offending term is not merely expensive but redundant. The parent's members are the union of its children's, so its column transmits what the decoder can already derive. Charging it only for the residual its children do not determine, which is empty when they partition it, removes exactly  $L_{\text{col}}(|M|, n)$  and leaves  $-\Delta$ , so the nested description wins at every  $n$  and by more as the stream grows, measured at 815 bits at  $n = 900$  and 3,326 at  $n = 27,000$  (E36). We report that as the repair the analysis prescribes rather than as a result, since the engine does not implement it and nothing in the paper needs it.

*And the specialisation edge is one of several, which is what makes the output a graph rather than a clustering.* If a general kind and a special one are separated by a count, so are the other relations a graph is made of, and each is an equality or a strict inequality with nothing to set. Two keys that never share a record while both specialising the same third key are **siblings**, which is what lets a taxonomy branch instead of chain. Two keys that never share a record and draw their values from one pool are **two names for one role**, which is the key-identity question answered without comparing key strings. And when a record's value on one key fixes its value on another, that is a **typed edge between values** with a cardinality attached, born in Paris implies citizen of France, which is what a knowledge graph is actually made of.

E40 plants all four in one stream and scores them: specialisation and the value edges recover at precision and recall 1.000 with the cardinality correct, and siblings and role identity at recall 1.000 and precision 0.750, the single error being the case the fixture plants on purpose (Appendix C.5).

**The experiment walk-throughs.** **E7: immediate mint is broken in both forms; deferral is not.** Both immediate arms mint zero times, which is what Lemma 1 says they must do. The record-seeded arm does it in all twenty cells of an alphabet by typo-rate sweep, because a whole typo record is worth 23.1 bits against a node price of 39.8. A constant price fails in the other direction. At sixteen bits every node it mints is seeded by a typo and then absorbs ordinary records, so the corruption becomes permanent structure. The deferred arm ends at  $K = 2$  on all five seeds, with exactly the two planted key sets as its supports, and cites no typo value at any mint (Appendix H.3).

**E8: no node is created on noise, and what the bound covers.** Table 1 shows the Ville bound holding with slack at every level on a standalone one-key block. That block is not the engine's accumulator, whose measured exceedance is the subject of Theorem C.1. The engine's own result is the practical one.  $K$  ends at 0 on 48 of 48 null streams, flat from 2,000 to 20,000 records, and on 300 further streams, with no significance level and no grace period anywhere in the rule. The streaming

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trees do not stay flat (Figure 3), and the strict setting that silences them costs 20,800 records of delay when structure is present and never splits at higher noise (Appendix H.5).

**E9, E13 and E5: the price behaves like a price, the bias is not the one Kearns predicts, and the order cost is measured.** A price should ask for more evidence when each record carries less. In all 12 cells of the  $(k, d)$  grid the cohort size at birth falls as  $k$  grows and never rises, and the overshoot at the mint is below one member’s average evidence in every cell (Table 9).

Read honestly, the table says this, and it begins with a falsifier of ours that fired. Before the experiment we wrote that if a distance rule on ESCROW’s own representation matched ESCROW, the criterion would not be what is doing the work. On all three streams one does.  $k$ -means with  $k$  chosen by silhouette, which consults no label and has no knob left to set, ties us at 1.0 on both synthetic streams and beats us on the encyclopedia stream. We report that the test fired rather than rewording it afterwards. What it rules out is an accuracy claim, which we therefore do not make. What it does not touch is what the rule is for: the baseline is batch and sees every record at once, it returns a partition rather than a cover, it names no reason for anything it does, and it is handed the whole vocabulary of the stream before it starts. On the two synthetic streams ESCROW recovers the truth exactly and so does an untuned baseline, so neither separates them. On the encyclopedia stream ESCROW does not win: a single run averages 0.8691 over sixty arrival orders and three restarts reach 0.9284, against 0.9898 for  $k$ -means whose  $k$  is chosen by silhouette without ever seeing a label, and that baseline returns the same answer on every arrival order. On this stream the key set nearly determines the category, and a batch method that sees every record at once exploits that better than a greedy one-pass search does. What separates the two is therefore not accuracy. It is that the baseline is batch and ESCROW reads the stream once, that the baseline returns a partition and ESCROW a cover, and that every node ESCROW creates carries a receipt in bits naming the key that paid for it. Transferred knobs, which is what a practitioner actually carries between datasets, still collapse to at most 0.0017 on Wikipedia and to zero on the synthetic streams, and ESCROW has no knob to transfer (Figure 8 and Table 12, Appendix H).

E25 asks whether multi-membership is load-bearing, which is the paper’s stated reason for needing a new code at all, and the answer it first gave was too flattering. On a fixture where records genuinely belong to several groups at once, ESCROW reaches an omega index of 0.8876 and a membership F1 of 0.9486, against 0.4411 and 0.7402 for the best partition baseline reading the same records, while on a single-membership control from the same generator both reach 1.0 and the gap is exactly zero. That isolates the gap to multi-membership, but it does not establish what it appears to, because every baseline in it is a partition method and a partition cannot place a record in two groups at all. Beating one on a cover metric is not evidence.

What survives is narrower and is the claim this paper is actually entitled to. Those baselines reach 1.0000 only while their component count and their cut are chosen on the test labels. Carry a knob across fixtures, which is what a practitioner does when a new stream arrives without labels, and the same factorisation falls to 0.1953 and 0.1483 where ESCROW is at 0.9332 and 0.7452. Fix the knob instead by a label-free rule declared in advance, an elbow in reconstruction error and a cut at half the row maximum, and it becomes erratic rather than good, 0.9058, 0.7168 and 1.0000 across the same three fixtures. So multi-membership does not make this rule the most accurate one available. It makes it the accurate one that has nothing to carry over.

E37 asks the same question on real records, since every number above comes from a fixture whose keys name the groups. The benchmark is 3,362 people from Wikidata, described by their properties, with their occupations as the label set, restricted to the twelve most frequent in a twenty thousand row sample so the groups are chosen by frequency rather than by us. Occupation is removed from every record, along with field of work, position held, professorship and affiliation, so nothing in the input names the answer. It is a genuine cover, 39.2 percent of records carrying more than one occupation at a mean of 1.74, and it is in scope by Section 5.5, since the most frequent characteristic pair of each label recurs between 282 and 1,033 times against a  $t^*$  of at most 24. A loss here could not be blamed on scope.

Proposition 1 names two conditions, and both are properties of the price rather than settings. E31 measures them on a stream built so that only one thing moves. Eight planted groups, four keys, two thousand records, every record carrying every key, so key presence carries no information about the

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group and the structure lives entirely in the values; each group then draws its values from a private pool whose size is swept, which sweeps the seed cohort and nothing else.

It holds. The smallest seed cohort in the stream goes 7, 11, 24, 51, 87 across the ladder, and the number of types that clear  $t^*$  goes 4, 5, 8, 8, 8 out of eight, turning over at  $n = 1,280$ , which is exactly where the smallest cohort reaches the  $t^*$  ceiling of 24. Agreement over eight arrival orders rises from 0.8753 to 0.9190 and the spread across those orders falls from 0.1704 to 0.1098.

**Further detail from Section 5.** Table 1 shows the Ville bound holding with slack on a standalone one-key block, which is not the engine’s accumulator; that is Theorem C.1.

One caution belongs with it: deferring the repair pass helps on a hierarchical stream and is a disaster here, which Section 7 states in full.

The silver truth carries 759 types over 1,000 records and 614 of them are singletons, so the index is uninformative and near zero for every method, which leaves the failure modes as the thing to measure.

The comparison on this stream is a loss, and not only against an oracle:  $k$ -means at the true count,  $k$ -means with  $k$  by silhouette, and HDBSCAN at its shipped default are all ahead of us at most sizes, and none shows a trend in the stream length (Appendix G).

The two synthetic streams carry characteristic pairs recurring 121 and 117 times, and 49 to 59 times, and both are solved exactly.

The encyclopedia stream carries them 39, 48 and 71 times, so its groups are reachable, but 92 percent of its pairs are seen exactly once, so a node is born and then has almost nothing with which to recruit, which is the spread of Section 5.2 rather than a second effect.

Lazada carries them in the thousands and yields nodes of 2,719, 1,574 and 1,364 records.

MusicBrainz 20K asks for 10,000 groups from 19,375 records and ReVerb45K for 6,061 from 12,327, so a characteristic pair cannot recur more than about twice in either, which is below the most favourable  $t^*$  we have measured, and Equation 3 separately demands 13.66 and 13.00 bits from each of two records.

## The longer form of Section 5.

### G.1 EXPERIMENTAL DESIGN

No price, cut-off or level is set per run, the rule is told nothing about the domain, and every input ESCROW receives, every baseline receives. Two quantities belong to the run protocol rather than to the criterion, and both did vary: the repair cadence is every 100 records except on three benchmark runs on a shared machine, which ran every 200, and the candidate pool budget is 4,096 except on MusicBrainz 20K, raised to 32,768 after a crash. Table 6 states the question, design, control, baselines, tuning, metric and falsifier for each experiment, and Appendix H.2 records the protocol with each run and defines the *oracle*, *transferred* and *handed the true K* conditions. An audit of our implementation before these runs found three defects, the worst of which coded some cells twice; every number below comes from the corrected engine.

**E9, E13 and E5: the price behaves like a price, and the order cost is measured.** A price should ask for more evidence when each record carries less, and in all 12 cells of the  $(k, d)$  grid the cohort at birth falls as  $k$  grows and never rises (Table 9). The creation bias runs opposite to the one Kearns et al. (1997) prove for penalty-based rules, which is the direction that had to be re-derived rather than inherited. The batch objective is a function of the state and not of the arrival order, which the tests assert, but the greedy sequence reaching those states is not: over 20 orders both synthetic streams return ARI 1.0 and the exact  $K$  every time, while Wikipedia averages 0.8438 and ranges from 0.6389 to 0.986 with  $K$  from 3 to 6 (Table 11). E28 asks what that spread is made of, since a modelling failure needs the theory changed and a search failure needs more search. The correlation between agreement and description length over sixty orders is  $-0.787$ , so the objective ranks the good runs above the bad, and running  $k$  orders and keeping the shortest description lifts the mean from 0.8691 at  $k = 1$  to 0.9284 at  $k = 3$  and 0.9666 at  $k = 40$ , adding no calibrated quantity (Appendix G).

It begins with a falsifier of ours that fired. Before the experiment we wrote that if a distance rule on ESCROW’s own representation matched ESCROW, the criterion would not be what is doing the work. On all three streams one does:  $k$ -means with  $k$  chosen by silhouette, consulting no label and with no knob left to set, ties us at 1.0 on both synthetic streams and reaches 0.9898 against our 0.8691 on the encyclopedia stream. We report that the test fired rather than rewording it. What it rules out is an accuracy claim, which we therefore do not make. What it does not touch is what the rule is for: that baseline is batch and sees every record at once, it returns a partition rather than a cover, it names no reason for anything it does, and it is handed the whole vocabulary before it starts. Transferred knobs, which is what a practitioner carries between datasets, collapse to at most 0.0017 on Wikipedia and to zero on the synthetic streams, and ESCROW has none to transfer.

E25 asks whether multi-membership is load-bearing, which is this paper’s stated reason for needing a new code, and the answer it first gave was too flattering. On a cover fixture ESCROW reaches an omega index of 0.8876 against 0.4411 for the best partition baseline, and on a single-membership control from the same generator the gap is exactly zero. That isolates the gap to multi-membership and establishes nothing, because a partition cannot place a record in two groups at all. Against methods that can express a cover it reverses, an oracle-tuned factorisation reaching 1.0000. What survives is the claim this paper is entitled to: those baselines reach 1.0000 only while their component count and cut are chosen on the test labels, and carrying a knob across fixtures drops the same factorisation to 0.1953 and 0.1483 where ESCROW is at 0.9332 and 0.7452. Multi-membership does not make this rule the most accurate available. It makes it the accurate one with nothing to carry over.

The condition also predicts our own losses without changing datasets, which E32 tests by holding the encyclopedia stream fixed in source, categories and parser and moving only its size. It holds: the smallest seed cohort goes 7, 11, 24, 51, 87 across the ladder and the types clearing  $t^*$  go 4, 5, 8, 8, 8 out of eight, turning over at  $n = 1,280$ , exactly where the smallest cohort reaches the  $t^*$  ceiling of 24. Agreement rises from 0.8753 to 0.9190 and the spread across arrival orders falls from 0.1704 to 0.1098. The comparison on this stream is still a loss, and not only against an oracle. Both clauses are necessary and neither is sufficient: E34 gives a stream meeting both with room to spare on which the engine still mints three nodes where the truth has eight, and localises where it goes wrong without claiming to explain why (Appendix G).

## G.2 REAL-WORLD STANDING

Four public streams, one arrival order each (Table 13). **Wikipedia** settles at  $K = 5$  with purity 0.9665, its large types splitting differently across orders, which is why  $K$  ranges from 3 to 6. **Lazada**, 21,365 listings over 3,033 raw keys, settles at  $K = 68$ , where the largest nodes cross-list specification templates and case drift is a stated negative. **MusicBrainz 20K** and **ReVerb45K** are where version 1 is weak, for two reasons given before the runs, of which the second binds. Identity in both is a string one token apart, which a categorical facet cannot represent; and both ask for groups of two, which Proposition 1 prices out of a stream of that length whatever the facets say. The second matters because the first invites the reply that the representation should be fixed, and fixing it would not help: at a group size of two the membership column alone demands 13.66 and 13.00 bits from each of the two records. These are not benchmarks the rule does badly on. They are benchmarks its price excludes, and we report them for that reason rather than dropping them. On the first, the five categorical fields are null for every method and under the token bridge ESCROW reaches 0.0011 pairwise F1 where oracle-tuned embedding baselines reach 0.61 to 0.78; on the second its macro F1 is 0.016. Appendices H.17 and H.18 carry both in full, with every baseline and the benchmark’s own no-merge floor.

**E22 and E23 in full.** E22 runs BP-means, whose hand-set penalty this price replaces: it does not reach ESCROW at any price, and on the two synthetic streams no constant on the grid gives it a usable feature count at all. E23 chooses  $k$  by a two-part description length instead, matching us on the two-group stream and collapsing to a single cluster on the other two (Appendix G).

**The four relations, defined.** *And that edge is one of several, which makes the output a graph rather than a clustering.* Each of the others is an equality or a strict inequality on counts. Two keys that never share a record while specialising the same third are **siblings**, which lets a taxonomy branch instead of chain; two that never share a record and draw from one value pool are **two names**

for one role, the key-identity question answered without comparing key strings; and when a record's value on one key fixes its value on another, that is a **typed edge between values** with a cardinality attached. E38 and E40 plant all four and score them: specialisation and the typed edges recover at precision and recall 1.000, the two resting on disjointness at 0.750 precision, the single error being the case the fixture plants on purpose. On 3,362 real Wikidata people the same counting returns 538 specialisations, 1,189 sibling pairs and 2,076 typed edges (Appendix C.5).

**Real-world standing, the longer form.** Four public streams, one arrival order each (Table 13). **Wikipedia** settles at  $K = 5$  with purity 0.9665; **Lazada**, 21,365 listings over 3,033 raw keys, at  $K = 68$ . **MusicBrainz 20K** and **ReVerb45K** are where version 1 is weak, for two reasons given before the runs, of which the second binds. Identity in both is a string one token apart, which a categorical facet cannot represent; and both ask for groups of two, which Proposition 1 prices out of a stream of that length whatever the facets say. The second matters because the first invites the reply that the representation should be fixed, and fixing it would not help: at a group size of two the membership column alone demands 13.66 and 13.00 bits from each record. These are not benchmarks the rule does badly on but ones its price excludes, and we report them for that reason rather than dropping them. Under the token bridge ESCROW reaches 0.0011 pairwise F1 on the first where oracle-tuned baselines reach 0.61 to 0.78, and its macro F1 on the second is 0.016.

**E25 and E33, the longer form.** E25 asks whether multi-membership is load-bearing, and its first answer was too flattering: ESCROW reaches omega 0.8876 on a cover fixture against 0.4411 for the best partition baseline, with the gap exactly zero on a single-membership control. That isolates the gap to multi-membership and establishes nothing, since a partition cannot place a record in two groups at all. Against methods that can express a cover it reverses, an oracle-tuned factorisation reaching 1.0000, though only while its components and cut are chosen on the test labels: carried across fixtures it drops to 0.1953 and 0.1483 where ESCROW is at 0.9332 and 0.7452 (Appendix G).

**E22 and E23.** E22 runs BP-means, whose hand-set penalty this price replaces, and E23 chooses  $k$  by a two-part description length: neither reaches ESCROW, and on the two synthetic streams no constant on BP-means's grid gives it a usable feature count at all (Appendix G).

**Records per label, in full.** One number per benchmark, the records per label its gold set asks for, orders every result we report: 1,000 and 375 on the synthetic streams, 107 on the encyclopedia stream, 3.85 for the Lazada silver types, 1.94 on MusicBrainz 20K and 2.03 on ReVerb45K. We are at or near 1.0 on the first three and at zero on the last three, with  $t^*$  between them.

**Real-world standing, in full.** Four public streams, one arrival order each (Table 13). **Wikipedia** settles at  $K = 5$  with purity 0.9665 and **Lazada**, 21,365 listings over 3,033 raw keys, at  $K = 68$ . **MusicBrainz 20K** and **ReVerb45K** are where version 1 is weak, for two reasons given before the runs, of which the second binds. Identity in both is a string one token apart, which a categorical facet cannot represent; and both ask for groups of two, which Proposition 1 prices out of a stream of that length whatever the facets say. The second matters because the first invites the reply that the representation should be fixed, and fixing it would not help: at a group size of two the membership column alone demands 13.66 and 13.00 bits from each record. These are not benchmarks the rule does badly on but ones its price excludes, and we report them for that reason rather than dropping them. Under the token bridge ESCROW reaches 0.0011 pairwise F1 on the first where oracle-tuned baselines reach 0.61 to 0.78, and its macro F1 on the second is 0.016.

**The Lazada graph a prompted model builds (Section 5.4).** A catch-all node holds 23 to 85 percent of the records, and in one seed 77 percent of the nodes are named after a value copied from a record. With no rule for when a node should exist the graph comes out non-deterministic, catch-all and value-named at scale.

**The spread over arrival orders (Section 5.2).** E28 asks what the spread over arrival orders is made of, since a modelling failure needs the theory changed and a search failure needs more search. The correlation between agreement and description length over sixty orders is  $-0.787$ , so the objective ranks the good runs above the bad, and keeping the shortest of  $k$  orders lifts the mean from 0.8691 to 0.9284 at  $k = 3$  and 0.9666 at  $k = 40$ , adding no calibrated quantity.

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**One number orders every result (Section 5.5).** One number per benchmark, the records per label its gold set asks for, orders every result we report: 1,000 and 375 on the synthetic streams, 107 on the encyclopedia stream, 1.94 on MusicBrainz 20K and 2.03 on ReVerb45K. We are at or near 1.0 on the first three and at zero on the last two, with  $t^*$  between them.

**How a constant price fails on the same sweep (Section 5.2).** A constant DP-means-style price mints on typos and then absorbs ordinary records, so the corruption becomes permanent structure.

**Holding the stream and moving only its size (Section 5.5).** Over the five sizes E32 runs, the smallest seed cohort goes 7, 11, 24, 51, 87 and the number of types clearing  $t^*$  goes 4, 5, 8, 8, 8, turning over at  $n = 1,280$ . The order spread falls from 0.1704 to 0.1098 across the same range.

**The other floors on the encyclopedia stream (Section 5.3).** Term frequency over the record read as text also reaches 1.0000, the key set under cosine 0.9898, key presence and the matched multi-hot 0.9797, and the values alone 0.1205, so what separates these three types is which keys a record carries and not what its values say (Table 15). E22 runs BP-means, whose hand-set penalty this price replaces, and E23 chooses  $k$  by a two-part description length.

**The two protocol quantities that varied (Section G.1).** The repair cadence was every 100 records except on three benchmark runs that used 200, and the candidate pool budget was 4,096 except on MusicBrainz 20K. Table 6 gives the question, design, control, baselines, tuning, metric and falsifier for each experiment. An audit before these runs found three defects, the worst of which coded some cells twice; every number in Section 5 comes from the corrected engine.

## H ADDITIONAL TABLES AND EXHIBITS

Terms used here:  $K$  is the number of nodes and  $\hat{K}$  the number the engine ends with;  $t^*$  is the cohort size at the first mint and  $G$  the escrowed evidence;  $\delta$  is the significance level of a Hoeffding tree; ARI, F1 and the oracle and transferred conventions are as in Appendix A.

### H.1 RESULTS IN BRIEF

In brief: on 48 pure-noise streams of up to 20,000 records ESCROW creates no node, where VFDT and EFDT at  $\delta = 0.01$  grow to means of 19 and 91 nodes; it recovers two synthetic streams exactly in every one of 20 arrival orders, ARI (the adjusted Rand index, 1.0 a perfect match) of 1.0. On 320 raw Wikipedia infoboxes its agreement with the three category labels depends on arrival order, ARI 0.64 to 0.99 over 20 orders with mean 0.84, the measured cost of a one-pass greedy sequence; a language model on the same records scores 0.99 at about 46,000 tokens per run, and on 1,000 raw product listings returns 24, 78 and 28 nodes across three seeds where ESCROW returns the same six each time. On the free-text benchmark ReVerb45K it is within 0.01 of oracle-tuned embedding baselines on micro F1 with no embedding or knob, but leaves 7,009 of 12,327 subjects as singletons, so its macro F1 is 0.016.

### H.2 THE RUN PROTOCOL

Every ESCROW run in the mechanism tier, the demonstrations, E4, E5 and the language-model comparison uses one protocol: the batch objective installed on the engine, a repair pass every 100 records and one full pass at the end of the stream; the three runs on the shared GPU box, Lazada raw keys, MusicBrainz 20K and ReVerb45K, ran a repair pass every 200 records rather than every 100, and of the three only Lazada was made on an earlier engine state. An *oracle* knob was chosen on the test labels themselves; a *transferred* knob was tuned on another stream and carried over unchanged, the worse of two transfers reported;  $k$ -means *handed the true*  $K$  was given the true cluster count.

### H.3 E7: IMMEDIATE AGAINST DEFERRED MINTING

The unseeded immediate arm mints zero times in 6,000 attempts, at a mean deficit of 31.4 bits per attempt over five stream seeds (range 30.6 to 32.1): a fresh node’s uninformed guess loses to the

background on every record, as Lemma 1 says it must. The record-seeded arm under the same computed price also mints zero times, and a sweep over five alphabet sizes  $d \in \{2, 4, 8, 32, 100\}$  by four typo rates  $\{0, 0.01, 0.05, 0.2\}$  leaves that at zero in all twenty cells. The reason is arithmetic, not luck: the mean cost of a whole typo record under the background is 23.06 bits against a mean node price of 39.79, a typo cell is worth 9.75 bits against a clean cell's 9.33, and even the most expensive single record in the stream, at 33.71 bits, is 6.08 bits short of the price.

That settles which rule the typo objection belongs to. It belongs to the rule the method section describes and rejects, an immediate mint at a *constant* price  $\lambda$ , which E7 also runs on the same stream (59 typo records in 6,000). At  $\lambda = 39$  bits it mints nothing. At  $\lambda = 16$  it mints 5 nodes and every one of them is seeded by a typo, and none of the five is left as an orphan, so each has gone on to absorb ordinary records and the corruption is now permanent structure with members. At  $\lambda = 8$  it mints 200 nodes, of which 30 are typo-seeded and 26 of those are never used again. At  $\lambda = 4$  and  $\lambda = 2$  it mints 5,998 nodes on 6,000 records, one per record, none ever reused. A constant price therefore has no setting on this stream that both creates structure and refuses typos, which is the failure a computed price does not have.

The deferred arm ends at  $K = 2$  on all five stream seeds, with exactly the two planted key sets as supports and the supports identical across seeds, at a mean of 7.4 mint events (range 4 to 9) after repair consolidation, and none of those mints cites a typo value: typos are absorbed, structure is not.

#### H.4 THE EXPERIMENTAL DESIGN, PER EXPERIMENT

##### H.5 E8: THE TREE ARMS ON THE NULL STREAMS

The trees do not stay flat because their gate controls error per test and then tests without bound, so no fixed  $\delta$  holds at every length, and a  $\delta$  strict enough to silence them buys the silence with delay, since by the Hoeffding bound the records a split needs grow with  $\ln(1/\delta)$ ; ours covers the running maximum of a fair game, so re-testing after every record costs nothing.

The loose  $\delta$  invents structure on noise. The strict  $\delta$  is silent here, and  $\delta = 10^{-7}$  is river 0.26.1's own default, so the strict setting is the shipped one and the loose setting is the deviation. By the Hoeffding bound the records a split needs grow with  $\ln(1/\delta)$ , so that silence should be bought with detection delay when structure does arrive. The planted arm measures it (`results/e13_tree_arms.json`, same generator, seeds and  $\delta$  values as the null arm, so the two tables are two halves of one trade-off).

The delay is real and it is large where the setting matters: on planted structure the strict setting costs VFDT 20,800 records at noise 0, and at noise 0.2 and 0.4 it costs the split entirely. The practitioner must therefore choose  $\delta$  without knowing which regime the stream is in, and the two regimes want opposite values. EFDT does not pay the delay because it splits at its grace period whatever  $\delta$  says, but it pays on the other side: it ends with 48 to 311 nodes per fit, cell means 51.4 to 222.6, where eight groups were planted, and the strict setting makes that worse rather than better at both noisy rates. Our own rule re-tests after every record at no cost, because the bound covers the running maximum of a fair game rather than each test separately, and it has no grace period to split at.

##### H.6 E9: THE SUPPORT CURVE IN FULL

The cohort size at birth falls from 11 to 3 at  $d = 4$ , 21 to 3 at  $d = 8$  and 22 to 3 at  $d = 16$ .

##### H.7 E13: THE CREATION-BIAS GRID

The first version of this fixture could not fail: it moved only the value while the key name still named the group, so the key set identified the group at every noise rate and neither failure signature was reachable. The grid below drives value noise (a slot takes a value from another group's alphabet), key noise (a slot takes another group's key name) and a fraction of wholly random records from one dial, so both signatures are reachable, and it is run at five seeds per cell rather than one. Two lengths measured earlier,  $T = 30,000$  and  $T = 100,000$ , were dropped for runtime: one run at  $K^* = 8$  and noise 0.3 costs about 235 s on one core, and the cost concentrates in the middle of the noise range, where a half-broken group manufactures many candidate supports.



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Reading the grid. Ten rows converge on  $K^*$ , eight plateau below it, none grows with  $T$ , and no single run of the 270 anywhere in the grid ends above its own  $K^*$ : the largest counts seen are exactly 2, 8 and 20. The steepest slope of mean  $\hat{K}$  against  $T$  is 0.0021 nodes per record (against  $\log T$ , 8.71), and the six rows carrying it are all climbing towards  $K^*$  from below, which is recovery of planted structure rather than noise absorbed as structure. The breakdown noise, the first rate whose mean ARI at  $T = 10,000$  falls below 0.5, is 0.2, 0.3 and 0.4 at  $K^* = 2, 8$  and 20: it arrives later the more groups are planted, which is what key noise predicts, since with few groups a swapped key lands on one of very few alternatives and wipes the key set out faster. Past it the rule reaches no nodes at all, so the measured failure of this criterion is refusal, not invention.

## H.8 E5: ORDER DEPENDENCE

## H.9 E4: THE HEAD-TO-HEAD TABLE

Transferred knobs collapse item by item: the agglomerative cut-off is perfect on the two-group stream at 0.05 and 0.0 at 0.1, DP-means on Wikipedia falls from its oracle 0.7286 to 0.0017 with its penalty from either synthetic stream, and the one label-free self-tuner fails likewise,  $k$ -means picking two clusters on Wikipedia by the silhouette score for 0.502 against 0.9898 when handed the true count. The two baselines that touch the labels are  $k$ -means at 0.9898 handed the true count and agglomerative at 0.9645 with its cut-off chosen on the labels it is scored against.

## H.10 CONTROLS AND CODE-LEVEL GATES

Thirteen gate tests ship with the code and all pass (Appendix E states each). Seven sit at the code level and check that the arithmetic is a code at all: the column code and the value-block code each sum to exactly one over their alphabets (Kraft equality within  $10^{-12}$ ); the incremental cost update equals the block difference within  $10^{-9}$  over 20,000 random blocks; the price of a singleton block is exactly  $\log_2 d$ ; the escape distribution's total mass is one within  $10^{-12}$  at every step; and the mint price satisfies a Kraft upper bound, so it is a codelength and not a score. The rest sit at the engine level. The *positive control* is exact at every scale tested:  $K = 2$  with exactly the planted supports at  $(k, n)$  in (2, 1000), (2, 5000), (4, 200), (4, 1000), (4, 5000). The *null control*, 30 iid streams of 3,000 records, passes the  $\mathbb{E}[\#\text{spurious}] \leq 1$  gate of Theorem C.1. The *degenerate controls* hold: identical records yield at most one node; globally unique values yield  $K = 0$ ; a constant key and a uniformly random key present on every record never justify a mint. The *sparsity control* holds: record widths 2 to 12 from a single population yield at most one node, so mint decisions do not track record width. The *merge guard* holds: on a healthy two-node state the merge delta is strictly positive and the full repair pass keeps  $K = 2$ .

## H.11 RAW RECORDS: A CATALOGUE DEPARTMENT

We stream 993 records of the Footwear department of a raw e-commerce catalogue: keys exactly as published, values kept as opaque strings, no cleaning, no schema. Throughput is 303.4 records per second, with the loop that checks whether a record matches no existing candidate not yet optimised. Over 43 raw keys and 51 mint events the graph settles at  $K = 6$ , and the nodes are readable. The largest ( $t = 263$  members) is men's casual footwear: Ideal For is Men on 242 records against Women on 29; Occasion is Casual on 145 against Sports on 21 and Formal on 10; Outer Material splits Genuine Leather 64, Leather 53, Synthetic Leather 31. The second ( $t = 58$ ) is women's heeled sandals: Type is Wedges 21, Heels 18, Flats 5, under a Type key the men's node does not publish; its Weight values are quoted per single sandal (200 g on 16 records, 400 g on 14); its top Closure values are Slip on 4, Slip on 3, SLIP ON 3, the same value spelled three ways, surfaced by the node that carries it. Two further nodes isolate records that are not footwear at all: a leather polish (support Polish Care, Applicator Provided, Purpose, Suitable for) and incense sticks (Burning Time, Fragrance, Number of Sticks per Box), both misfiled listings whose supports share no key with the footwear nodes. Every mint receipt names its justifying key and its per-facet bits. No label, embedding or similarity function appears anywhere in the run.

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## H.12 REAL-WORLD STANDING AT A GLANCE

### H.13 WIKIPEDIA INFOBOXES: THE RUN IN DETAIL

We fetch 320 pages from three unrelated categories: 1994 films (120 records), English male film actors (113), and Alpine four-thousanders, the peaks of the Alps above 4,000 metres (87). From each page we take the raw top-level `key = value` parameters of its first infobox, with no cleaning and no schema, and shuffle everything into one stream of 121 distinct raw keys. Under the protocol of Section G.1, on the demonstration order, the result is five nodes: a film node of 130 records (119 film, 9 person, 2 mountain), a person node of 101 and a mountain node of 86, both pure, a second film node of 10, pure, and one politician record isolated; purity 0.9665 over 328 memberships, since 8 records belong to two nodes. The film node’s discovered support begins `alt_name`, `budget`, `caption`, `cinematography`, `company`, `country`, `director`, `distributor`, `editing`, `editor`, `executive_producer`, `genre`, `gross`, `language`; the person node’s is `birth_place`, `children`, `death_place`, `education`, `occupation`, `years_active`; the two mountain nodes share `easiest_route`, `elevation_m`, `first_ascent` and `image_caption`, and separate on `isolation`, `isolation_km`, `listing` and `highest` against `prominence_m`, `range`, `pushpin_map`, `pushpin_relief` and `map_caption`, two infobox templates for mountains (the results file records the first fourteen keys of each support in sorted order); the politician record carries `constituency`, `office`, `party`, `predecessor`, `successor`, `term_start`, `term_end`. This order is one instance: the E4 harness (Table 12) streams the same records under the same protocol on its own shuffle and scores ARI 0.6790 at  $K = 5$ ; the language-model comparison (Appendix H.15) streams them on a third order and scores 0.8744 at  $K = 5$ ; E5 (Table 11) gives the spread over 20 orders, ARI 0.6389 to 0.986 with mean 0.8438 and  $K$  from 3 to 6, and the common variation is how the mountains split into nodes. The stream publishes `distributor` on 46 records and `distributors` on one, and the film node’s own support holds both `editing` and `editor`: the raw key drift the designed key-identity operator would price.

### H.14 LAZADA RAW KEYS: THE RUN IN DETAIL

The stream is the Lazada marketplace listings released with AutoPKG (Hongwimol et al., 2026): 21,365 listings that carry a specification block, parsed to 3,033 raw keys exactly as sellers published them, shuffled once, no label fed. The graph settles at  $K = 68$  nodes covering 9,752 records (46%) at 1.73 records per second, with repair taking 12,282.0 of the 12,365.5 seconds. This run predates the unified protocol of Section G.1: its results file records a repair pass every 200 records (107 passes) and one at the end, on the batch objective without mint-time absorption, with a fast reassignment override. The largest node (1,510 members) is `grocery`: `dietary_needs` on 935 records, `packaging_type` on 892, `pack_type` on 888, `organic` on 804, `storage_type_gr` on 685. The second (1,332) is `fashion`: `clothing_material` on 821, `size` on 721, `color_family` on 700, `fa_pattern` on 697. The third and fourth (1,274 and 1,190) are two placeholder templates on `color_family`, `model` and `brand`, where the top value is `Not Specified` (712 and 957 records). The fifth (694) is `supplements and beauty`: `units_hb`, `formulation_supplement`, `ingredient`, `skin_type`. The case-variant survey folds keys by lowercasing and stripping every non-alphanumeric character, giving 167 families and 340 pairs of raw keys that differ only in case or spacing. Exactly 1 pair lands in a shared node (`Occasion`, 19 records, and `occasion`, 122, co-supported by the fashion node); 5 pairs have both keys supported somewhere but never together; in 82 pairs one key is supported and the other is not; in 252 pairs neither key is in any support. The file also scores the partition against the silver product type each listing carries in the released knowledge graph, itself model-derived: ARI 0.0013 over 21,260 scored records, with 5,518 silver types against 52 template-level nodes plus one background of 12,906 records. The top nodes do concentrate types (Children’s Clothing Set in the fashion node, Nutrient Supplement in the supplements node, Gift Card in a digital-goods node), but concentration is not partition agreement, and the score says so.

### H.15 A LANGUAGE MODEL BUILDING THE SAME GRAPH

Terms used here: determinism is the mean pairwise ARI between the three seeded runs of one method, 1.0 meaning the same graph every time; purity is the fraction of records whose node’s

majority type is their own type; the largest node share is the fraction of scored records in the largest node, the catch-all indicator; a value-named node is one whose name is a value copied from a member record, a value promoted to a node. On Wikipedia every model run formed the same four nodes, Film, Actor, Mountain and Animation, the last a second film node holding one or two records; seeds 1 and 2 each left one record unparsed and placed Animation differently, which is the 0.994. ESCROW formed a film node of 122 records (two mountain records inside), a person node of 112, a mountain node of 74, a second mountain node of 11, and a one-record politician node, the five-node structure of Appendix H.13 on a third arrival order; neither method made a wrong merge. On Lazada the three sampled runs gave  $K = 24, 78$  and  $28$  with pairwise ARI 0.069, 0.375 and 0.051 between them. Seeds 0 and 2 and the greedy run formed department nodes (Toys, Clothing, Electronics) beside a Miscellaneous node holding 403, 392 and 231 of the records and spanning 356, 348 and 217 silver types; seed 1 degenerated: a No\_Type node took 851 records spanning 659 silver types, and 60 of its 78 nodes (77 percent) were named after a value copied from one record, such as `Model_SYS_CARA7A67/PILVA`, 55 of them singletons. Same records, same order, same prompt, and the output changed in kind. The bill was 107,512 (greedy), 109,384, 157,821 and 110,772 tokens per 1,000 records and 1,603 to 1,859 s on one A100; scaled linearly to the 21,365-record stream, a lower bound because the prompt grows with the node list, that is 2.3M to 3.4M tokens and 9.5 to 11 A100-hours. ESCROW's six nodes are spec templates, each with a receipt naming its justifying key and its bits per key: `fashion` on `fa_season` (39.6 bits), `size`, `fa_general_styles`, `fa_pattern` and `clothing_material`; `grocery` on `dietary_needs`, `organic`, `pack_type`, `packaging_type` and `storage_type_gr`; a placeholder template on `model` (70.4 bits) and `color_family`; a second placeholder on `color_family` (122.1 bits), `model`, `brand` and `Variation`; `supplements` on `formulation_supplement` (48.2 bits), `ingredient` and `units_hb`; `books` and `media` on `language` (73.3 bits), `author`, `isbn_issn`, `media_format` and `artist`. The run made 24 mints and 18 merges in about 1.5 s; members were identical on all three runs, and 699 of the 1,000 records stayed in the background because their keys carry too little categorical signal for the price (covered-only ARI 0.0301): the model attaches every record it parses, but 23 to 85 percent of them into a catch-all node, which is a wrong merge under another name. Of 35 case-variant raw-key pairs such as `color` and `Color`, ESCROW covered both keys of 11 and placed 8 of those 11 in one node; the model placed 13, 29 and 14 of 35 together in the sampled runs and 9 of 35 in the greedy run.

## H.16 WHAT THE TWO BENCHMARKS A LANGUAGE MODEL WINS ACTUALLY RANK

Section 5.4 reports two losses to a prompted model. Neither is read here as evidence about language, because both benchmarks are cleared by methods with no model in them. The two tables below are those floors. Nothing in either is proposed as a contribution.

**The encyclopedia stream (E43).** The same 320 records the model read, in the same file. Each method chooses its own number of clusters by silhouette and consults no label; the last column takes the best number on the test labels and is an upper bound it could not have in use. The first two rows read the record as a string, an input the criterion never takes, so they are a floor on the benchmark and not matched baselines.

**Key identity on AutoPKG's Lazada keys (E44).** The gold of E20 marks a pair positive exactly when `norm_key(a) == norm_key(b)`, lowercase then strip non-alphanumeric, over 328 positive and 22,921 negative pairs with 12 collisions dropped. It is therefore the output of a string function. Swept rules are given their best knob on the test labels, stated as such.

## H.17 MUSICBRAINZ 20K, CONDITION A, IN FULL

Condition A hands every method only source, number, year, language and a length band of 30 seconds; title, artist and album are withheld. On the fixed engine ESCROW finds 4,594 nodes at pairwise F1 0.0001 (pairwise recall 0.1703); the run before the engine fixes found 3 nodes at pairwise F1 0.0, with a repair pass every 200 records, a forced pass whenever  $K$  exceeded 300 with at least 60 records since the last, and one final pass; one arrival order. Condition B, which adds presence facets for title, artist and album tokens seen at least twice, is pending.

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## H.18 REVERB45K UNDER CESI’S METRICS, IN FULL

Each record carries its relation phrases as keys, with the sorted objects as the value, plus presence facets for the subject’s surface tokens; every baseline renders the same strings to text. ESCROW ran once, on one arrival order, in 690.1 seconds with 105.9 seconds of repair, minting 75 nodes over 209 records with 12,118 records in the background, on the cadence its results file records: a repair pass every 200 records, a forced pass whenever  $K$  exceeded 300 with at least 60 records since the last, and one final pass. The embedding baselines were measured on the same records and protocol in the earlier run and were not recomputed, since they never touch the engine; the file records this. The knobs matter: carried to the worst setting of their own sweeps, HDBSCAN falls to 0.0025, agglomerative to 0.0243 and DP-means to 0.0267 pairwise F1, and single sweep points that go higher on macro alone collapse the other two metrics, HDBSCAN at minimum cluster size 5 reaching macro 0.8915 at micro 0.1140 and pairwise 0.0025. A secondary reading in the file pools the background into one cluster and reports macro 0.5499 at pairwise 0.0025; we do not use it, because one cluster of 12,118 records is not a partition. CESI’s own normaliser could not run on the Python on this machine, so the dataset’s `triple_norm` field is used for every method, which makes the string space slightly harder than CESI’s; the deviation is recorded in the file.

## H.19 THE REMAINING EXTERNAL TIER

Its protocol is the one the first tier already follows, fixed in advance. Every input our method receives, every baseline receives. Each knobbed baseline is run three ways: oracle-tuned on the test stream, transferred from another stream, and at shipped defaults. ESCROW runs once. The roster, each with its role:

- the anytime-valid betting repair of the Hoeffding gate (Amoukou et al., 2026) on the null streams of E8; the betting form is the same kind of wealth argument our own bound uses;
- the Alaska benchmark as the one-shot key-identity test set against the hand-curated Alaska gold (Crescenzi et al., 2021);
- AutoPKG (Hongwimol et al., 2026), head to head on its own product data under its own evaluation protocol; the opponent there is its decision agent, driven by a large language model; the Lazada run above is the structure exhibit on that data, not the head-to-head;
- the DBpedia mappings wiki, with its thousands of hand-written raw-key-to-ontology mappings, as the open-domain key-identity target at scale;
- MusicBrainz 2M, the full-scale registry behind the 20K benchmark, and voter registries with degenerate nulls, at record level;
- the nested stochastic block model (Peixoto, 2017), a random-graph model with planted communities inside communities, as the hardest structural objection, with HISEvent (Cao et al., 2024) beside it;
- the two adversarial arrival orders of PERCH (Kobren et al., 2017), sorted by gold type and round-robin over types, on the E4 streams, and random orders on the external streams; the random-order spread on the E4 streams is run (E5, Table 11);
- the encoding-stability sweep on which the hidden-knob objection turns is run (E12, Appendix D); its extension to the external streams is what remains designed.

WDC Products, the web product-matching benchmark, was probed and deferred: its identity signal lives in free-text titles, which version 1 quarantines rather than reads.

## H.20 LIMITATIONS, IN DETAIL

Honest scope is part of the claim, so we state it as measurements rather than caveats. **Categorical facets only.** The claim is scoped to discrete facets, and the evaluation matches. Free text is quarantined because any code for it must first choose a tokenisation, worth 5 to 75 bits per field with median 20 over 31 Wikipedia text fields, more than any constant the method removes, and no min-max code exists for an unbounded alphabet. A numeric code is specified, a Student- $t$  predictive whose resolution term cancels from every decision difference, and is quarantined until evaluated. Neither needs new machinery, because the code factorises over facets. **Detection latency.** Deferral

is bought with lag. In the two-key cell at eight values the online release fires at record 229, on 21 supporting records; at three keys five records suffice (Table 9). At fifty records the shipped batch objective already prefers the planted two-node structure, by 78.8 bits over the one-node-total code and 69.6 over the all-background code. One caveat belongs beside it: with a single key per group the objective prefers the degenerate code at every stream length, by 12 to 22 bits, which is exactly why the one-key control returns no nodes; a limit of the criterion, not of the search. On a structureless stream the mint time is infinite, which is the correct answer, not a failure mode. **The search is approximate; the criterion is not.** The exact argmax over candidates is a maximum edge biclique problem and NP-hard (Peeters, 2003), the same wall MDL4BMF names (Miettinen & Vreeken, 2014). The rule evaluates its criterion exactly over a generated pool, one candidate per residual (key, value) pair, under a pool budget that is computational and not part of the criterion. We state that budget as what it is, a declared operational parameter that can change the output, rather than as a formality. When the pool is full the engine evicts the escrow account with the lowest accrued evidence, so the budget decides which accounts survive long enough to be released, and the shipped test measures both endpoints on one stream: at a cap of one, on the shuffled planted stream, each record’s fresh candidates evict the previous record’s, no escrow ever accumulates and no node is minted at all; on the same records sorted so that a group arrives together, the surviving candidate mints; and at a cap of eight the planted  $K^* = 8$  is recovered in full (`code/tests/test_batch.py`, the pool-eviction test). The budget binds on both large real streams, not only in this constructed test: the Lazada run hit the default cap of 4,096 at about record 3,600, where an eviction of a candidate touched by the current record raised a `KeyError` at release, and the MusicBrainz run raised the cap eightfold to 32,768 after the same crash, which removes eviction entirely on that dataset because its whole (key, value) signature space is about 18,300. The engine now never evicts a candidate touched by the current record. The curve between those endpoints is E19, reported in Section 7: sixteen budgets from 1 to 32,768 on four streams at five seeds each, where the output stops moving at 128, 256, 2,048 and 3,072 and is identical at every larger budget, so a minted set that moved with the budget would have traded the removed constants for one, and it does not. The code-choice curve beside it is E12, which is run. Absence evidence costs  $O(|\mathcal{A}|)$  work per record in version 1, with  $\mathcal{A}$  the set of keys seen so far. **Order dependence.** The batch objective is a function of the state, not of the arrival order: the test suite asserts its invariance to relabelling the nodes (difference 0.0 over 20 relabellings, `code/tests/test_batch.py`). The greedy pass is not order-invariant: which candidate clears escrow first can depend on arrival order, and 20 orders that reach the same member partition can still differ by up to 225 bits because the sequence assigns cell ownership differently. E5 measures the spread of the final structure over 20 arrival permutations (Table 11): none on the two planted streams, and ARI 0.6389 to 0.986 with  $K$  from 3 to 6 on Wikipedia. The PERCH adversarial orders (Kobren et al., 2017) are designed, and the one-pass-to-batch gap of Definition C.2, computable on every run with no ground truth, is 9,966 bits and 0.50 per record on the twenty-node stream of Appendix C.7, though the shipped runners do not yet record it per run. **Cost and scale, measured.** The claim that the cost per record does not grow with the node count holds only under a condition, and E14 states it (`results/e14_cost_and_scale.json`; every cell in its own child process, the headline statistic the median over the last decile of the stream, where the graph is at full size). With disjoint supports the insertion step is flat: 48.58 microseconds per record at  $K = 8$  and 44.98 at  $K = 128$ . With every node supporting every key, so that candidate retrieval returns the whole graph, it is not: 113.0 microseconds at 5 nodes and 349.17 at 20, and at planted  $K \in \{32, 128\}$  in that regime the repair pass did not finish the stream inside a 180 second budget. Cost per record is flat in stream length, 33.37 microseconds at  $n = 3,000$  and 35.08 at  $n = 10^5$ , but the repair pass is not: it takes 22.7 percent of wall time at  $n = 10^3$  and 95.81 percent at  $n = 10^5$ , so end-to-end throughput falls from 7,891.5 to 908.8 records per second over that range at a constant node count. That is the mechanism behind the twentyfold spread between our real-world runs, 303.4 records per second on the footwear catalogue against 1.73 on the Lazada stream. Peak resident memory is 242.45 MB at  $n = 10^5$ . **Key identity.** A key enters a support only through a minted node; the operator that prices two keys as one under the same code is designed, not run in this version. **Replication.** E4 synthetic and E9 are one seed each, and every real-world run is one arrival order; E5 (20 orders), E7 (five seeds) and E13 (five seeds per cell) report their spread. **Inherited from the price.** The model must not condition on the number of attachments per record; that breaks the complete randomness the martingale rests on, and in the conditioned model the rate’s scale cannot be identified even in principle (Doshi-Velez & Williamson, 2015). And  $\hat{\gamma}$  sees evidence growing like  $\log t$ , so after a regime change in the mint rate the price stays miscalibrated for a long stretch. **Scope of the evidence.** The

external tier of Appendix H.19 is designed, not run, and on free-text identity version 1 leaves most subjects as singletons (ReVerb45K macro F1 0.016 against micro 0.7777).

## H.21 FIGURE SOURCES

Every figure is drawn from a results file by a script under `code/experiments/`; the captions give the headline numbers and the file, and the full data paths are these. Figure 2: the receipt under step 7 is `results/e9_support_curve.json`, cell  $k = 4$ ,  $d = 8$ , keys `t_star`, `G_at_mint` and `price_at_mint`. Figure 1: `results/llm_graph_formation.json`, keys `datasets.lazada.llm_determinism` (node counts 24, 78, 28; mean pairwise ARI 0.1647), the `largest_node_share`, `tokens_total` and `wall_s` of the four Lazada model runs (0.4038, 0.8527, 0.3928, 0.2315; 109,384, 157,821, 110,772, 107,512; 1,641.5, 1,858.9, 1,629.9, 1,603.2 s), `datasets.lazada.escrow_determinism` and the ESCROW `wall_s` (2.39, 2.41, 2.39 s); column A stands for the agglomerative cut-off and cosine components of E4, column B for DP-means and BP-means, and column C for AutoSchemaKG, AutoPKG and the Qwen2.5-14B run; the cost row for A and B reads not run because those rules were not timed on the Lazada records. Figure 6: `results/e9_support_curve.json`, every cell. Figure 3: `results/e8_false_mint_null.json` (the plateau) and `results/e8_tree_arms.json` (five-seed means; per-seed lists in the file). Figure 8: `results/e4_first_tier.json`, per stream the keys `kmeans_oracleK.ARI` and `kmeans_silhouette.ARI`, and for HDBSCAN, agglomerative, cosine components and DP-means `oracle_ARI` and the minimum over transferred `ARI`; ESCROW’s marks are the per-stream `ARI.mean`, `ARI.min`, `ARI.max` and `K` of `results/e5_order_dependence.json`. Figure 9: node counts after each chunk from `results/llm_gf_runs/` (the four Lazada model runs, `chunks[].records` and `chunks[].nodes_after`); ESCROW’s final count from `escrow_lazada_r0.json` to `r2.json`; seed agreement, background share and the Wikipedia scores from `results/llm_graph_formation.json`. No node count over  $n$  is stored for ESCROW, so no ESCROW curve is drawn. Figure 7: the same four Lazada model run files (`chunks[].tokens_in`, `tokens_out` and `seconds`), `escrow_lazada_r0.json` (`wall_s`), and `datasets.lazada.total_stream` and `reading.cost` of `results/llm_graph_formation.json`, against which the script checks the extrapolation. Figure 5: the per-record steps are `code/escrow/engine.py` (`process` and `_mint`); the operator order and the full-pass rule are `code/escrow/batch.py` (`repair`); the price is Equation D.10.

## H.22 WHAT BUILDING THE GRAPH COSTS

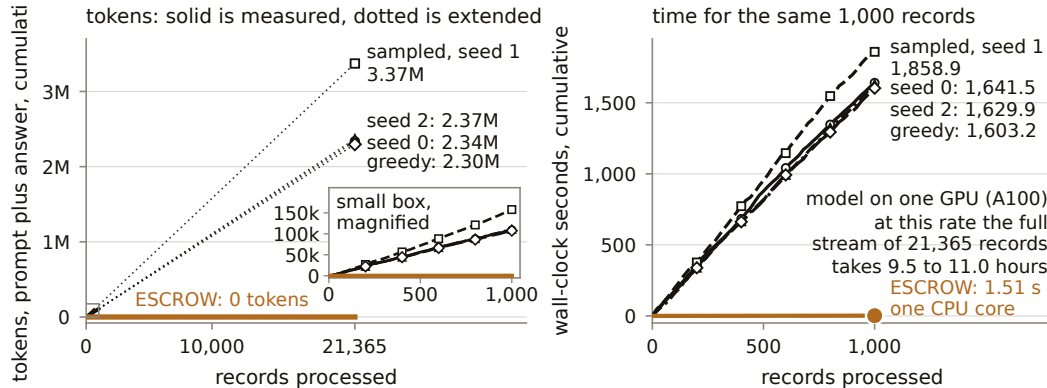


Figure 7: What building the graph costs. Left: cumulative tokens for the four model runs on the first 1,000 Lazada listings, 107,512 to 157,821 (the box at the origin, magnified in the inset), extended at each run’s rate to the full stream of 21,365 records, 2.30M to 3.37M tokens, a lower bound. Right: cumulative seconds on one A100, 1,603 to 1,859 s per 1,000 records and 9.5 to 11 hours for the full stream, against about 1.5 s for ESCROW on one CPU core (`results/llm_gf_runs/`, `results/llm_graph_formation.json`).

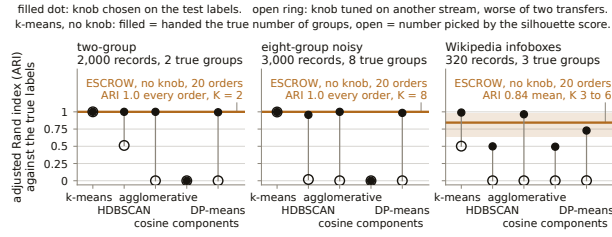


Figure 8: Tuned baselines against ESCROW on three streams, scored by ARI. Filled markers are oracle scores, with the knob chosen on the test labels, and open markers are transferred scores, with the knob tuned on another stream, the worse of two transfers. The grey line is the drop between them: to 0.0 for agglomerative and DP-means on the synthetic streams, and to at most 0.0017 for every transferred knob on Wikipedia.  $k$ -means has no knob, so filled is handed the true  $K$  and open picks  $K$  by silhouette. ESCROW (gold) has nothing calibrated and no embedding (results/e4\_first\_tier.json, results/e5\_order\_dependence.json).

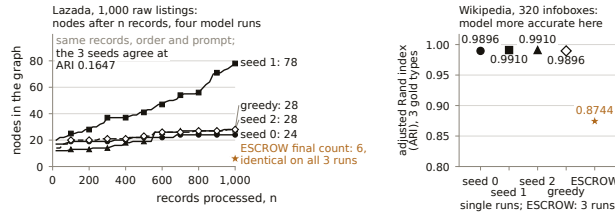


Figure 9: A language model building the same graph on identical records. Left: the first 1,000 raw Lazada listings. Four runs of Qwen2.5-14B-Instruct saw the same records, order and prompt, yet end with 24, 78, 28 and 28 nodes, agreeing at a mean pairwise ARI of 0.1647; the gold star is ESCROW's 6 nodes on all three runs, with 699 records left in its background. Right: Wikipedia on one arrival order, where the model is more accurate (results/llm\_graph\_formation.json, results/llm\_gf\_runs/).

## I FIGURES REFERENCED FROM THE MAIN TEXT

## J FIGURES REFERENCED FROM THE MAIN TEXT

| Exp.       | Question; design (data, order, what is planted or held out)                                                                                                                                                                                                                                                                                                                                                                              | Control; baselines and how each was tuned                                                                                                                                                                                                                                                                                                                                                                      | Metric; falsifier                                                                                                                                                                                                                       |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E7         | Can a rule mint the moment a record fails to fit? 6,000 records, two planted groups of three keys, alphabet 8; one percent of records carry a corrupted value that is seen once, ever.                                                                                                                                                                                                                                                   | Four arms of the mint rule on the same stream, five stream seeds: unseeded immediate; record-seeded immediate under the computed price, swept over five alphabet sizes by four typo rates; record-seeded immediate under a constant price $\lambda$ , the DP-means shape, swept over $\lambda \in \{2, 4, 8, 16, 39\}$ bits; deferred, ESCROW as shipped. No external baseline.                                | Mints, final $K$ , typo-seeded nodes, and how many of those are never reused. Falsifier: an immediate arm that recovers the planted structure without minting on typos.                                                                 |
| E8         | Does the lifetime false-mint bound hold on a standalone block, and does anything grow on noise? 20,000 null candidate trajectories on one key of alphabet 10, 200 steps, background warmed on 200 draws; null streams of independent uniform draws over four keys, alphabet 10, 12 seeds at each of 2,000, 5,000, 10,000 and 20,000 records.                                                                                             | The null itself: every node created is false. VFDT and EFDT, the Very Fast and Extremely Fast Decision Trees, predicting one key from the other three at $\delta = 0.01$ and at $\delta = 10^{-7}$ , which is river 0.26.1's own default, five seeds per cell.                                                                                                                                                 | Empirical crossing rate against $2^{-b}$ ; final node count by length. Falsifier: a crossing rate above $2^{-b}$ at any $b$ , or a node count that grows with the stream.                                                               |
| E9         | Does the price behave like a price? Two-group fixture over $k \in \{2, 3, 4, 6\}$ keys per group and $d \in \{4, 8, 16\}$ values per key; the first mint in each cell.                                                                                                                                                                                                                                                                   | The directional prediction is the control: $t^*$ must fall as $k$ grows. No baseline.                                                                                                                                                                                                                                                                                                                          | Cohort size at birth $t^*$ ; evidence $G$ against price at the mint. Falsifier: $t^*$ not falling with $k$ , or $G$ overshooting the price by more than one member's evidence.                                                          |
| E13        | How biased is the creation rule? $K^* \in \{2, 8, 20\}$ planted groups of three keys, alphabet 8; lengths 1,000, 3,000 and 10,000 by six noise rates 0 to 0.5, five seeds per cell, 270 runs. One dial drives three corruptions together: a slot's value drawn from another group's alphabet, a slot's key name replaced by another group's key, and a fraction of records drawn wholly at random and labelled as belonging to no group. | The planted $K^*$ . No baseline; this answers Kearns et al. (1997) by measurement.                                                                                                                                                                                                                                                                                                                             | Final $\hat{K}$ and ARI against the planted labels. Falsifier: $\hat{K}$ growing with length (noise coded as structure) or plateauing below $K^*$ (structure written off).                                                              |
| E13, trees | What does the strict $\delta$ cost when structure is really there? The same planted streams at $K^* = 8$ and noise 0, 0.2, 0.4, with the planted group label as the target, supervision ESCROW never receives; VFDT to $T = 30,000$ , EFDT to $T = 2,000$ , five seeds per cell.                                                                                                                                                         | The two $\delta$ settings against each other on the same records; river 0.26.1's own default is the strict $\delta = 10^{-7}$ , and the E8 null arm is the other half of the same trade.                                                                                                                                                                                                                       | Index of the first split; final node count. Falsifier: a strict $\delta$ that costs no delay on planted structure.                                                                                                                      |
| E5         | How much does record order change the graph? The E4 streams, two-group (2,000), eight-group noisy (3,000) and the 320 Wikipedia infoboxes, each streamed in 20 random orders under the one protocol.                                                                                                                                                                                                                                     | The synthetic streams, where the truth is planted, are the control. No baseline.                                                                                                                                                                                                                                                                                                                               | ARI and $K$ : mean, standard deviation, minimum and maximum over the 20 orders. Falsifier: $K$ or ARI drifting on planted structure.                                                                                                    |
| E4, Wiki   | What does a method with nothing to tune give up, and does the unmodified engine recover types from raw infoboxes? Two-group (2,000 records), eight-group noisy (3,000), raw Wikipedia infoboxes (320 pages from three unrelated categories: 120 films, 113 actors, 87 Alpine peaks; raw first-infobox parameters, shuffled, 121 raw keys); category labels held out.                                                                     | Five baselines on the same frozen MiniLM sentence embedding, a small pretrained text encoder: $k$ -means (handed the true $K$ , or chosen without labels by the silhouette score, a measure of cluster separation), HDBSCAN, agglomerative, cosine components (records linked when their embeddings are close enough), DP-means, each at oracle and transferred knobs. ESCROW reads raw records, no embedding. | ARI; on Wikipedia also purity per node and $K$ . Falsifier: baselines keeping their oracle scores under transfer, or ESCROW trailing an honestly run baseline; mixed nodes, or a node count unrelated to the categories.                |
| Lazada     | Does it find the spec templates a marketplace is built from, and does it group case-drifted keys? 21,365 spec-bearing listings, 3,033 raw keys as published, shuffled once, no labels.                                                                                                                                                                                                                                                   | A structure exhibit by protocol, no embedding baseline; a case-variant survey of 340 raw-key pairs; the released silver product type (a model-derived label) as a sanity score.                                                                                                                                                                                                                                | $K$ , coverage, pairs in a shared node. Falsifier: $K$ in the hundreds or unreadable supports; or drifted keys landing in shared supports as a matter of course.                                                                        |
| MB20K A    | What happens where the visible fields carry no identity signal? 19,375 songs, five categorical fields (source, number, year, language, length band), 10,000 true entities; title, artist and album withheld.                                                                                                                                                                                                                             | The null-signal input. $k$ -means handed $K = 10,000$ ; HDBSCAN, agglomerative, DP-means at oracle knobs on the same embedding.                                                                                                                                                                                                                                                                                | Pairwise F1, ARI, $K$ . Falsifier: a node count that grows to fill the record count.                                                                                                                                                    |
| ReVerb     | Can relation and token facets alone recover entity identity? 12,327 subject strings, 36,814 mentions, 6,061 gold clusters, CESI's granularity and metrics code.                                                                                                                                                                                                                                                                          | The no-merge floor: same-string mentions are one cluster for every method by construction. $k$ -means handed 6,061; HDBSCAN, agglomerative, DP-means oracle; CESI's published numbers as context, with side information no method here sees.                                                                                                                                                                   | Macro, micro, pairwise F1 (averaged over clusters, over records, over record pairs); pairwise precision. Falsifier: not a low score, but merges that pull pairwise precision below the no-merge floor, meaning they cross entity lines. |
| LLM        | Does a language model that builds the graph directly form it well at demo scale and at production scale? The 320 Wikipedia infoboxes (3 gold types, 121 raw keys) and the first 1,000 listings of the Lazada stream (643 raw keys), each in one fixed arrival order shared by both methods; the Lazada silver truth is AutoPKG's model-derived product type, 759 types over 1,000 records, 614 of them singletons.                       | ESCROW on the identical records and order, three repetitions, under the one protocol of Section G.1. Qwen2.5-14B-Instruct (bf16, one A100) prompted to create or attach: chunks of 20, the current node list in the prompt, a JSON answer per record, no cluster count given. Not tuned: sampled at temperature 0.7 with seeds 0, 1, 2, and once greedy.                                                       | ARI, purity, $K$ ; determinism (mean pairwise ARI across seeds), largest node share, value-named nodes, tokens, wall time. Falsifier: a model that is deterministic across seeds and forms no catch-all node at scale.                  |

Table 6: Experimental design, one row per experiment. VFDT and EFDT are Domingos & Hulten (2000) and Manapragada et al. (2018); HDBSCAN is hierarchical density-based clustering; DP-



|        | $\delta$  | $T=2,000$ | $T=5,000$ | $T=10,000$ | $T=20,000$ |
|--------|-----------|-----------|-----------|------------|------------|
| VFDT   | 0.01      | 1.0       | 1.0       | 1.0        | 19.0       |
| EFDT   | 0.01      | 15.0      | 73.0      | 73.0       | 91.0       |
| VFDT   | $10^{-7}$ | 1.0       | 1.0       | 1.0        | 1.0        |
| EFDT   | $10^{-7}$ | 1.0       | 1.0       | 1.0        | 1.0        |
| ESCROW | none      | 0         | 0         | 0          | 0          |

Table 7: E8, tree arms on the iid null streams: mean final node count over five seeds per cell; the per-seed EFDT values at  $\delta = 0.01$  and  $T = 20,000$  are 111, 111, 111, 11 and 111, so the maximum is 111. The two scales differ by one because a silent tree still has its root: tree silence reads 1, and our silence reads 0 minted nodes. Every tree node beyond the root is false structure. Our row is the 48-of-48 result of `results/e8_false_mint_null.json` (12 seeds per length).

|                    | noise | first split, record index |                    | final nodes     |                    |
|--------------------|-------|---------------------------|--------------------|-----------------|--------------------|
|                    |       | $\delta = 0.01$           | $\delta = 10^{-7}$ | $\delta = 0.01$ | $\delta = 10^{-7}$ |
| VFDT, $T = 30,000$ | 0.0   | 8,400                     | 29,200             | 20.2            | 9.0                |
|                    | 0.2   | 9,400                     | never              | 129.0           | 1.0                |
|                    | 0.4   | 8,800                     | never              | 129.0           | 1.0                |
| EFDT, $T = 2,000$  | 0.0   | 200                       | 200                | 51.4            | 51.4               |
|                    | 0.2   | 200                       | 200                | 101.4           | 177.6              |
|                    | 0.4   | 200                       | 200                | 131.8           | 222.6              |

Table 8: The planted arm of the tree comparison,  $K^* = 8$  planted groups, the planted label as the target, five seeds per cell; means over seeds. *Never* means no split inside the stream on any of the five seeds. VFDT’s first split at noise 0 is record 8,400 on all five seeds at the loose setting and 29,200 on all five at the strict one, so the strict setting costs 20,800 records of delay; at noise 0.2 and 0.4 it never splits at all inside 30,000 records where the loose setting splits at 9,400 and 8,800. EFDT’s setting buys nothing on this stream: it splits at record 200, its grace period, in all thirty fits at both settings. VFDT runs to  $T = 30,000$  and EFDT to  $T = 2,000$  because one EFDT fit costs about 79 s at  $T = 2,000$  and did not finish inside twenty five minutes at  $T = 10,000$  on one core.

| $k$ | $d$ | $t^*$ | $n$ at mint | $G$ at mint | price (bits) | bits/member |
|-----|-----|-------|-------------|-------------|--------------|-------------|
| 2   | 4   | 11    | 46          | 55.458      | 54.775       | 5.042       |
| 2   | 8   | 21    | 229         | 122.987     | 122.960      | 5.857       |
| 2   | 16  | 22    | 548         | 158.305     | 156.739      | 7.196       |
| 3   | 4   | 6     | 20          | 40.497      | 37.129       | 6.750       |
| 3   | 8   | 5     | 33          | 41.177      | 40.807       | 8.235       |
| 3   | 16  | 4     | 23          | 36.389      | 35.131       | 9.097       |
| 4   | 4   | 4     | 12          | 32.924      | 32.033       | 8.231       |
| 4   | 8   | 4     | 20          | 38.085      | 36.505       | 9.521       |
| 4   | 16  | 4     | 41          | 47.629      | 42.489       | 11.907      |
| 6   | 4   | 3     | 8           | 37.854      | 32.972       | 12.618      |
| 6   | 8   | 3     | 10          | 43.035      | 34.594       | 14.345      |
| 6   | 16  | 3     | 12          | 38.849      | 35.885       | 12.950      |

Table 9: E9, the support curve: cohort size at birth  $t^*$ , stream position  $n$ , accumulated evidence  $G$  and price at the first mint, per  $(k, d)$  cell;  $k$  is keys per planted group and  $d$  the alphabet size; one seed per cell.  $t^*$  falls as  $k$  grows, and the overshoot  $G$  minus price is below one member’s average evidence (last column) in every cell. Across  $d$  both competing effects are visible: at  $k = 2$  thinner values raise the price and  $t^*$  rises (11, 21, 22); at  $k = 3$  they make the background’s predictions worse and  $t^*$  falls (6, 5, 4); at  $k \geq 4$  the two cancel.

| $K^*$ | noise | $T=1,000$ |       | $T=3,000$ |       | $T=10,000$ |       | regime        |
|-------|-------|-----------|-------|-----------|-------|------------|-------|---------------|
|       |       | $\hat{K}$ | ARI   | $\hat{K}$ | ARI   | $\hat{K}$  | ARI   |               |
| 2     | 0.0   | 2.0       | 1.000 | 2.0       | 1.000 | 2.0        | 1.000 | converges     |
| 2     | 0.1   | 1.0       | 0.530 | 1.0       | 0.547 | 1.0        | 0.553 | plateau below |
| 2     | 0.2   | 0.0       | 0.000 | 0.0       | 0.000 | 0.0        | 0.000 | plateau below |
| 2     | 0.3   | 0.0       | 0.000 | 0.0       | 0.000 | 0.0        | 0.000 | plateau below |
| 2     | 0.4   | 0.0       | 0.000 | 0.0       | 0.000 | 0.0        | 0.000 | plateau below |
| 2     | 0.5   | 0.0       | 0.000 | 0.0       | 0.000 | 0.0        | 0.000 | plateau below |
| 8     | 0.0   | 8.0       | 1.000 | 8.0       | 1.000 | 8.0        | 1.000 | converges     |
| 8     | 0.1   | 8.0       | 0.753 | 8.0       | 0.854 | 8.0        | 0.895 | converges     |
| 8     | 0.2   | 8.0       | 0.457 | 8.0       | 0.563 | 8.0        | 0.621 | converges     |
| 8     | 0.3   | 2.0       | 0.081 | 7.4       | 0.397 | 8.0        | 0.443 | converges     |
| 8     | 0.4   | 0.6       | 0.012 | 1.0       | 0.019 | 2.8        | 0.017 | plateau below |
| 8     | 0.5   | 0.0       | 0.000 | 0.0       | 0.000 | 0.0        | 0.000 | plateau below |
| 20    | 0.0   | 20.0      | 1.000 | 20.0      | 1.000 | 20.0       | 1.000 | converges     |
| 20    | 0.1   | 20.0      | 0.674 | 20.0      | 0.827 | 20.0       | 0.893 | converges     |
| 20    | 0.2   | 15.6      | 0.283 | 20.0      | 0.496 | 20.0       | 0.679 | converges     |
| 20    | 0.3   | 3.6       | 0.038 | 19.6      | 0.409 | 20.0       | 0.504 | converges     |
| 20    | 0.4   | 0.0       | 0.000 | 8.0       | 0.130 | 20.0       | 0.461 | converges     |
| 20    | 0.5   | 0.0       | 0.000 | 0.2       | 0.004 | 1.8        | 0.025 | plateau below |

Table 10: E13, creation bias: mean final  $\hat{K}$  and mean ARI against the planted labels over five seeds, for three planted counts by six noise rates by three stream lengths, 270 runs. The regime column applies the rule declared in the results file: *converges* is mean  $\hat{K}$  within 0.5 of  $K^*$  at the longest stream, *plateau below* is mean  $\hat{K}$  below  $K^* - 0.5$  there; no row meets the growing-with- $T$  rule (slope on  $\log T$  at least 0.5 and mean  $\hat{K}$  above  $K^* + 1$ ), and no row sits above  $K^*$  at all.

| stream              | $n$   | $K^*$ | ARI mean | sd     | min    | max   | $K$ : mean (min to max) |
|---------------------|-------|-------|----------|--------|--------|-------|-------------------------|
| two-group           | 2,000 | 2     | 1.0      | 0.0    | 1.0    | 1.0   | 2 (2 to 2)              |
| eight-group noisy   | 3,000 | 8     | 1.0      | 0.0    | 1.0    | 1.0   | 8 (8 to 8)              |
| Wikipedia infoboxes | 320   | 3     | 0.8438   | 0.1029 | 0.6389 | 0.986 | 4, sd 0.77 (3 to 6)     |

Table 11: E5, order dependence: the same records streamed in 20 random orders under the one protocol (seeds 1000 to 1019). Planted structure is recovered exactly on every order; on the real stream the greedy one-pass sequence is order-dependent and the spread is its measured cost; this spread is ESCROW’s Wikipedia result. The 0.6790 of E4’s order (Table 12) and the 0.8744 of the demonstration order (Table 14) are two instances of it.

| method                               | two-group<br>( $n=2,000$ , $K^*=2$ ) | eight-group noisy<br>( $n=3,000$ , $K^*=8$ ) | Wikipedia infoboxes<br>( $n=320$ , $K^*=3$ ) |
|--------------------------------------|--------------------------------------|----------------------------------------------|----------------------------------------------|
|                                      | oracle / transferred                 | oracle / transferred                         | oracle / transferred                         |
| $k$ -means (handed $K$ / silhouette) | 1.0 / 1.0                            | 0.9977 / 0.9977                              | 0.9898 / 0.502                               |
| HDBSCAN                              | 1.0 / 0.5121                         | 0.9565 / 0.0184                              | 0.4989 / 0.0                                 |
| agglomerative                        | 1.0 / 0.0                            | 1.0 / 0.0                                    | 0.9645 / 0.0002                              |
| cosine components                    | 0.0 / 0.0                            | 0.0 / 0.0                                    | 0.4938 / 0.0002                              |
| DP-means                             | 0.994 / 0.0                          | 0.9845 / 0.0                                 | 0.7286 / 0.0017                              |
| ESCROW (ours), 20 orders             | 1.0, $K=2$                           | 1.0, $K=8$                                   | 0.8438, 0.64 to 0.99, $K$ 3 to 6             |

Table 12: E4, head to head: ARI against the true labels, as oracle / worst transferred. Oracle tunes the knob on the test labels; transferred tunes it on another stream and reports the worse of the two transfers.  $k$ -means has no knob: its first value is handed the true cluster count, its second picks the count by the label-free silhouette score. All baselines run on the same frozen MiniLM embedding, each a single run on one fixed embedding; ESCROW runs on the raw records with no embedding and no calibrated parameter, and its row is the mean and range over the 20 arrival orders of Table 11 (the synthetic entries hold on all 20), of which E4’s own order gives 0.6790 at  $K = 5$ .

| stream (records)                                                            | ESCROW, run once                                                                                                                                                                                                                   | strongest comparison without labels                                                                                                                                                                                                                                                           | strongest comparison touching labels                                                                                                                                             |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wikipedia infoboxes (320)                                                   | over 20 orders ARI 0.8438 mean, 0.6389 to 0.986, $K$ 3 to 6 (E5). On the demonstration order $K = 5$ : film 130 (119 film, 9 person, 2 mountain), person 101, mountain 86, a second film node of 10, one politician; purity 0.9665 | $k$ -means with silhouette-chosen $K = 2$ : ARI 0.502; every transferred knob scores 0.0 to 0.0017                                                                                                                                                                                            | $k$ -means handed $K = 3$ : 0.9898; agglomerative at oracle threshold 0.7: 0.9645                                                                                                |
| Lazada raw keys (21,365)                                                    | $K = 68$ covering 9,752 records (46%); 2 of 340 case-variant key pairs in a shared node, 260 with neither key in any support                                                                                                       | none by protocol: a structure exhibit                                                                                                                                                                                                                                                         | silver product type, itself model-derived: ARI 0.0013 against 5,518 types                                                                                                        |
| MusicBrainz 20K, condition A (19,375)                                       | $K = 4,594$ ; pairwise F1 0.0001, ARI about zero                                                                                                                                                                                   | none run without labels                                                                                                                                                                                                                                                                       | $k$ -means handed $K = 10,000$ : 0.0016; HDBSCAN oracle 0.0075; agglomerative oracle 0.0003; DP-means oracle 0.0005                                                              |
| ReVerb45K test (12,327)                                                     | 75 nodes over 209 records, 12,118 in the background; macro F1 0.0160, micro 0.7777, pairwise 0.7160 at pairwise precision 0.7690                                                                                                   | the no-merge floor, any method that merges nothing: pairwise 0.7677 at precision 0.9035                                                                                                                                                                                                       | DP-means oracle: micro 0.7871, pairwise 0.7708; $k$ -means handed $K = 6,061$ : macro 0.3537; CESI as published, with side information: macro 0.627, micro 0.844, pairwise 0.819 |
| Wikipedia (320) and Lazada first 1,000, against a language model (Table 14) | Wikipedia: ARI 0.8744, purity 0.9656, $K = 5$ , identical on three runs, 0 tokens, 0.17 s. Lazada: $K = 6$ spec templates, identical on three runs, 699 of 1,000 records in the background, 0 tokens, 1.5 s                        | Qwen2.5-14B-Instruct building the graph by prompting, no labels fed: Wikipedia ARI 0.9905, purity 1.0, $K = 4$ , determinism 0.994, about 46,400 tokens and 365 s per run; Lazada determinism 0.1647, $K$ of 24, 78 and 28, largest node 0.5498 of records, 107,512 to 157,821 tokens per run | the Lazada silver product type on the same 1,000 records, 759 types: ARI 0.0002 (ESCROW), 0.002 (model, sampled mean), 0.0094 (model, greedy)                                    |

Table 13: Real-world standing at a glance. Every number is copied from the results file named in the source comment; the runs are detailed in the subsections that follow, and each is one arrival order. A transferred knob is the worse of two transfers; an oracle knob is chosen on the test labels.

| stream                                 | method                          | ARI    | purity | $K$  | determinism | largest node | tokens  | wall (s) |
|----------------------------------------|---------------------------------|--------|--------|------|-------------|--------------|---------|----------|
| Wikipedia (320 records, 3 gold types)  | model, sampled, mean of 3 seeds | 0.9905 | 1.0    | 4    | 0.994       | 0.3709       | 46,384  | 364.5    |
|                                        | model, greedy                   | 0.9896 | 1.0    | 4    | run once    | 0.3688       | 46,420  | 353.2    |
|                                        | ESCROW, three runs              | 0.8744 | 0.9656 | 5    | 1.0         | 0.4062       | 0       | 0.17     |
| Lazada, first 1,000 (759 silver types) | model, sampled, mean of 3 seeds | 0.002  | 0.0778 | 43.3 | 0.1647      | 0.5498       | 125,992 | 1,710.1  |
|                                        | model, greedy                   | 0.0094 | 0.0802 | 28   | run once    | 0.2315       | 107,512 | 1,603.2  |
|                                        | ESCROW, three runs              | 0.0002 | 0.0371 | 6    | 1.0         | 0.0701       | 0       | 2.40     |

Table 14: A language model building the same graph, against ESCROW on the identical records in the identical order. The model is Qwen2.5-14B-Instruct (Qwen Team et al., 2024) in bf16 on one A100, prompted to create or attach: chunks of 20 records in arrival order, the current node list in the prompt, a JSON answer per record naming an existing node or a new one, no cluster count given; sampled at temperature 0.7 with seeds 0, 1, 2 (the mean is reported; the sampled-mean  $K$  of 43.3 is 24, 78 and 28) and once greedy, which ran once by design. ESCROW ran three times on one arrival order under the one protocol of Section G.1. ESCROW’s background (745 of the 1,000 Lazada records, none on Wikipedia) is not a node and is not counted in its largest node share. The Lazada silver truth is AutoPKG’s model-derived product type, 759 types over 1,000 records, 614 of them singletons, so ARI against it is near zero for every method. Tokens are prompt plus completion under the model’s own tokenizer; wall time is per run, the model on one A100 and ESCROW on one CPU core including repair. Every number is from `results/llm_graph_formation.json`.

| representation                                       | $k$ chosen with no label | ARI    | at an oracle $k$ |
|------------------------------------------------------|--------------------------|--------|------------------|
| term frequency over the record as text               | 3                        | 1.0000 | 1.0000           |
| character 3 to 5 grams                               | 3                        | 1.0000 | 1.0000           |
| key set, cosine                                      | 3                        | 0.9898 | 0.9898           |
| key presence, multi-hot                              | 3                        | 0.9797 | 0.9797           |
| (key, value) and key presence, the E21 matched input | 3                        | 0.9797 | 0.9797           |
| (key, value) pairs only                              | 2                        | 0.1205 | 0.3930           |
| Qwen2.5-14B-Instruct, sampled mean                   | (none to choose)         | 0.9905 |                  |
| ESCROW                                               | (none to choose)         | 0.8744 |                  |

Table 15: A code over character n-grams cannot represent a word and it separates the three types perfectly, so this stream ranks methods by vocabulary. The last row of the upper block is the useful one: the values alone reach 0.1205, so what separates these types is which keys a record carries.

| rule                                              | knob                         | precision | recall | F1     |
|---------------------------------------------------|------------------------------|-----------|--------|--------|
| <code>norm_key</code> , the gold’s own definition | fixed                        | 0.9266    | 1.0000 | 0.9619 |
| lowercase, drop non-alphanumeric                  | fixed                        | 0.9266    | 1.0000 | 0.9619 |
| lowercase, drop whitespace                        | fixed                        | 1.0000    | 0.7043 | 0.8265 |
| <code>str.lower()</code> and compare              | fixed                        | 1.0000    | 0.6921 | 0.8180 |
| token Jaccard                                     | 1.0                          | 0.9938    | 0.9848 | 0.9893 |
| diff lib edit ratio                               | 0.85                         | 0.9116    | 0.9116 | 0.9116 |
| character trigram cosine                          | 0.8                          | 0.9558    | 0.7256 | 0.8250 |
| shared prefix                                     | 3                            | 0.3461    | 0.9360 | 0.5053 |
| AutoPKG’s decision agent                          | prompt, $k = 10$ , $T = 0.7$ | 0.6497    | 0.6220 | 0.6355 |
| ESCROW                                            | (none to choose)             | 0.0005    | 0.0030 | 0.0008 |

Table 16: Six of the eight string rules score above the language model, and plain lowercasing beats it by 18 F1 points. What the gold does not record is two different names for one role, which is the decision Section 5.4 says is open for this construction.

| method                              | pairwise F1 | clusters    |
|-------------------------------------|-------------|-------------|
| ESCROW (ours), run once             | 0.0001      | 4,594 nodes |
| $k$ -means, handed $K=10,000$       | 0.0016      | 10,000      |
| HDBSCAN, oracle min size 3          | 0.0075      | 7,327       |
| agglomerative, oracle threshold 0.1 | 0.0003      | 1,924       |
| DP-means, oracle $\lambda=0.1$      | 0.0005      | 1,990       |

Table 17: MusicBrainz 20K, condition A: five categorical fields, no text, one arrival order. Pairwise F1 against the 10,000 true entities over 19,375 records. Every method is null. ESCROW mints 4,594 nodes at pairwise F1 0.0001; each baseline emits thousands of clusters of noise. Baseline knobs are chosen on the test labels.

| method                                 | macro  | micro  | pair   | pair prec. | clusters |
|----------------------------------------|--------|--------|--------|------------|----------|
| ESCROW (ours), run once                | 0.0160 | 0.7777 | 0.7160 | 0.7690     | 12,192   |
| HDBSCAN, oracle min size 50, no merges | 0.0138 | 0.7830 | 0.7677 | 0.9035     | 12,327   |
| agglomerative at 0.1, best pairwise F1 | 0.0391 | 0.7849 | 0.7679 | 0.9036     | 12,203   |
| agglomerative at 0.2, best macro F1    | 0.1002 | 0.7863 | 0.7192 | 0.7287     | 11,822   |
| DP-means, oracle $\lambda=0.25$        | 0.0504 | 0.7871 | 0.7708 | 0.9034     | 12,142   |
| $k$ -means, handed $K=6,061$           | 0.3537 | 0.6559 | 0.3729 | 0.2388     | 6,061    |
| CESI, published, side information      | 0.627  | 0.844  | 0.819  |            |          |

Table 18: ReVerb45K test set under CESI’s metrics code, one arrival order: macro, micro and pairwise F1, pairwise precision, and cluster count over 12,327 subject strings (6,061 gold clusters), background records counted as singletons. A method that merges nothing scores pairwise 0.7677 at precision 0.9035 because same-string mentions are one cluster by construction; ESCROW’s 75 nodes lower precision to 0.7690, so some merges still cross entity lines, and 7,009 subjects end as singletons, which is why macro F1 is 0.016. Only  $k$ -means, handed the gold count, exceeds macro 0.35. CESI’s published row uses side information no method here sees.